

**VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING**



**REPORT
SPECIALIZED PROJECT**

Building a software system managing tours

Major: Computer Science

COMMITTEE : Computer Science – 09 CLC

SUPERVISORS : Dr. Truong Tuan Anh

Mr. Mai Duc Trung, MSc

— o0o —

STUDENT 1 : Nguyen My Khanh **ID:** 2052525

STUDENT 2 : Pham Quynh Tran **ID:** 2053518

STUDENT 3 : Tran Gia Linh **ID:** 2053182

HO CHI MINH CITY, DECEMBER 2023

**VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING**



**REPORT
SPECIALIZED PROJECT**

Building a software system managing tours

Major: Computer Science

COMMITTEE : Computer Science – 09 CLC

SUPERVISORS : Dr. Truong Tuan Anh

Mr. Mai Duc Trung, MSc

— o0o —

STUDENT 1 : Nguyen My Khanh **ID:** 2052525

STUDENT 2 : Pham Quynh Tran **ID:** 2053518

STUDENT 3 : Tran Gia Linh **ID:** 2053182

HO CHI MINH CITY, DECEMBER 2023

Declaration of Authenticity

Our group declares that this Specialized Project in the field of Software Engineering, majoring in Computer Science, with the topic “Building a software system managing tours” has been written by ourselves under the supervision of Dr. Truong Tuan Anh and Mr. Mai Duc Trung, MSc.

In the process of researching and implementing this Specialized Project, we have referenced multiple studies and documents from other authors. All of them have been fully and clearly stated in the References section.

The authors,

Nguyen My Khanh

Pham Quynh Tran

Tran Gia Linh

Acknowledgement

In completing this specialized project we have been fortunate to have help, support and encouragement from many people. I would like to acknowledge them for their cooperation.

First, we would like to express our thanks and gratitude to Mr. Truong Tuan Anh, D.Sc and Mr. Mai Duc Trung, MSc - our project advisors, for guiding us through each and every step of the process with knowledge and support. Thank you for your advice, guidance and assistance throughout the time.

We would also like to thank the teachers of the University of Technology - National University of Ho Chi Minh City who taught us throughout our study at school. Especially, the teachers of the Department of Computer Science have brought us extremely valuable knowledge, not only the foundation for the project research process but also a valuable package for us to apply into reality firmly and confidently.

Due to limited time and our own abilities, we still have many shortcomings and limitations. We look forward to receiving your contributions to improve the project.

Finally, we would like to sincerely thank and wish you good health, success and happiness!

Ho Chi Minh City, December 2023

The authors,

Nguyen My Khanh

Pham Quynh Tran

Tran Gia Linh

Abstract

Nowadays, tourism is one of the fastest growing industries. It is undeniable that traveling has become much more popular as compared to the past. As technology advances more and more with each passing year, the concept of traveling is also changing along with it. People have become more familiar with and comfortable using the Internet. They want to be in control of finding “experiences” and creating their reservations. An automated online booking system will allow them to easily and quickly make bookings.

Our group decided to implement a tourism website that provides travelers in Vietnam with the best experience in booking domestic tours. Tourists have a variety of choices for tours and they can easily and quickly book a tour and track their transactions. Tourists can discover the highlights of Vietnam and plan their trip with the suggestion on Vietnam's best destinations, what to see and do there. user-friendly interfaces along with useful and helpful accessibility. Because there are so many competing websites nowadays, it's very important to have user-friendly interfaces along with useful and helpful accessibility.

Besides, we also aim to build a tour management system. Developing a system for both the staff and the managers required us to come up with tailored solutions for each party. For managers, we will give them the maximum decision-making power and assist them with data visualizations or the dashboard to help them in making efficient decisions regarding the company's strategic plan. For staff, they will be assigned a list of tours and their responsibilities are to create and manage the tours. Of course, any changes need to be sent to the managers for approval. Different from the system for customers, we strive to deliver a system that is simple, focused on functions, reliable and efficient.

Contents

Abstract	3
List of Tables	12
Chapter 1 Introduction	16
1.1. Motivation	16
1.2. The Project's Scope and Objectives	16
1.3. Project structure	17
Chapter 2 Requirement elicitation.....	20
2.1. Related works	20
2.1.1. Tourism websites.....	20
2.1.1.1. Vietnam Booking.....	20
2.1.1.2. Saigontourist.....	22
2.1.2. Service management software.....	26
2.1.2.1. Microsoft Dynamics 365	26
2.1.2.2. HubSpot-CRM.....	27
2.2. Stakeholders	29
2.3. Functional and non-functional requirements.....	30
2.3.1. Functional requirements	30
2.3.2. Non-Functional requirements.....	32
Chapter 3 System design.....	34
3.1. Use Case Diagrams.....	34
3.1.1. Guest/Customer Use Case	36
3.1.2. Staff Use Case	45
3.1.3. Manager Use Case	51
3.2. Activity Diagrams	57
3.2.1. Customer and Guest Activity Diagrams.....	57
3.2.2. Customer Activity Diagrams.....	60
3.2.3. Staff Activity Diagrams.....	63
3.2.4. Manager Activity Diagrams	65
3.4. Database design	75
3.4.1. Enhanced Entity Relationship Diagram (EERD)	75

3.4.2. Relational mapping.....	78
3.5. Architecture design.....	78
3.5.1. Architectural pattern.....	78
3.5.2. Architecture diagram.....	79
3.6. Component diagram	80
3.7. Class diagram	84
3.7.1. View layer	85
3.7.2. Controller layer.....	85
3.7.3. Service layer	86
3.7.4. Repository layer.....	87
3.7.5. Model layer.....	88
Chapter 4 Technology research	90
4.1. Front-end development research	90
4.1.1. AngularJS	90
4.1.2. Vue.....	92
4.1.3. React.....	93
4.1.3.1. Component	95
4.1.3.2. React Router Dom	95
4.1.3.3. React Memo.....	96
4.1.3.4. React Redux.....	96
4.1.3.5. Axios.....	98
4.1.4. CSS framework	98
4.2. Server development research.....	99
4.2.1. Node.js (Express.js).....	100
4.2.2. Ruby on Rails	101
4.2.3. Django	101
4.3. Database management system research	103
4.3.1. Postgres.....	103
4.3.2. MySQL	103
4.4. Testing technology research.....	104
4.4.1. API testing	104
4.4.1.1. REST Assured	104

4.4.1.2. Google Apigee.....	106
4.4.1.3. Postman	107
4.4.2. Performance testing	109
4.4.2.1. Artillery	109
4.4.2.2. Apache Jmeter	110
4.4.2.3. Autocannon.....	110
Chapter 5 System implementation.....	112
5.1. Wireframe design	112
5.1.1. Wireframe for Customer View.....	112
5.1.1.1. UI Flow Diagram.....	112
5.1.1.2. Detailed UI design.....	113
5.1.2. Wireframe for Staff View.....	134
5.1.2.1. UI Flow Diagram.....	134
5.1.2.2. Detailed UI design.....	134
5.1.3. Wireframe for Manager View	146
5.1.3.1. UI Flow Diagram.....	146
5.1.3.2. Detailed UI design.....	146
5.2. Application building.....	160
Chapter 6 Summary and Future plan	165
6.1. Task evaluation.....	165
6.2. Future plan.....	165
Appendix.....	170
1. Work contribution	170
2. Work process	172

List of Figures, Diagrams

Figure 1. Vietnam Booking payment methods.....	21
Figure 2. Vietnam Booking search box.....	22
Figure 3. Saigontourist website interface	23
Figure 4. Saigontourist search bar	24
Figure 5. MS Dynamics 365 report view	27
Figure 6. HubSpot service management view.....	28
Figure 8. General use-case diagram for guest and customer.....	36
Figure 9. Use-case diagram for customer/guest - login module.....	37
Figure 10. Use-case diagram for customer - personal profile management module.....	37
Figure 11. Use-case diagram for customer/guest - tour browsing module	38
Figure 12. Use-case diagram for customer/guest - news browsing module.....	38
Figure 13. Use-case diagram for customer/guest - booking module.....	39
Figure 14. General use-case diagram for staff.....	45
Figure 15. Use-case diagram for staff - tour management module	46
Figure 16. General use-case diagram for manager.....	51
Figure 17. Use-case diagram for manager - staff management module.....	52
Figure 18. Use-case diagram for manager - tour management module	52
Figure 19. Activity diagram for guest/customer online tour booking function	58
Figure 20. Activity diagram for guest/customer viewing location information function	59
Figure 21. Activity diagram for customer login function	60
Figure 22. Activity diagram for customer ratings and reviews function	62
Figure 23. Activity diagram for staff login function	63
Figure 24. Activity diagram for staff tour management function	65
Figure 25. Activity diagram for staff account deactivation function	66

Figure 26. Activity diagram for manager tour management function.....	67
Figure 27. Sequence Diagram for Customer Login and Signup	68
Figure 28. Sequence Diagram for Customer Viewing Detailed Information of a Tour.....	69
Figure 29. Sequence Diagram for Customer Booking and Paying for a Tour	70
Figure 30. Sequence Diagram for Customer View News and Tours Suggestions for a Place.....	70
Figure 31. Sequence Diagram for Staff Login	71
Figure 32. Sequence Diagram for Staff Making a Request.....	72
Figure 33. Sequence Diagram for Staff Adding a Customer into a Tour.....	72
Figure 34. Sequence Diagram for Manager Logging in the system.....	73
Figure 35. Sequence Diagram for Manager Viewing and Determining a Request.....	74
Figure 36. Sequence Diagram for Manager Deactivating an Account	74
Figure 37. Enhanced Entity Relationship Diagram (EERD).....	77
Figure 38. Relational Mapping of EERD	78
Figure 39. MVCS pattern	79
Figure 40. Architecture Diagram for the system.....	80
Figure 41. Component Diagram for Customer.....	81
Figure 42. Component diagram for Staff and Manager	83
Figure 43. Class diagram.....	84
Figure 44. Class diagram in Repository Layer.....	85
Figure 45. Class diagram in Controller Layer	86
Figure 46. Class diagram in Service Layer	86
Figure 47. Class diagram in Repository Layer.....	88
Figure 48. Class diagram in Model Layer	89
Figure 49. Redux architecture	97
Figure 50. UI Flow Diagram for Customer View	112
Figure 51. Design for header	114

Figure 52. Design for footer	115
Figure 53. Design for registration form.....	116
Figure 54. Design for login form.....	117
Figure 55. Design for forgot password form	117
Figure 56. Design for password reset form	118
Figure 57. Design for Guest and Customer homepage (1).....	119
Figure 58. Design for Guest and Customer homepage (2).....	120
Figure 59. Design for Guest and Customer homepage (3).....	121
Figure 60. Design for Guest and Customer homepage(4).....	121
Figure 61. Design for domestic tour list page	122
Figure 62. Design for tour details page (1)	123
Figure 63. Design for tour details page (2)	124
Figure 64. Design for average ratings and reviews section.....	125
Figure 65. Design for related tour suggestion section.....	126
Figure 66. Design for booking step 1 - Filling in booking information.....	127
Figure 67. Design for booking step 2 - Make payment.....	127
Figure 68. Design for booking step 3 - Review booking	128
Figure 69. Design for booking history section	129
Figure 70. Design for feedback form	129
Figure 71. Design for password change section.....	130
Figure 72. Design for contact information page.....	131
Figure 73. Design for tourist attraction news page	132
Figure 74. Design for detailed article section.....	133
Figure 75. UI Flow Diagram for Staff View	134
Figure 76. Design for Header in staff view	134
Figure 77. Design for staff and manager login form.....	135
Figure 78. Change password form for the first login	136

Figure 79. Design for password change	137
Figure 80. Design for tour list management page	137
Figure 81. Design for adding a new tour function	138
Figure 82. Design for view of tour details.....	139
Figure 83. Design for add-new-customer form	140
Figure 84. Design for tour cancellation request view	141
Figure 85. Design for tour edit request view.....	142
Figure 86. Design for request management page	143
Figure 87. Design for view of a sent request (cancellation request)	143
Figure 88. Design for view of a sent request (edit request)	144
Figure 89. Design for editing a sent request (edit request)	145
Figure 90. UI Flow Diagram for Manager View.....	146
Figure 91. Design for Header in manager view	146
Figure 92. Design for password change	148
Figure 93. Design for staff management page	148
Figure 94. Design for adding a new staff member function.....	149
Figure 95. Design for staff information view (case 1)	149
Figure 96. Design for staff information view (case 2)	150
Figure 97. Design for editing staff information	150
Figure 98. Design for tour list management page	151
Figure 99. Design for view of tour details.....	152
Figure 100. Design for tour details editing form.....	153
Figure 101. Design for requests management page	154
Figure 102. Design for request details view (edit request).....	155
Figure 103. Design for view to reject an edit request	156
Figure 104. Design for request details view (cancellation request)	157
Figure 105. Design for view to reject a cancellation request	158

Figure 106. Design for report and statistics page.....	159
Figure 107. Login form implementation	160
Figure 108. Registration form implementation	161
Figure 109. Forgot password form implementation	161
Figure 110. Reset password form implementation for the first login of Staff	162
Figure 111. Header and Footer implementation.....	162
Figure 112. Homepage implementation - First section	163
Figure 113. Homepage implementation - Second section.....	163
Figure 114. Homepage implementation - Second section.....	164
Figure 115. Homepage implementation - Fourth section.....	164

List of Tables

Table 1. Comparison of existing systems and our system	25
Table 2. Summary table of all Use-cases of the System	34
Table 3. Description table for Login use-case.....	39
Table 4. Description table for Logout use-case.....	40
Table 5. Description table for Signup use-case	41
Table 6. Description table for Operate with history page use-case.....	41
Table 7. Description table for Change information use-case	42
Table 8. Description table for Browse all tours use-case	43
Table 9. Description table for Browse all destination news use-case	43
Table 10. Description table for Book tour use-case	44
Table 11. Description table for Login use-case.....	46
Table 12. Description table for Logout use-case.....	47
Table 13. Description table for View assigned tours use-case.....	48
Table 14. Description table for Operate with assigned tours use-case.....	48
Table 15. Description table for View list of requests use-case	49
Table 16. Description table for Operate with requests use-case	49
Table 17. Description table for View rating & reviews use-case.....	50
Table 18. Description table for Login use-case.....	53
Table 19. Description table for Logout use-case.....	53
Table 20. Description table for View staff list use-case.....	54
Table 21. Description table for Operate with staff list use-case	54
Table 22. Description table for View Tours use-case	55
Table 23. Description table for Operate with tours use-case	56
Table 24. Description table for Operate with requests use-case	56
Table 25. Description table for Operate with Dashboard use-case	57

Table 26. Some features of Aritllery testing tool	109
Table 27. Implementation plan for the next phase	166
Table 28. Work contribution	170

List of symbols and abbreviations

1. AJAX - Asynchronous Javascript and XML
2. AWS - Amazon Web Services
3. API - Application Programming Interface
4. CI/CD - Continuous Integration / Continuous Delivery
5. CLI - Command Line Interface
6. CSS - Cascading Style Sheet
7. CRUD - Create, Read, Update, Delete
8. DOM - Document Object Model
9. GUI - Graphical User Interface
10. HTML - Hyper Text Markup Language
11. HTTP - HyperText Transfer Protocol
12. HTTPS - Hypertext Transfer Protocol Security
13. JS - Javascript
14. JSP - Java Server Pages
15. MPL-2.0 - Mozilla Public License 2.0
16. MVC - Model View Controller
17. MVCS - Model View Controller Service
18. REST - Representational State Transfer
19. SEO - Search Engine Optimization
20. SQA - Software Quality Assurance
21. UI - user interface
22. URL - Uniform Resource Locator

GLOSSARY OF TERMS

Term	Definition
Framework	A pre-built set of libraries, software tools, and best practices that enable developers to create applications more quickly and efficiently.
Rendering	The process of generating HTML markup to display web pages in the browser
Responsive design	Responsive design refers to a site or application design that responds to the environment in which it is viewed. [1]
Data binding	Data binding is the process that couples two data sources together and synchronizes them [2]
Memoization	In computing, memoization is used to speed up computer programs by eliminating the repetitive computation of results, and by avoiding repeated calls to functions that process the same input.
Benchmark	The act of running a computer program, a set of programs, or other operations, in order to assess the relative performance of an object, normally by running a number of standard tests and trials against it.
Database	An organized collection of data, generally stored and accessed electronically from a computer system.
Interface	A shared boundary across which two or more separate components of a computer system exchange information.

Chapter 1

Introduction

The goal of Chapter 1 is to introduce the research topic “Building a software system managing tours”, the topic motivation, the scope along with the project's objectives, and the project structure.

1.1. Motivation

In the era of technology 4.0, which has been evolving today, along with "life in the fast lane" influenced by intense competition between many different economies. Besides the positive effect that the quality of life of the vast majority of people has improved, even become prosperous. People are gradually immersed in work. Previously, the average working time of one person was eight hours per day. Currently, people tend to work more than the standard time by working overtime or finding a part-time job after work to earn as much income as possible. Although that is not bad, spending most of the time jumping head first into work means that the rest time for relaxation is reduced, or even a deficit. Excessive work and stress at work for a long time can easily cause damage to mental health, thereby affecting daily life. Understanding the importance of balancing material and spiritual life, many people no longer impose too much on themselves at work. Instead, they choose to refresh themselves after exhausted working time via many kinds of outdoor activities. And one of them that is impossible not to mention is tourism.

1.2. The Project's Scope and Objectives

With the aim of providing travelers with the experience of exploring all the attractive places in Vietnam, our team decided to develop a software system which is a domestic tour booking website with a user-friendly interface along with useful and helpful accessibility. Although the idea of developing a web application for travel services is no longer new, our team's application will bring a completely different experience compared to other web versions by optimizing and overcoming limited functions or exploring and integrating those that have not been widely used.

For users, our team's web application allows them to access, search, book tours, or check related information without logging in to their account. Users can create their accounts, log in to save their transaction history, and give feedback on the booked tours.

For management, only managers and employees - those on duty - can access. Each account will access a different tour management interface, depending on whether the account belongs to a manager or an employee. If the users are managers, they are responsible for approving edit/create/remove requests of tours and managing staff (view list, add, delete). Besides, they can also view the dashboard to update the situation of the business and print it out as a file if they need it. If the users are employees, they are subject to more constraints. They cannot view the entire tours of all other staff, instead, they can only view their created tours. Even if staff want to adjust the tour information, they cannot arbitrarily edit or change anything unless their edit request proposed to the manager is approved.

The above approach of organizing and operating our group system helps clearly distinguish the authorities and duties of each position - between managers and employees. Simultaneously, this helps better and more closely manage business information and data.

In general, it can be said that our team's web application is not the most perfect version. However, we will try to apply all the knowledge that we learned and do our best to bring users an experience that does not disappoint them. Hopefully, this web application of our group will be known and trusted by many users.

1.3. Project structure

Below is a summary of the entire project's content. The content of each chapter is listed in detail as follows:

Chapter 1: Introduction

- Part 1: Present the reasons why we chose the “Tourism management system” topic.

- Part 2: Summarize the scope as well as objectives of the project - Which problems the project handles, Who the project aims at, Which functions the system provides, and How users interact with it.

Chapter 2: Requirement elicitation

- Part 1: Present some current tourism websites as well as some field service management software - Overview, advantages, and disadvantages.
- Part 2: List out related users.
- Part 3: Present functional requirements for each specific user type and non-functional requirements for the whole system.

Chapter 3: System design

- Part 1: Present Use Case Diagram Figures along with explanations and Use Case Scenarios corresponding to each kind of user.
- Part 2: Present Activity Diagram Figures along with explanations corresponding to each kind of user.
- Part 3: Present Sequence Diagram Figures along with explanations corresponding to each kind of user.
- Part 4: Present and explain the Database design for the whole system - EERD and relational mapping.
- Part 5: Present the Architecture design Figures of the project along with explanations.
- Part 6: Present the Component diagram Figures for the whole system along with explanations.
- Part 7: Present the Class diagram Figures along with explanations corresponding to each kind of layer.

Chapter 4: Technology research

- Part 1: Present some front-end frameworks, compare them, and indicate one chosen framework along with explanations.

- Part 2: Present some back-end technologies, compare them, and indicate one chosen technology along with explanations.
- Part 3: Present some database management software, compare them, and indicate one chosen software along with explanations.
- Part 4: Present some testing tools for API and Performance - compare them and indicate one chosen tool along with explanations.

Chapter 5: System implementation

- Part 1: Present UI Flow Diagram Figures, Detailed UI design Figures, and explain step-by-step operations corresponding to each kind of user.
- Part 2: Present some user interface figures that were implemented.

Chapter 6: Summary and Future plan

- Part 1: Present what has been achieved during this specialized project.
- Part 2: Present the plan table for the project in the next phase.

Appendix: Present the contribution as well as the implementation plan of each member in this project.

Chapter 2

Requirement elicitation

2.1. Related works

2.1.1. Tourism websites

2.1.1.1. Vietnam Booking

Vietnam Booking is a top search website providing information on booking domestic and international tours with specific information and prices.

Functionality

- **Account registration:** Vietnam Booking allows account registration for customers. Customers who already have an account will be able to log in or reset their password if they forget their password.
- **Tour browsing:** Customers can view domestic and international tour lists.
- **Tour searching:** Customers can search for a tour by choosing the trip destinations, departure dates, etc.
- **Tour detail viewing:** Customers can click on a tour to view its detailed information.
- **Online tour booking:** Customers can book the tour online.
- **Location news browsing:** Customers can view the articles about some tourist attractions.
- **Transaction history:** After login, customers can view all the tours they have booked in their personal profile.

Strengths

- **Responsive design:** The website interface is a responsive design that looks good on all devices.

- **User-friendly interface:** users can easily understand and navigate through the application in an efficient way.
- **Multiple customer support methods:** The website provides a variety of communication and support methods for customers such as live chat, chat through zalo/messenger, call with Whatsapp, help center and contact information.
- **Diverse images:** Vietnam Booking provides numerous real-life images of each location in the tour. These are images depicting scenic landscapes, entertainment activities and local people.
- **Multiple payment methods:** Vietnam Booking integrates an online tour booking feature with multiple payment methods available for customer convenience, as you can see in Figure 1.
- **Well divided search dropdown:** In Figure 2, when a customer enters into the search bar, the box shows a quite complete and detailed dropdown menu, divided into two parts: the destinations and the tours.

The screenshot displays the 'Thanh toán' (Payment) page on the Vietnam Booking website. At the top, there are three progress indicators: 'Điền thông tin' (Fill in information), 'Thanh Toán' (Payment), and 'Xác nhận' (Confirmation). The main content area is divided into three sections:

- Thanh toán qua** (Payment via): A sidebar on the left lists various methods including 'Thẻ ATM nội địa', 'Thẻ tín dụng quốc tế', 'ZaloPay', 'Chuyển khoản', 'Tại cửa hàng', 'Tại văn phòng', and 'Tại nhà'.
- Thanh toán qua thẻ ATM nội địa** (Payment via domestic ATM card): The main section shows instructions for using a domestic ATM card, including a warning to check the card's validity and a 'XÁC NHẬN' (Confirm) button.
- Tour summary**: A sidebar on the right provides details about the selected tour, including the tour name, departure date, number of guests, and the total price.

The tour summary details are as follows:

Item	Value
Mã đơn hàng	VNBKTS749948179
Mã tour	TOHCMCHAHAQ5NSD-MBQK-437910
Ngày khởi hành	28/08/2023
Số khách	1 khách
Giá 1 khách	22,690,000 VND
Tổng tiền	22,690,000 VND

At the bottom, there is a 'Security Payment' logo and a contact number: 'Gọi 1900 3398 để được hỗ trợ 24/7'.

Figure 1. Vietnam Booking payment methods

Limitations

- **Too complicated tour naming:** In Figure 2, we can see that the tour names are quite complicated and do not follow a certain rule. Therefore, it is very hard and time-consuming for customers to find the desired tours.

- **Limited in search and filter options:** The limitation in terms of search and filter options such as departure date, number of days, and transportation will make it difficult for customers to find the desired tour.

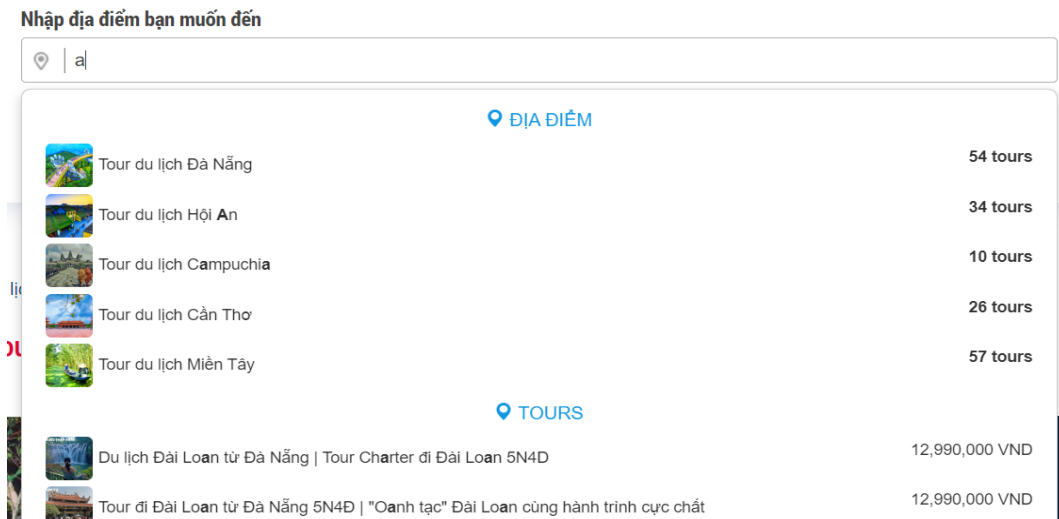


Figure 2. Vietnam Booking search box

2.1.1.2. Saigontourist

Saigontourist is also a popular travel website that provides domestic and international tours and an online booking feature with many payment methods.

Functionality

- **Account registration:** Customers can register a new account and login or they can log in with Google or Facebook. Saigontourist also supports customers with the password reset function if they forgot their password.
- **Tour browsing:** Customers can view domestic and international tour lists.
- **Tour searching:** Customers can search a tour by the name, departure dates, etc.
- **Tour filtering:** Customers can filter the tour list by date and price.
- **Tour detail viewing:** Customers can click on a tour to view its detailed information.
- **Online tour booking:** Customers can book the tour online.
- **Transaction history:** After login, customers can view all the tours they have booked in their personal profile.

Strengths

- **Responsive design:** The website interface is a responsive design that looks good on all devices.
- **Well-organized and user-friendly interface:** The website layout is well-organized, user-friendly, and clear so that users can quickly understand and traverse through the application in an efficient way.
- **Multiple customer support methods:** The website provides a variety of communication and support methods for customers such as live chat, help center and contact information.
- **Attractive images:** As you can see in Figure 3, Saigontourist provides many realistic, beautiful, and attractive images of the tourist attractions.
- **Multiple payment methods:** Vietnam Booking integrates an online tour booking feature with multiple payment methods available for customer convenience.



Figure 3. Saigontourist website interface

Limitations

- **Limited in tour information:** The information of a tour that the website provides is quite limited. For example, some information such as remaining available seats or the average ratings for that tour.

- **Difficult and time-consuming account registration:** Customers have to fill in a lot of information to register an account which will make them irritated.
- **Limited in search and filter options:** The website provides only one search box for customers to input the tour name. Entering the tour name will make it harder to search for the tours than entering some separate options such as departure point and destination points. Moreover, lack of options in the tour filter, such as transportation or number of days, can make customers spend more time finding the tour they want.
- **No dropdown when searching:** When you look at the Figure 4, you can see there is no dropdown menu when searching for items, so customers will not have suggestions for their search.

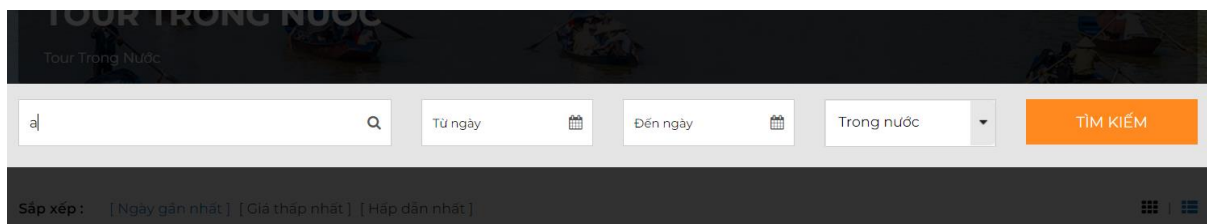


Figure 4. Saigontourist search bar

To conclude our research, we present a table to summarize each system's strengths and limitations.

Table 1. Comparison of existing systems and our system

	Account registration	Tour searching	Tour filtering	Location news	Online tour booking	Ratings & reviews	Transaction history	Responsive design
Vietnam Booking	✓	✓	✗	✓	✓	✗	✓	✓
Saigontourist	✓	✓	✓	✗	✓	✗	✓	✓
KBDulich (Our system)	✓	✓	✓	✓	✓	✓	✓	✓

2.1.2. Service management software

2.1.2.1. Microsoft Dynamics 365

Microsoft Dynamics is a line of integrated, adaptable business management solutions that enable your people to make important business decisions with greater confidence. Microsoft Dynamics works like and with familiar Microsoft software adoption, reducing the risks inherent in implementing a new solution.

Functionality

- Microsoft Dynamics 365 CRM helps businesses manage customer interactions and relationships. It includes modules for sales, marketing, customer service, and field service.
- Microsoft Dynamics 365 enables staff manage tasks and information.
- Microsoft Dynamics 365 helps tracking task performance through productivity charts and reports.
- Microsoft Dynamics 365 provides tools for customer support, case management, and knowledge base management.
- Microsoft Dynamics 365 supports HR processes, such as employee onboarding, talent management, and workforce planning.

In the Figure 5, it is shown that staff can view his/her tasks in the dashboard, information on service management, productivity chart, and reports.

Strengths

- Microsoft Dynamics 365 integrates both CRM and ERP applications.
- Dynamics 365 uses a common data model, ensuring that data consistency is maintained across different modules and business processes.
- Dynamics 365 offers the advantages of accessibility, scalability, and automatic updates. Organizations can access their data and applications from anywhere with an internet connection.
- Dynamics 365 is highly customizable, allowing businesses to adapt the system to their unique needs.

- The user interface of Dynamics 365 is designed to be user-friendly, with a look and feel similar to other Microsoft products.

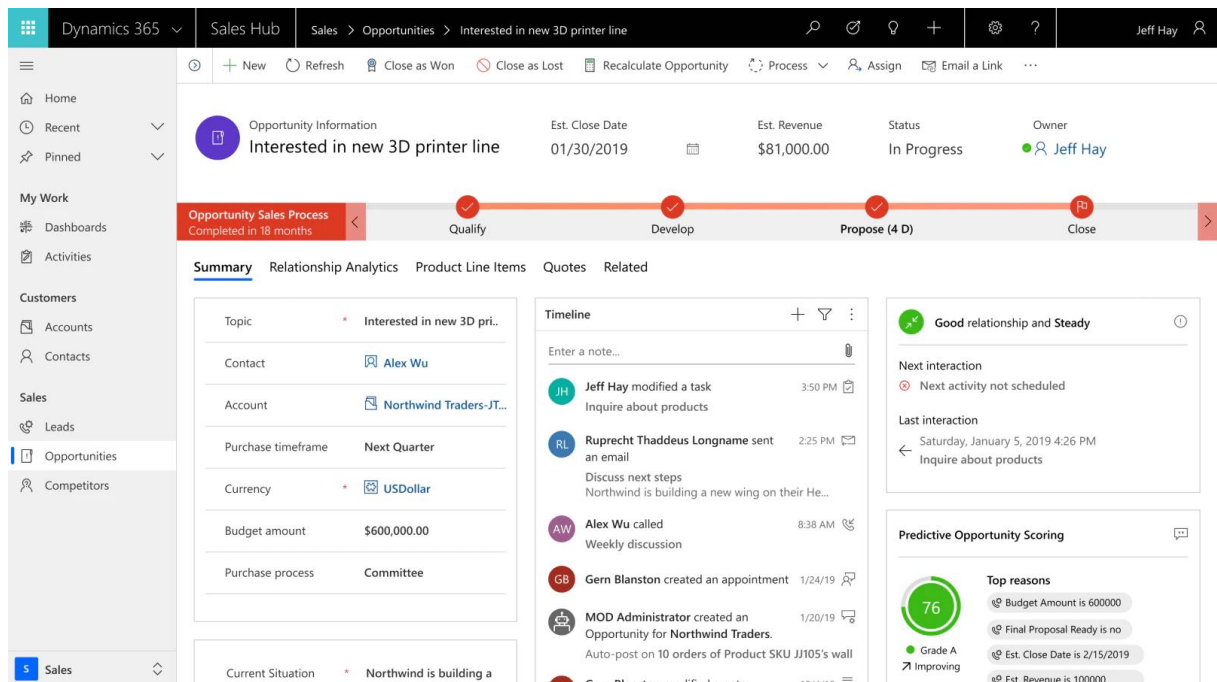


Figure 5. MS Dynamics 365 report view

Limitations

- The cost of implementing and licensing Dynamics 365 can be significant.
- Dynamics 365 relies on internet connectivity, as it is a cloud-based solution. Organizations in areas with unreliable or slow internet may experience limitations in accessibility and performance.
- Dynamics 365 is tightly integrated with other Microsoft products and services, which can be a strength but also a limitation for organizations that prefer or rely on non-Microsoft solutions.

2.1.2.2. HubSpot-CRM

HubSpot customer relationship management (CRM) software is built for businesses that want to spend less time on logging data.

Functionality

- HubSpot CRM provides a centralized database for managing contacts. Users can store and organize contact information, communication history, and interactions in one place.
- Users can schedule tasks, appointments, and follow-up activities within HubSpot CRM.
- HubSpot CRM integrates with email platforms, allowing users to log and track email communications directly within the CRM system.
- The CRM includes reporting and analytics features that provide insights into sales performance, lead sources, and other key metrics.
- HubSpot CRM supports automation through workflows.

With the view shown in Figure 6, a staff can manage all services that he/she is assigned. The design has some general features similar to our design. However, a plus point is that all services can be managed via the filter bars provided by the service management features.

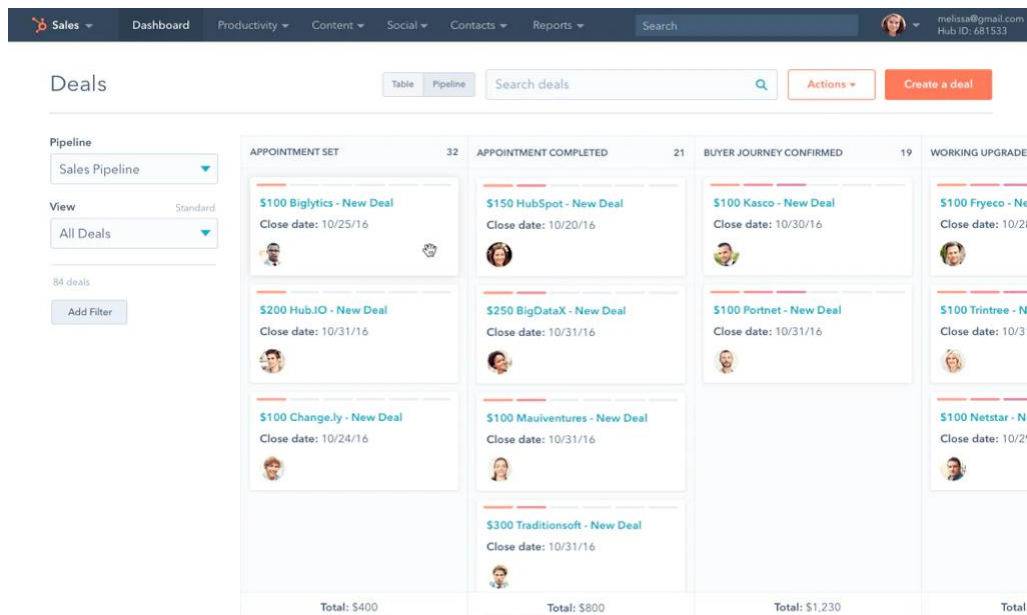


Figure 6. HubSpot service management view

Strengths

- HubSpot CRM is known for its intuitive and user-friendly interface.
- HubSpot CRM offers a free version that provides essential CRM functionalities at no cost.

- The CRM provides robust contact and lead management capabilities, allowing users to store, organize, and track information about their contacts and leads in a centralized database.
- The CRM integrates with email platforms, enabling users to log and track email communications directly within the system.

Limitations

- HubSpot CRM is well-suited for small to mid-sized businesses, but larger enterprises with complex sales processes or extensive customization needs may find that the platform lacks certain advanced functionalities.
- HubSpot CRM may not be as flexible or offers as many customization options as some other enterprise-level CRM solutions.
- Organizations with specific data residency or compliance requirements may need to carefully evaluate HubSpot's data storage and processing policies to ensure compliance with local regulations.

2.2. Stakeholders

The system includes the following user types:

- **Consumers:** Consumers include **Guests** and **Customers**. The main difference between them is that Guest are users who do not have any accounts and Customers are users who have created their accounts. Both Guests and Customers are users who have demands for finding desired tours. They will have countless choices among hundreds of tours provided by our web. Just with a few simple order steps, they can easily purchase the tours they want .
- **Staff:** Staff are responsible for managing all of their created tours. If they have any problems with the tours such as creating, editing or deleting, they can contact or exchange information with their manager.
- **Manager:** Manager is responsible for managing all the staff and tours created by the corresponding staff. Although the manager is not directly related to tours as the staff, the manager also has some right and knows the information clearly in

order to make the right decisions as well as solving the staff's problems satisfactorily.

Each type of user will have roles and perform different functions detailed in the Functional requirements section.

2.3. Functional and non-functional requirements

2.3.1. Functional requirements

General functional requirements for guests

- Account registration function.
- View highly recommended destinations.
- Search and filter the tours:
 - Search or filter the tours to find the desired tours.
 - Search for the tours by specifying main options such as departure point, destination point, departure date and other options like number of days, number of people, price range or transportation.
 - Filter the tour lists by price, ratings or departure date.
- View a tour's detailed information:
 - View basic information: the title, images, prices, schedule, transportation and a summary of the trip.
 - View additional information: notes, services included and other departure dates of the tour.
 - View other tour departure dates.
 - View ratings and reviews of the tours by previous customers.
 - View related tour suggestions.
- Book a tour: Customer can choose to:
 - Book online by filling in contact information and make payment.
 - Book directly at the store.
- See travel news:
 - See articles about some tourist attractions, the article may include information about locations, culture and recreational activities.

- View suggested tours for the locations.
- Contact the company by email, phone or go directly to the company address.

Functional requirements for customers with accounts

In addition to general functions for guests, customers with accounts also have the following specific functions:

- Log in and log out the system.
- Manage personal profile:
 - When the customer logs in the system, their booking history will be saved.
 - Rate and review for successfully booked tours when they end.
 - Change their password for security reasons or reset it if they forget it.

Functional requirements for staff

- Log in, log out the system.
- Each staff member is allocated one account. They are required to change their password after they log in to this account for the first time.
- Personal account management: Staff can change their password.
- Staff are assigned to manage certain tours by the managers.
- Tour management:
 - View all the tours they created, use search tools or filters to find a tour.
 - View detailed information of a tour type.
 - Request creating a new tour by filling tour information including: ID, name, vehicle, schedule, departure date, tour notes or services, etc.
 - Request editing or canceling tours for some reasons such as the tours do not have enough passengers.
 - Manage customers in a tour:
 - View the list of customers who have booked tours
 - Add new customers to the list.
 - View customers' ratings and reviews for the tour.
- Request management:

- Staff views the request history including all requests they sent to the manager.
- Staff can duplicate an old request version, edit it and submit.

Functional requirements for the manager

- Log in and log out of the system.
- Personal account management: Managers can change their password.
- Staff management:
 - Create new accounts for staff according to their information.
 - View the staff list and view a staff member's details.
 - Can edit staff information.
 - Activate and deactivate staff accounts (all tasks must be assigned to another staff before deactivating a staff).
- Tour management:
 - View a list of tours, search for a tour.
 - View the tour details.
 - Edit and cancel tours.
 - Assign a tour to another staff (when the current staff is unavailable).
- Staff request management: See the requests to create, edit or cancel tours from the staff and decide whether to accept or not.
- Data analysis: have reports and charts about the tours and customers to come up with new plans and strategies for the company. They can also export data to Excel if they want.

2.3.2. Non-Functional requirements

- Usability:
 - The website has a user-friendly interface that is easy to navigate. Users take no more than three steps to perform a functionality.
 - Responsive design that will automatically adjust for different screen sizes and viewports on laptop, tablets and mobile devices.
- Security of system:

- All "sensitive" user data such as passwords, phone numbers, ID cards, etc., must be encrypted.
- The system requires users to create highly complex passwords, specifically as follows:
 - Use a minimum of 8 characters and a maximum of 15 characters.
 - Include at least an English lowercase letter (a - z), an English uppercase letter (A - Z) and one digit (0 - 9) or a special character.
- Performance: System should respond to user request within an acceptable time:
 - Page loading (1-3 seconds)
 - Search response (1-2 seconds)
 - Payment response (5-7 seconds)
 - User registration and authentication (1-3 second)
- System must be able to handle up to 10000 concurrent requests without significant decrease of performance
- Availability of system: The system should be available 24/7.

Chapter 3

System design

3.1. Use Case Diagrams

Table 2. Summary table of all Use-cases of the System

ID	Use-case	Actors	Description
UC-1.1	Login	Customer	As a customer, I would like to log in and use the services of the system.
UC-1.2	Logout	Customer	As a customer, I do not need to use the web anymore and would like to log out of the system.
UC-1.3	Sign up	Guest	As a guest, I would like to sign up and use more services of the system.
UC-1.4	Operate with history page	Customer	As a customer, I would like to have some operations with booking history.
UC-1.5	Change information	Customer	As a customer, I would like to update my new information.
UC-1.6	Browse all tours	Customer, Guest	As a customer or a guest, I would like to browse all tours and have more operations with those tours.
UC-1.7	Browse all destination news	Customer, Guest	As a customer or a guest, I would like to browse all destination news and have more operations.
UC-1.8	Book tour	Customer, Guest	As a customer or a guest, I would like to book a tour.
UC-2.1	Login	Staff	As a staff, I would like to log in to the system to use the functions of the system.

UC-2.2	Logout	Staff	As a staff, I completed my work and would like to log out of the system.
UC-2.3	View assigned tours	Staff	As a staff, I would like to view all assigned tours.
UC-2.4	Operate with assigned tours	Staff	As a staff, I would like to operate with each of the assigned tours.
UC-2.5	View list of requests	Staff	As a staff, I would like to view all sent requests.
UC-2.6	Operate with requests	Staff	As a staff, I would like to operate with requests to do more things.
UC-2.7	View ratings & reviews	Staff	As a staff, I would like to view all ratings and reviews.
UC-3.1	Login	Manager	As a manager, I would like to log in and use the functions of the system.
UC-3.2	Logout	Manager	As a manager, I complete my work and would like to log out of the system.
UC-3.3	View staff list	Manager	As a manager, I would like to view the list of staff.
UC-3.4	Operate with staff list	Manager	As a manager, I would like to operate with the list.
UC-3.5	View Tours	Manager	As a manager, I would like to view all tours.
UC-3.6	Operate with tours	Manager	As a manager, I would like to operate with tours.
UC-3.7	Operate with Requests	Manager	As a manager, I would like to operate with requests.
UC-3.8	Operate with Dashboard	Manager	As a manager, I would like to operate with the dashboard.

3.1.1. Guest/Customer Use Case

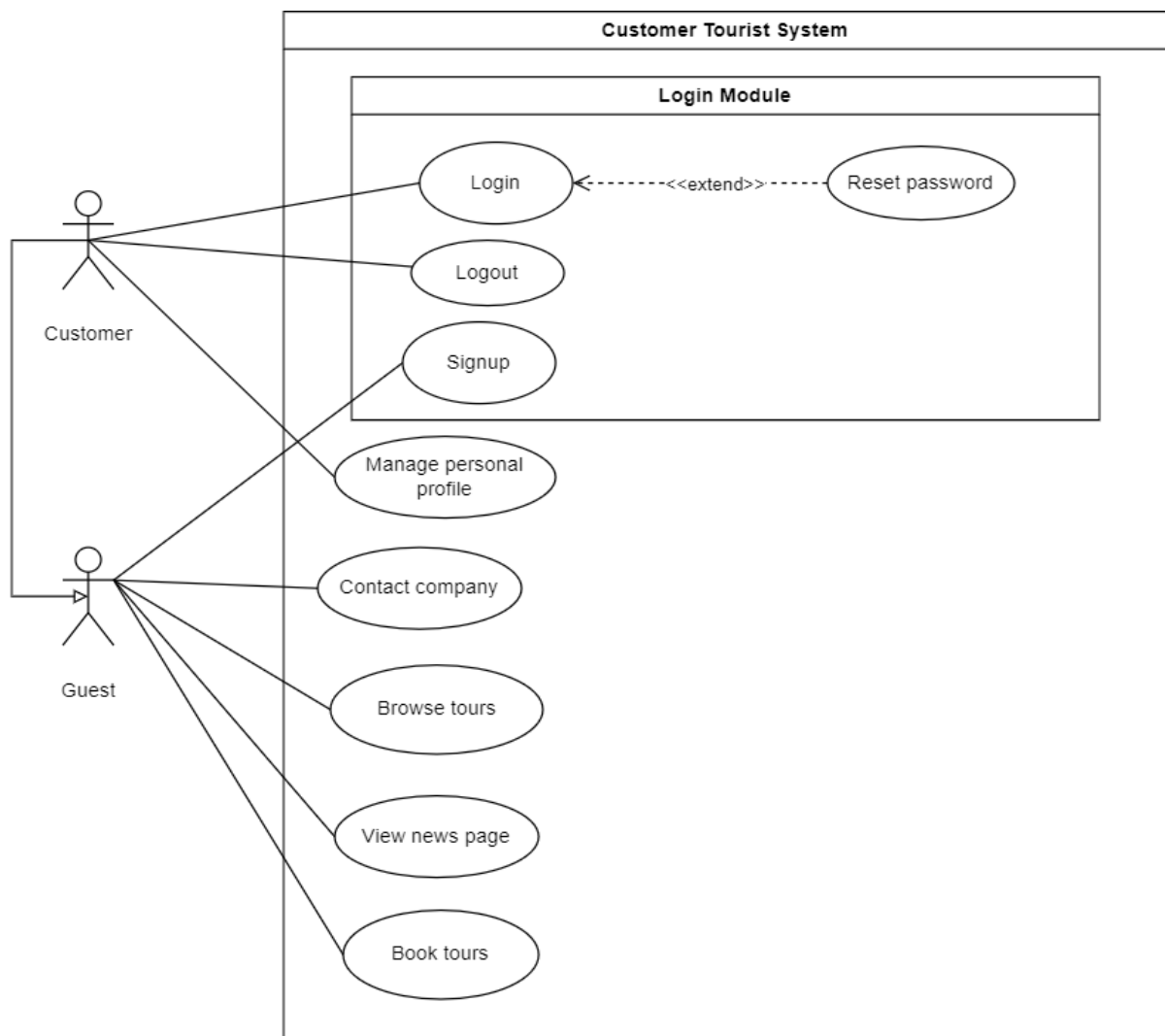


Figure 7. General use-case diagram for guest and customer

The Figure 7 illustrates all main operations a consumer can have with the web page in general. In the case that if the consumer is a Customer, after logging in successfully, a Customer can reset his/her password as he/she wants after that. For the remain operations, we will explain below.

The Figure 8 shows that if the user is a guest, the user is required to fill-in personal information form if the user would like to sign up. If the user is a customer – already have a created account, the user is required to log in the account successfully in order to use more services of the web page. In the case that the Customer forgets the password, the Customer has an option to reset the password and is required to fill-in email address.

If the Customer does not want to use the web page anymore, the Customer can log out to exit the web page.

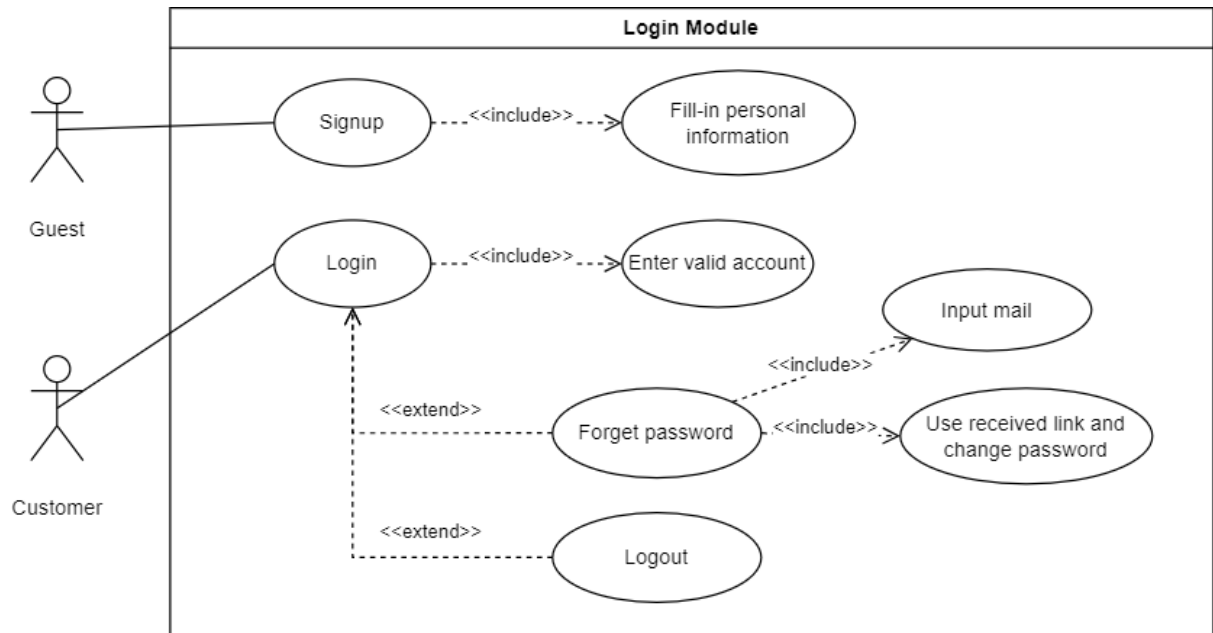


Figure 8. Use-case diagram for customer/guest - login module

The Figure 9 shows that the Customer can view all their booked tours as well as giving rating along with reviews by operating with his/her history page. Furthermore, the Customer can also update his/her personal information as the Customer wants.

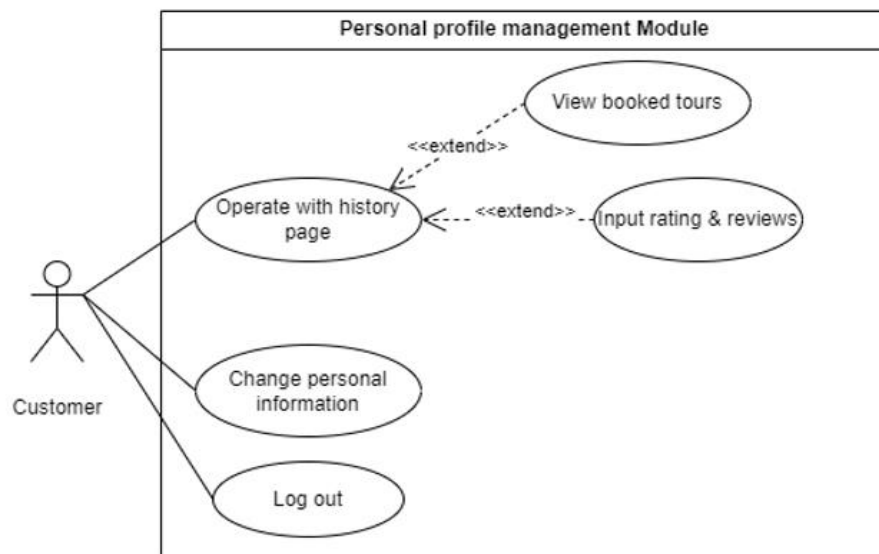


Figure 9. Use-case diagram for customer - personal profile management module

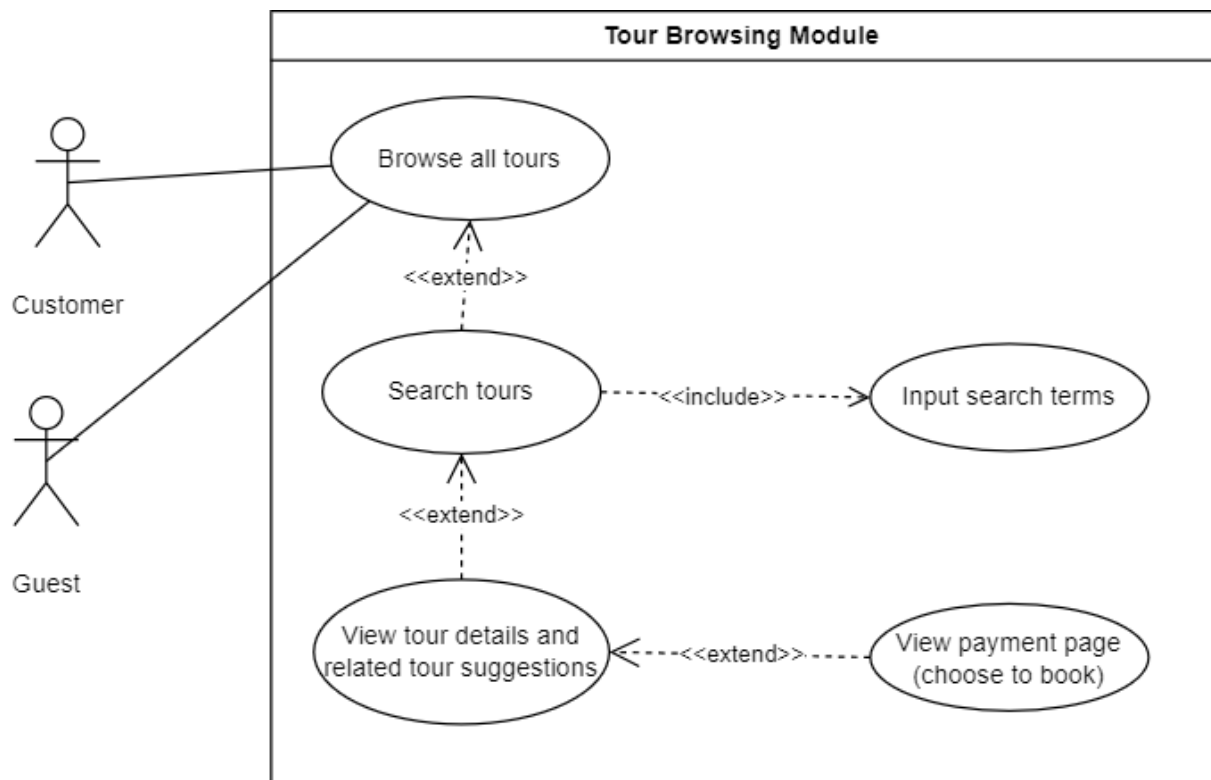


Figure 10. Use-case diagram for customer/guest - tour browsing module

The Figure 10 shows that both Customer and Guest can surf all the tours. To search more easily, they have to enter the term that they want. After searching, along with searching results, the web page will suggest some related tours. They can read to get more details. If they are satisfied, they can proceed with payment if they want.

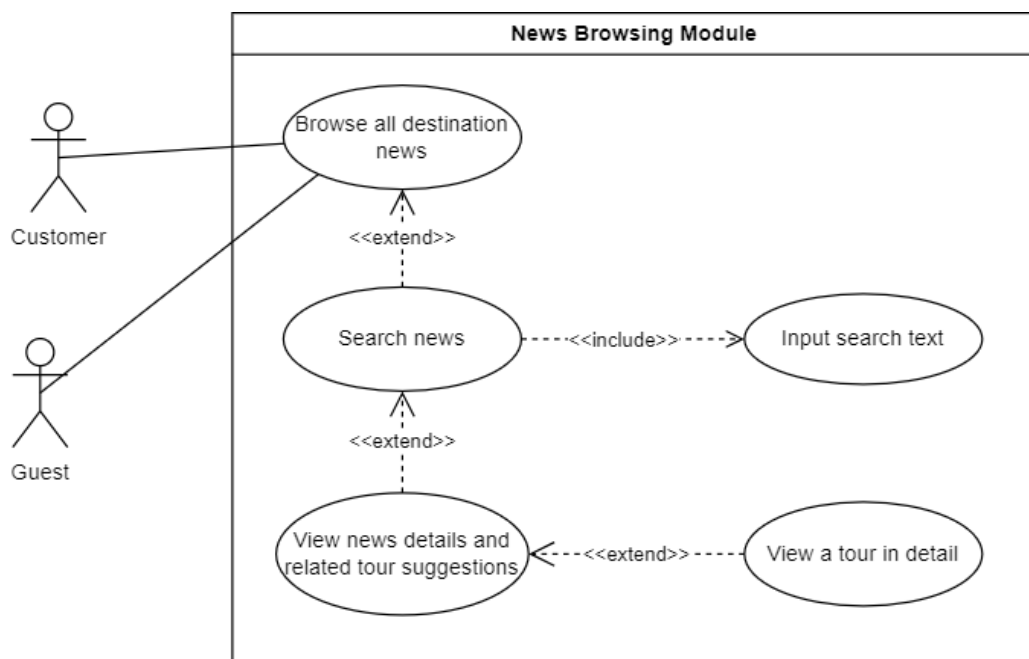


Figure 11. Use-case diagram for customer/guest - news browsing module

The Figure 11 shows that both Customer and Guest can surf all destination news. To search more easily, they have to enter the name of destinations that they want. After searching, along with searching results, the web page will suggest some related tours. They can operate with related tours as with tour browsing module we just mentioned above.

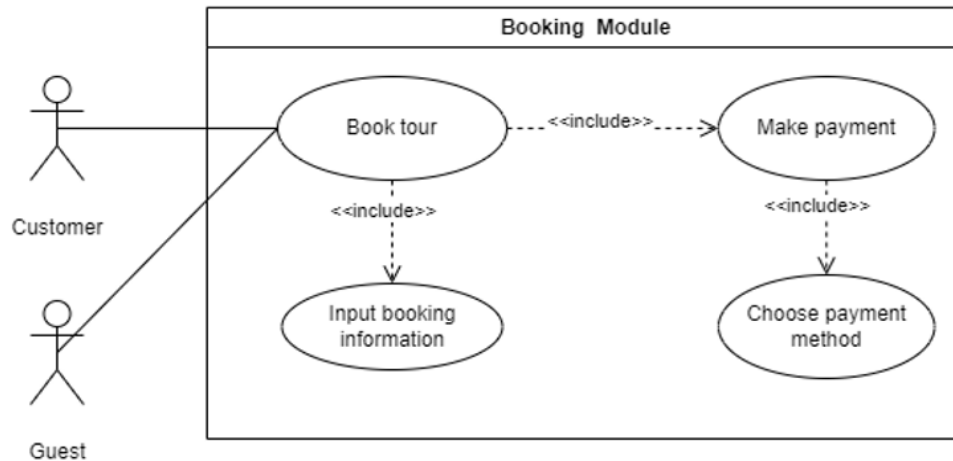


Figure 12. Use-case diagram for customer/guest - booking module

The Figure 12 shows that if Customer or Guest would like to book a tour, they are required to fill-in information as well as proceed with payment.

Guest/Customer Use Case Table Description

Table 3. Description table for Login use-case

Name	Login	Use-case ID: UC-1.1
Actors	Customer	
Trigger Condition	Customer would like to log in to the system.	
Description	As a customer, I would like to log in and use the services of the system.	
Pre-Conditions	- Customer clicks on the "Login" option. - Login page is shown. - Customer's device has accessed the Internet.	
Post-Conditions	Customer logs in successfully.	

Main Flow	1. The customer clicks on the "Login" button. 2. System shows the Login page. 3. Customer inputs username and password. 4. The system authenticates the information. 5. Customer logs in successfully.
Alternative Flow	No.
Exception	- In step 4, if the account is invalid, the system will alert the customer and ask if the customer wants to reset the password. - If the Internet is poorly connected, the customer will fail to log in to the system.

Table 4. Description table for Logout use-case

Name	Logout	Use-case ID: UC-1.2
Actors	Customer	
Trigger Condition	The Customer would like to log out of the system.	
Description	As a customer, I do not need to use the web anymore and would like to log out of the system.	
Pre-Conditions	- Customer logged in to the system. - Customer's device has accessed the Internet.	
Post-Conditions	Customer logs out successfully.	
Main Flow	1. The customer clicks on the "Sign out" option. 2. The system allows the customer to exit his/her account. 3. The customer is out of the system.	
Alternative Flow	No.	
Exception	If the Internet is poorly connected, the customer will fail to log out of the system.	

Table 5. Description table for Signup use-case

Name	Sign up	Use-case ID: UC-1.3
Actors	Guest	
Trigger Condition	The Guest would like to sign up for the system.	
Description	As a guest, I would like to sign up and use more services of the system.	
Pre-Conditions	- Guest clicks on the "Sign up" option. - Signup page is shown. - Guest's device has accessed the Internet.	
Post-Conditions	Guest signs up successfully.	
Main Flow	1. The guest clicks on the "Sign up" option. 2. System shows the Signup page. 3. Guest fills in information. 4. The system checks the information. 5. Guest signs up successfully.	
Alternative Flow	No.	
Exception	If the Internet is poorly connected, guests will fail to sign up for the system.	

Table 6. Description table for Operate with history page use-case

Name	Operate with history page	Use-case ID: UC-1.4
Actors	Customer	
Trigger Condition	The Customer would like to operate with the history page.	
Description	As a customer, I would like to have some operations with booking history.	
Pre-Conditions	- Customer clicks on the "History" option. - History page is shown. - Customer's device has accessed the Internet.	
Post-Conditions	Customers can operate with the history page.	

Main Flow	1. The customer clicks on the "History" option. 2. System shows the history page. 3. Customer can operate with the page: view booking tours, give rating and reviews for booked tours.
Alternative Flow	No.
Exception	If the Internet is poorly connected, customer's operations will fail.

Table 7. Description table for Change information use-case

Name	Change information	Use-case ID: UC-1.5
Actors	Customer	
Trigger Condition	The Customer would like to change his/her information.	
Description	As a customer, I would like to update my new information.	
Pre-Conditions	- Customer clicks on the "My Account" option. - My account page is shown. - Customer's device has accessed the Internet.	
Post-Conditions	Customer's information is changed.	
Main Flow	1. The customer clicks on the "My Account" option. 2. System shows my account page. 3. The customer clicks on the "Edit" option change any information he/she wants and clicks on the "Save" option to update the new one. 4. Information is changed successfully.	
Alternative Flow	No.	
Exception	- If the Internet is poorly connected, the customer's operations will fail. - If the information has no change, the "Save" option will not be activated.	

Table 8. Description table for Browse all tours use-case

Name	Browse all tours	Use-case ID: UC-1.6
Actors	Customer, Guest	
Trigger Condition	Customer and Guest would like to browse tours.	
Description	As a customer or a guest, I would like to browse all tours and have more operations with those tours.	
Pre-Conditions	- Customer and Guest login to the system successfully. - Main page is shown. - Customer and Guest's devices have accessed the Internet.	
Post-Conditions	Customers and Guests can browse all tours as well as operate with those tours.	
Main Flow	1. Customer and Guest are on the main page. 2. System shows all tours existing. 3. Customer and Guest can view tours or operate more with those tours such as searching for a specific tour, viewing details of a tour and related tours by choosing a specific tour.	
Alternative Flow	In step 3, Customer and Guest can search for a specific tour by combining with the filtering toolbar.	
Exception	If the Internet is poorly connected, customer and guest operations will fail.	

Table 9. Description table for Browse all destination news use-case

Name	Browse all destination news	Use-case ID: UC-1.7
Actors	Customer, Guest	
Trigger Condition	Customer and Guest would like to browse all destination news.	
Description	As a customer or a guest, I would like to browse all destination news and have more operations.	
Pre-Conditions	- Customer and Guest click on the "News" option. - News page is shown. - Customer and Guest's devices have accessed the Internet.	

Post-Conditions	Customer and Guest can browse all destination news as well as operate with the News page.
Main Flow	1. Customer and Guest click on the "News" option. 2. System shows the News page. 3. Customer and Guest can view all destination news or operate more with a News page such as searching for specific news, viewing details of news and related tours by choosing specific news.
Alternative Flow	No.
Exception	If the Internet is poorly connected, customer and guest operations will fail.

Table 10. Description table for Book tour use-case

Name	Book tour	Use-case ID: UC-1.8
Actors	Customer, Guest	
Trigger Condition	Customer and Guest would like to book a tour.	
Description	As a customer or a guest, I would like to book a tour.	
Pre-Conditions	- Customer and Guest click on a specific tour. - Customer and Guest's devices have accessed the Internet.	
Post-Conditions	Customer and Guest can book a tour.	
Main Flow	1. Customers and Guests click on a specific tour. 2. The system shows all the information about the tour. 3. Customer and Guest can book a tour by completing payment and entering booking information.	
Alternative Flow	No.	
Exception	If the Internet is poorly connected, customer and guest operations will fail.	

3.1.2. Staff Use Case

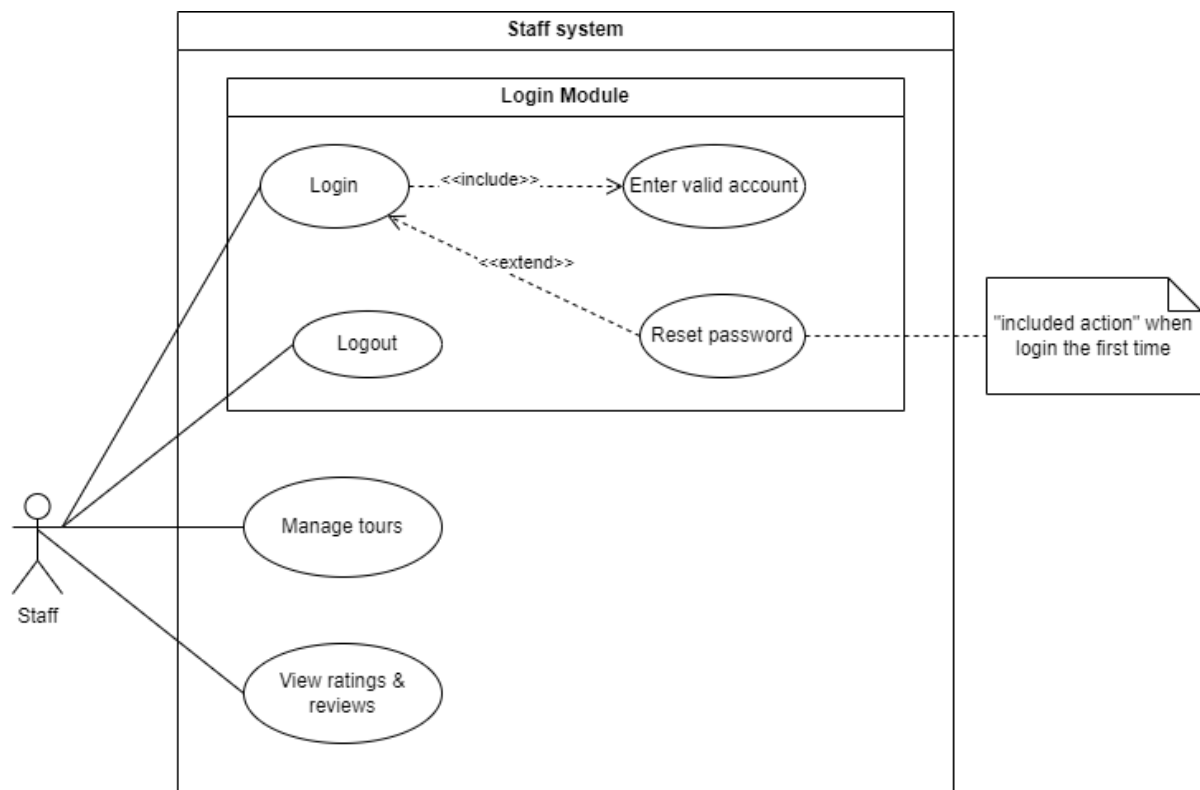


Figure 13. General use-case diagram for staff

The Figure 13 illustrates the main operations a staff can have with the management web page in general. A staff can log in the page and reset his/her password whenever they want after that. After logging in, if a staff does not want to use the system anymore, he/she can log out to exit the system. Besides, a staff can view given comments as well as rating from the customers. For the tours management, we will explain below.

The Figure 14 shows that a staff can view all his/her created tours. Furthermore, if a staff would like to add to new tour, edit or delete existing tours, a staff have to send requests to the manager. After sending requests, a staff can review them to check whether the manager responds or not. A staff can also edit the sent edit requests.

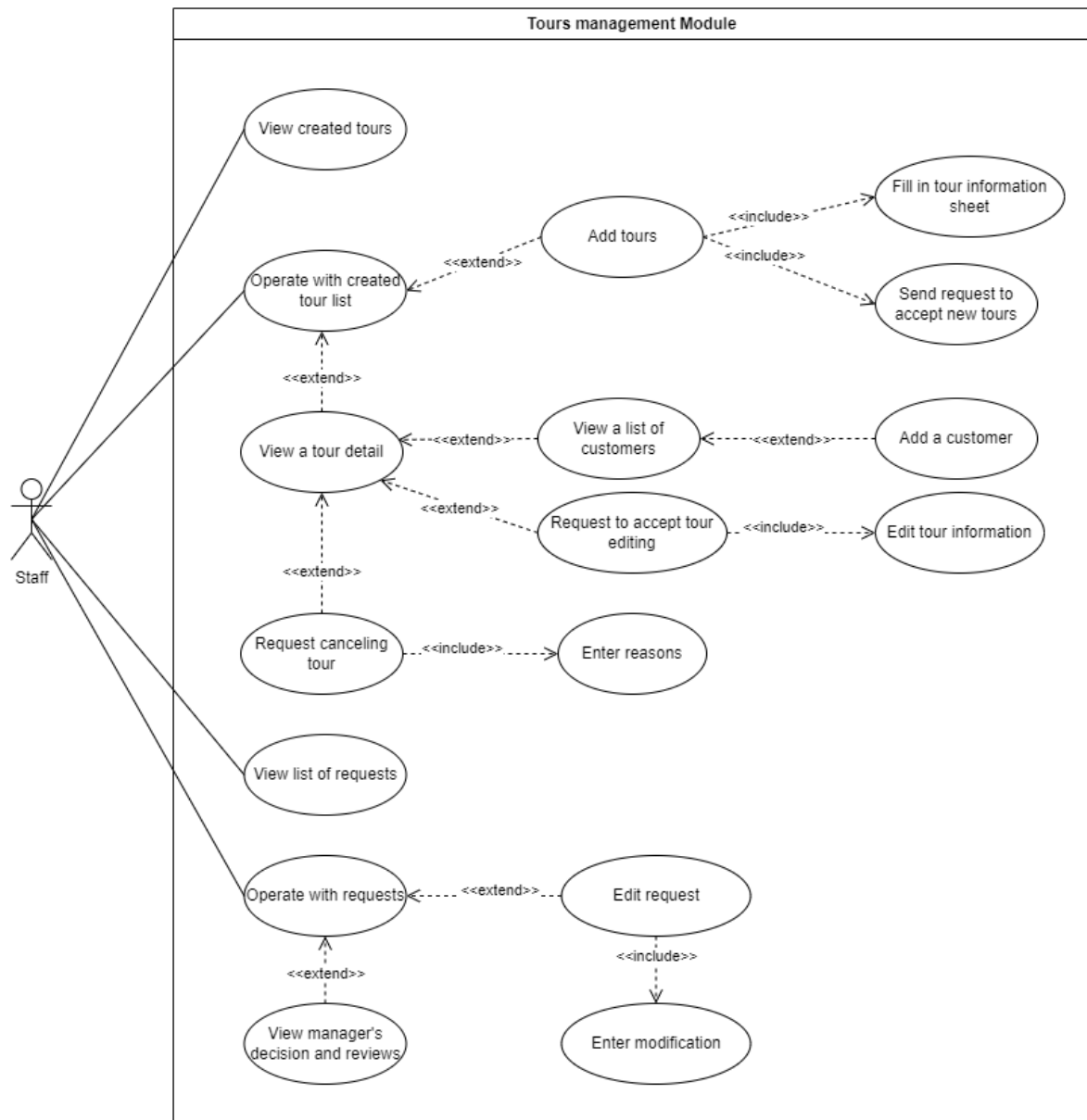


Figure 14. Use-case diagram for staff - tour management module

Staff Use Case Table Description

Table 11. Description table for Login use-case

Name	Login	Use-case ID: UC-2.1
Actors	Staff	
Trigger Condition	Staff would like to log in the system.	
Description	As a staff, I would like to log in the system in order to use the functions of the system.	

Pre-Conditions	- Staff has an existing account. - Staff's device has accessed the Internet.
Post-Conditions	Staff logs in the system successfully.
Main Flow	1. Staff clicks on the "Login" button. 2. System shows the Login page. 3. Staff inputs username and password. 4. System authenticates the input. If the account is activated for the first time using, staff will have to re-input a different password (his/her own password). 5. Staff logins successfully.
Alternative Flow	No.
Exception	- In step 4, if the account is invalid, system will alert staff and ask if staff wants to reset the password. - If the Internet is poorly connected, staff will fail to log in the system.

Table 12. Description table for Logout use-case

Name	Logout	Use-case ID: UC-2.2
Actors	Staff	
Trigger Condition	Staff would like to log out the system.	
Description	As a staff, I complete my work and would like to log out the system.	
Pre-Conditions	- Staff logged in the system. - Staff's device has accessed the Internet.	
Post-Conditions	Staff logs out the system successfully.	
Main Flow	1. The staff clicks on the "Sign out" option. 2. The system allows the staff to exit his/her account. 3. The staff is out of the system.	
Alternative Flow	No.	
Exception	If the Internet is poorly connected, staff will fail to log out the system.	

Table 13. Description table for View assigned tours use-case

Name	View assigned tours	Use-case ID: UC-2.3
Actors	Staff	
Trigger Condition	Staff would like to view all assigned tours.	
Description	As a staff, I would like to view all assigned tours.	
Pre-Conditions	- Staff clicks on the "List of Tours" option. - Staff's device has accessed the Internet.	
Post-Conditions	List of assigned tours is shown.	
Main Flow	1. The staff clicks on "List of Tours" option. 2. System shows tours that have been assigned to staff. 3. Staff can view, or search for a specific tour.	
Alternative Flow	No.	
Exception	If the Internet is poorly connected, the system will fail to load the list of tours.	

Table 14. Description table for Operate with assigned tours use-case

Name	Operate with assigned tours	Use-case ID: UC-2.4
Actors	Staff	
Trigger Condition	Staff would like to operate with assigned tours.	
Description	As a staff, I would like to operate with each of assigned tours.	
Pre-Conditions	- List of assigned tours is shown. - Staff's device has accessed the Internet.	
Post-Conditions	Staff operates with assigned tours.	
Main Flow	1. The system shows a list of assigned tours. 2. Staff clicks on a specific tour that he/she wants. 3. Staff operates with tours: add a new tour, send request to accept new tours, view information of tours, view/add customers, edit tours and then send request to accept	

	modification/cancellation.
Alternative Flow	No.
Exception	If the Internet is poorly connected, staff's operations will fail.

Table 15. Description table for View list of requests use-case

Name	View list of requests	Use-case ID: UC-2.5
Actors	Staff	
Trigger Condition	Staff would like to view sent requests.	
Description	As a staff, I would like to view all sent requests.	
Pre-Conditions	- Staff clicks on the "List of Sent Requests" option. - Staff's device has accessed the Internet.	
Post-Conditions	Staff can view all of the sent requests.	
Main Flow	1. Staff clicks on the "List of Sent Requests" option. 2. The system will show a list of sent requests. 3. Staff can view, or search.	
Alternative Flow	No.	
Exception	If the Internet is poorly connected, the system will fail to load the list of sent requests.	

Table 16. Description table for Operate with requests use-case

Name	Operate with requests	Use-case ID: UC-2.6
Actors	Staff	
Trigger Condition	Staff would like to operate with sent requests.	
Description	As a staff, I would like to operate with requests to do more things.	

Pre-Conditions	- List of sent requests is shown. - Staff's device has accessed the Internet.
Post-Conditions	Staff can operate with sent requests.
Main Flow	1. Staff clicks on a specific request. 2. The system will show the related information of that request. 3. Staff can operate with the request: edit, view manager's decision/reviews.
Alternative Flow	No.
Exception	If the Internet is poorly connected, staff's operations will fail.

Table 17. Description table for View rating & reviews use-case

Name	View ratings & reviews	Use-case ID: UC-2.7
Actors	Staff	
Trigger Condition	Staff would like to view ratings and reviews of all tours.	
Description	As a staff, I would like to view all ratings and reviews.	
Pre-Conditions	- Staff clicks on the "Ratings and Reviews" option. - List of reviews and ratings is shown. - Staff's device has accessed the Internet.	
Post-Conditions	List of ratings and reviews is shown.	
Main Flow	1. Staff clicks on "Ratings and Reviews" option. 2. List of ratings and reviews is shown. 3. Staff can view the list.	
Alternative Flow	No.	
Exception	If the Internet is poorly connected, the system will fail to load ratings and reviews.	

3.1.3. Manager Use Case

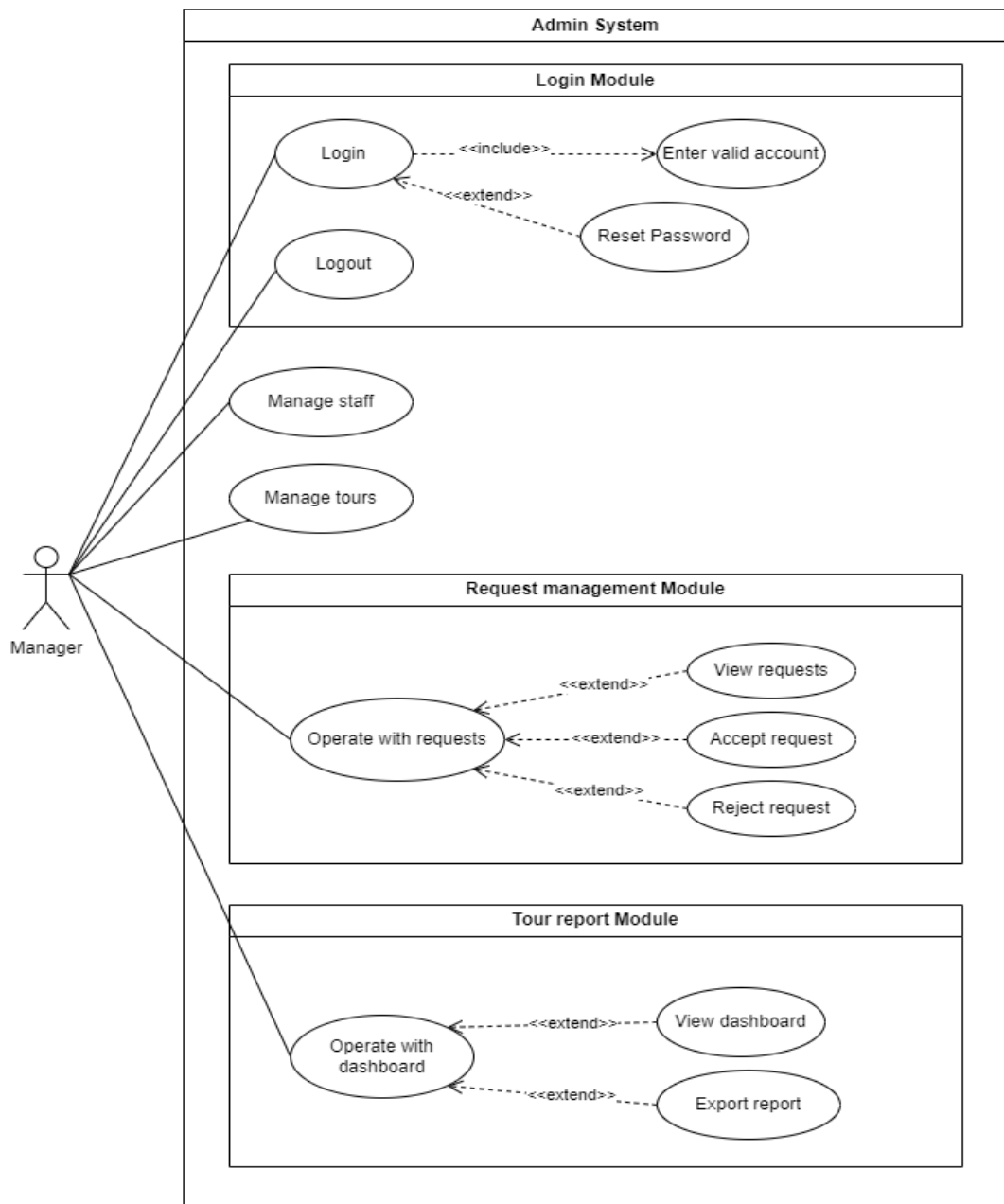


Figure 15. General use-case diagram for manager

The Figure 15 illustrates the main operation a manager can have with the management web page in general. The manager can log in the page and reset his/her password whenever he/she wants after that. After logging in, if the manager does not want to use the system anymore, he/she can log out to exit the system. With requests management, the manager can view, accept or reject. With the dashboard, the manager can view or export it as a hard copy. For staff management and tour management, we will explain below.

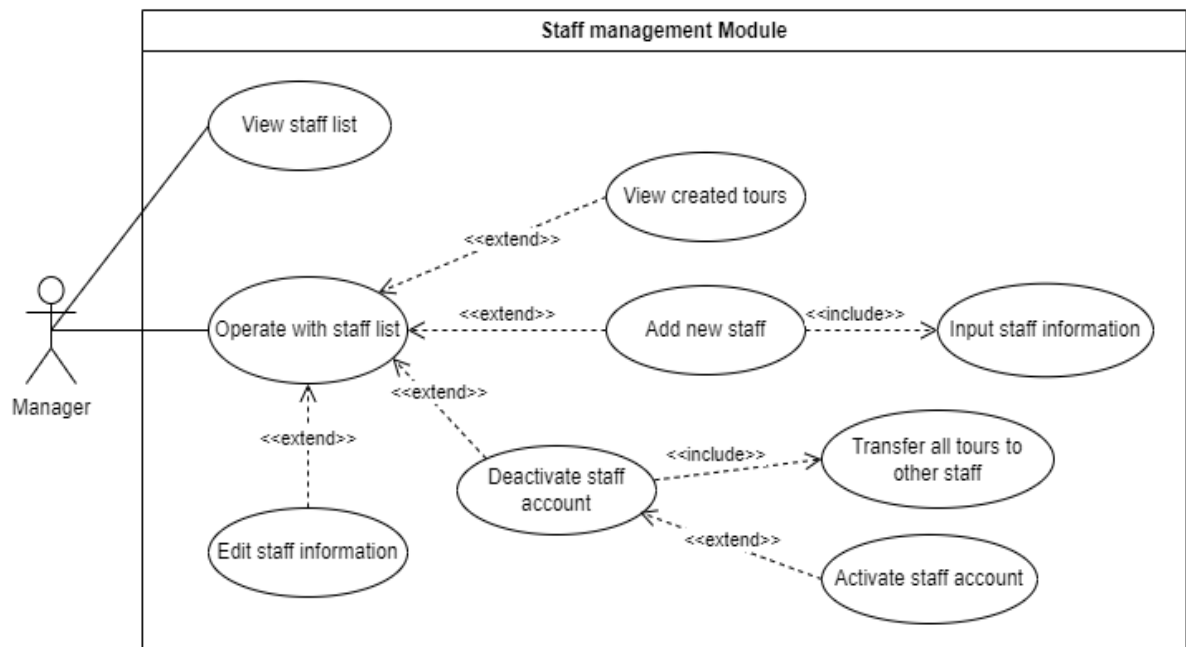


Figure 16. Use-case diagram for manager - staff management module

The Figure 16 shows that with staff management, the manager can view all staff as well as view all created tour by corresponding staff. The manager can also edit, add or deactivate any staff accounts he/she wants.

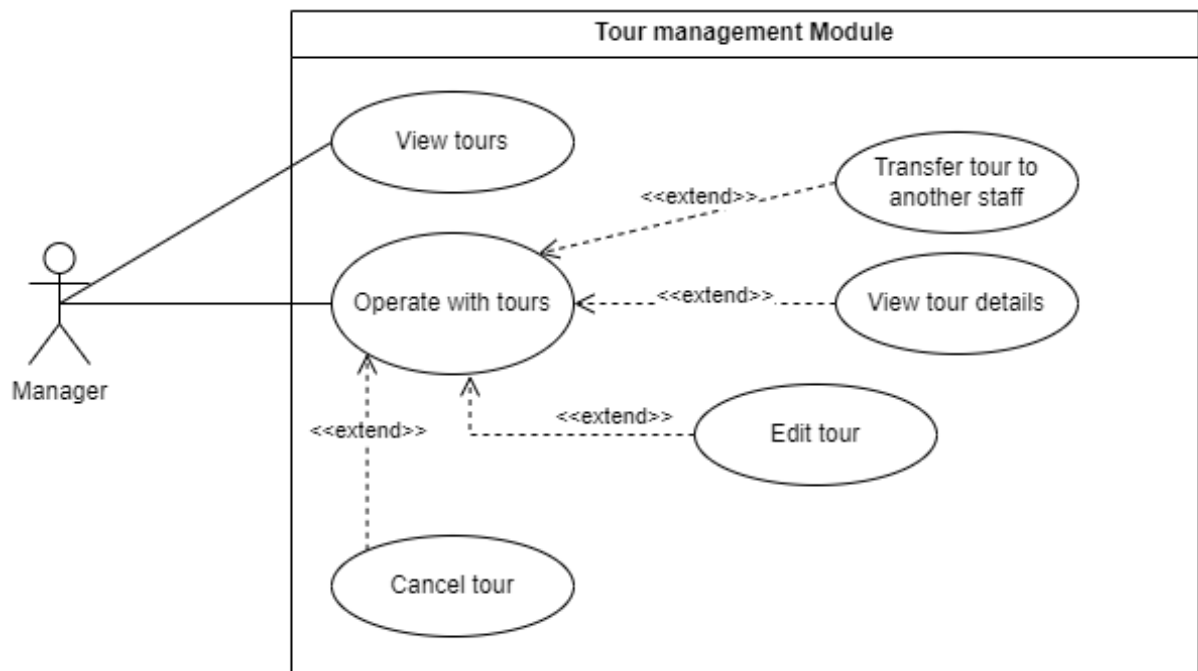


Figure 17. Use-case diagram for manager - tour management module

The Figure 17 shows that with tour management, the manager can view all of created tour as well as has the right to edit, delete or assign the tours to another staff directly.

Manager Use Case Table Description

Table 18. Description table for Login use-case

Name	Login	Use-case ID: UC-3.1
Actors	Manager	
Trigger Condition	Manager would like to log in the system.	
Description	As a manager, I would like to log in and use functions of the system.	
Pre-Conditions	- Manager has an existing account. - Manager's device has accessed the Internet.	
Post-Conditions	Manager logins successfully.	
Main Flow	1. Manager clicks on the "Login" button. 2. System shows the Login page. 3. Manager inputs username and password. 4. System authenticates the input information. 5. Manager logs in successfully.	
Alternative Flow	No.	
Exception	- In step 4, if the account is invalid, system will alert the manager and ask if manager wants to reset the password. - If the Internet is poorly connected, manager will fail to log in the system.	

Table 19. Description table for Logout use-case

Name	Logout	Use-case ID: UC-3.2
Actors	Manager	
Trigger Condition	Manager would like to log out of the system.	
Description	As a manager, I complete my work and would like to log out the system.	
Pre-Conditions	- Manager logged in the system. - Manager's device has accessed the Internet.	

Post-Conditions	Manager logs out successfully.
Main Flow	1. The manager clicks on the "Sign out" option. 2. The system allows the manager to exit his/her account. 3. The manager is out of the system.
Alternative Flow	No.
Exception	If the Internet is poorly connected, manager will fail to log out the system.

Table 20. Description table for View staff list use-case

Name	View staff list	Use-case ID: UC-3.3
Actors	Manager	
Trigger Condition	Manager would like to view the list of staff.	
Description	As a manager, I would like to view the list of staff.	
Pre-Conditions	- Manager clicks on "List of Staffs" option. - Manager's device has accessed the Internet.	
Post-Conditions	The system shows the list of staff.	
Main Flow	1. The manager clicks on the "List of Staff" option. 2. The system shows the list of staff. 3. The manager can view all the staff.	
Alternative Flow	No.	
Exception	If the Internet is poorly connected, the system will fail to load the list of staff.	

Table 21. Description table for Operate with staff list use-case

Name	Operate with staff list	Use-case ID: UC-3.4
Actors	Manager	

Trigger Condition	Manager would like to operate with the staff list.
Description	As a manager, I would like to operate with the list.
Pre-Conditions	- List of staffs is shown. - Manager's device has accessed the Internet.
Post-Conditions	The manager can operate with the list.
Main Flow	1. The system shows the list of staff. 2. The manager can operate with the staff list: add new staff, deactivate/re-activate staff's accounts, view all tours of any staff.
Alternative Flow	No.
Exception	If the Internet is poorly connected, manager's operations will fail.

Table 22. Description table for View Tours use-case

Name	View Tours	Use-case ID: UC-3.5
Actors	Manager	
Trigger Condition	Manager would like to view all tours.	
Description	As a manager, I would like to view all tours.	
Pre-Conditions	- Manager clicks on “Tours” option. - Manager's device has accessed the Internet.	
Post-Conditions	The system shows the list of tours.	
Main Flow	1. The manager clicks on “Tours” option. 2. The system shows the list of tours. 3. The manager can view all tours.	
Alternative Flow	No.	
Exception	If the Internet is poorly connected, the system will fail to load the list of tours of one tour type.	

Table 23. Description table for Operate with tours use-case

Name	Operate with tours	Use-case ID: UC-3.6
Actors	Manager	
Trigger Condition	Manager would like to operate with tours.	
Description	As a manager, I would like to operate with tours.	
Pre-Conditions	- Tour list is shown. - Manager's device has accessed the Internet.	
Post-Conditions	The manager can operate with tours.	
Main Flow	1. The manager clicks on a specific tour. 2. The system shows the related information of the tour. 3. The manager can operate with tours: view details, edit/delete.	
Alternative Flow	No.	
Exception	If the Internet is poorly connected, manager's operations will fail.	

Table 24. Description table for Operate with requests use-case

Name	Operate with requests	Use-case ID: UC-3.7
Actors	Manager	
Trigger Condition	Manager would like to operate requests.	
Description	As a manager, I would like to operate with requests.	
Pre-Conditions	- Manager clicks on the "List of Requests" option. - List of requests is shown. - Manager's device has accessed the Internet.	
Post-Conditions	The manager can operate with the list.	
Main Flow	1. The manager clicks on the "List of Requests" option. 2. The system shows the list of requests. 3. The manager can operate with requests: view/accept/reject.	

Alternative Flow	No.
Exception	If the Internet is poorly connected, manager's operations will fail.

Table 25. Description table for Operate with Dashboard use-case

Name	Operate with Dashboard	Use-case ID: UC-3.8
Actors	Manager	
Trigger Condition	Manager would like to operate with dashboard.	
Description	As a manager, I would like to operate with dashboard.	
Pre-Conditions	- Manager clicks on "Dashboard" option. - Dashboard is shown. - Manager's device has accessed the Internet.	
Post-Conditions	The manager can operate with the dashboard.	
Main Flow	1. The manager clicks on the "Dashboard" option. 2. The system shows the report of tours. 3. The manager can operate with dashboard: view/export.	
Alternative Flow	No.	
Exception	If the Internet is poorly connected, manager's operations will fail.	

3.2. Activity Diagrams

3.2.1. Customer and Guest Activity Diagrams

Online tour booking function

The diagram in Figure 18 models the activity flows of guest or customer online tour booking function. Firstly, user will see all the tours supplied. User can choose to view a tour from the displayed list or search for a desired tour and view the results. User then chooses a tour, views it in detail and clicks the “Book tour” button. System will redirect user to the booking page. In the booking page, user is required to fill in the information

and make payment. Finally, the system will save customer's booking details and display them on the staff view.

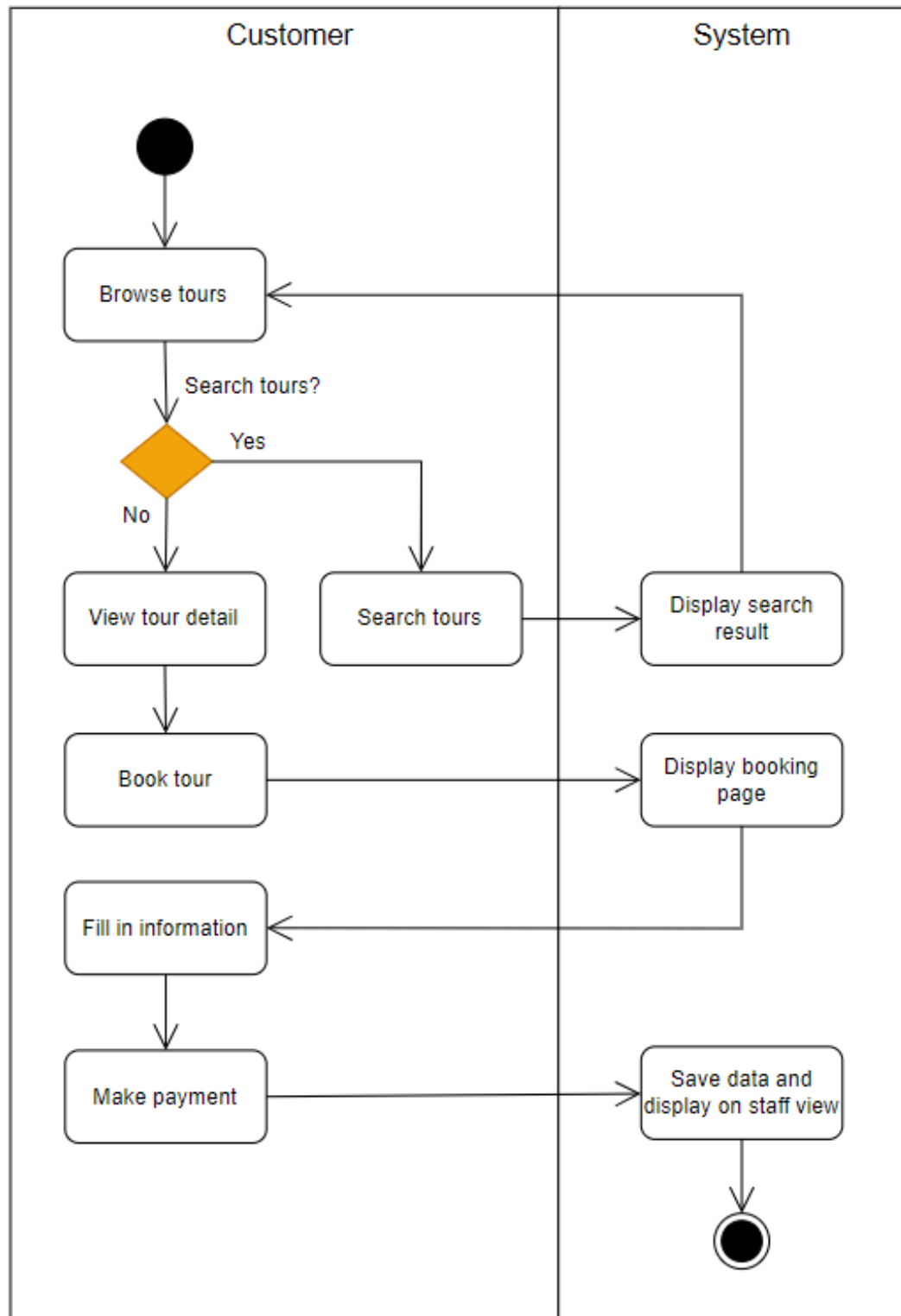


Figure 18. Activity diagram for guest/customer online tour booking function

Viewing location information function

The diagram in Figure 19 models the flows of guest or customer viewing location information function. Firstly, the user goes to the news page. Here, users can see the articles about all the attractions where the company organizes tours. An article is a general introduction about the landscape, people, activities and festivals in this location.

Users can search for a location by typing into the search bar and view the results. Next, user selects an article to view details. When an article is selected, the system will display the location information and suggest tours related to that location.

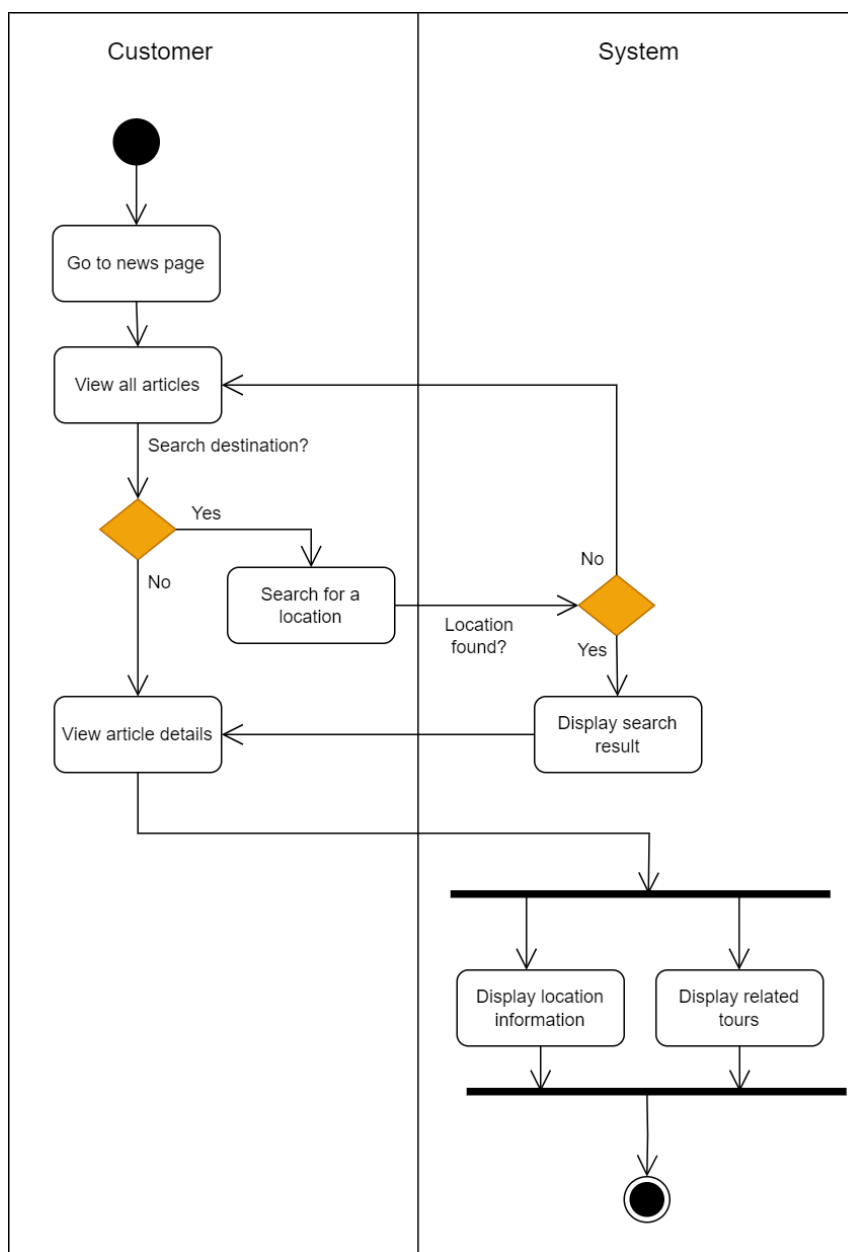


Figure 19. Activity diagram for guest/customer viewing location information function

3.2.2. Customer Activity Diagrams

Login function

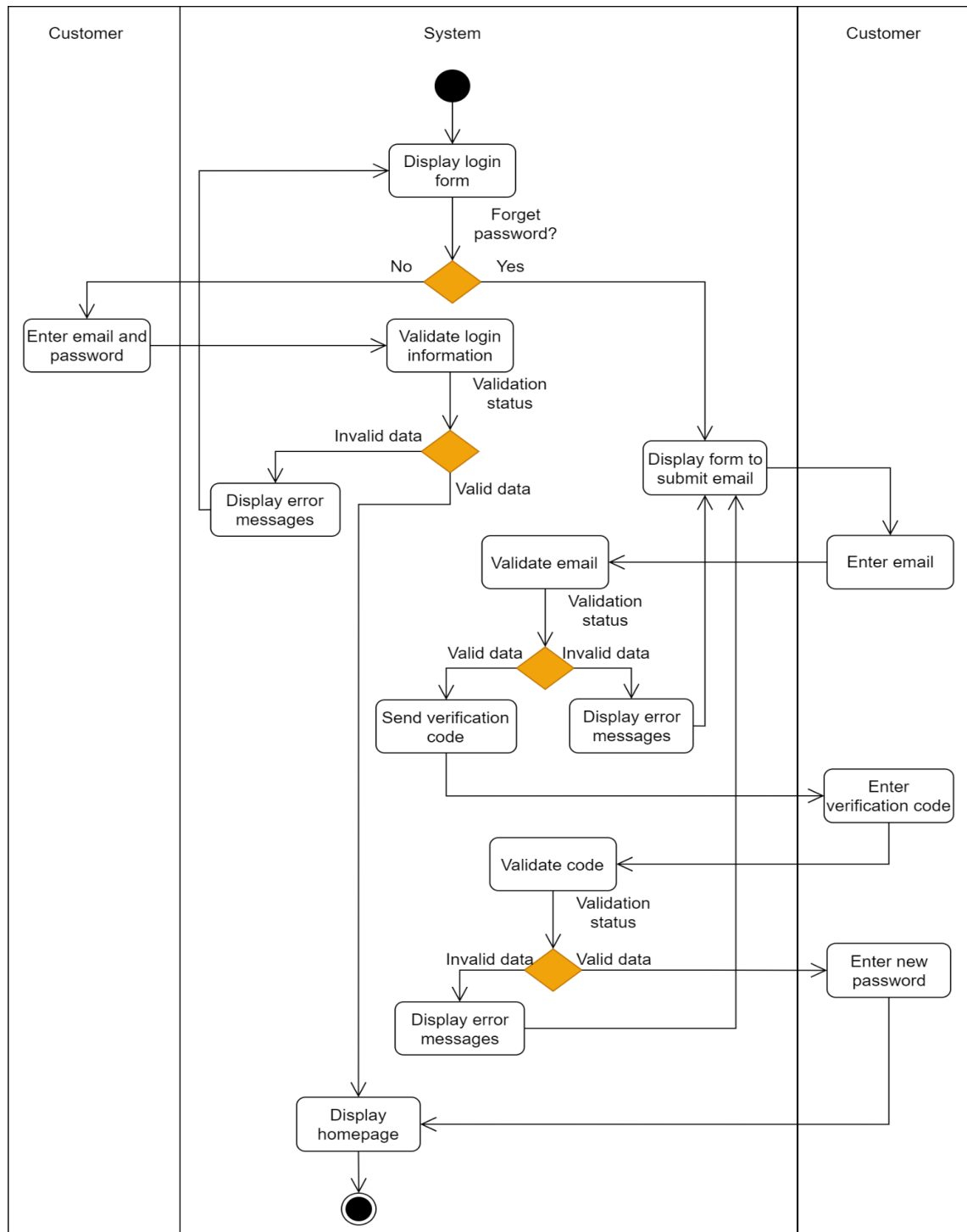


Figure 20. Activity diagram for customer login function

The diagram in Figure 20 models the activity flows of customer login function. When entering the login page, the customer will see a login form:

- Customer enters email, password into the form and submit it. The system then receives the form and validates the information. If the information is valid, the customer will be redirected to the Homepage. If the information is invalid, the system gives an error message and requires the customer to re-enter.
- On the other hand, if the customer forgets the password, the system will display a form for the customer to enter their email. After that, the customer receives a link containing a confirmation code and enters a new password. When the password is successfully changed, the customer will be redirected to the Homepage.

Ratings and reviews function

The diagram in Figure 21 models the activity flows of customer ratings and reviews function. Firstly, customer needs to login and go to the personal profile management page. In this page, customer will see the booking history and they can choose a tour to give a rating. The system will then display a form for customer to give ratings and write reviews for the tour. When the customer submits a valid form, the system will display the customer's comments in the tour detail page.

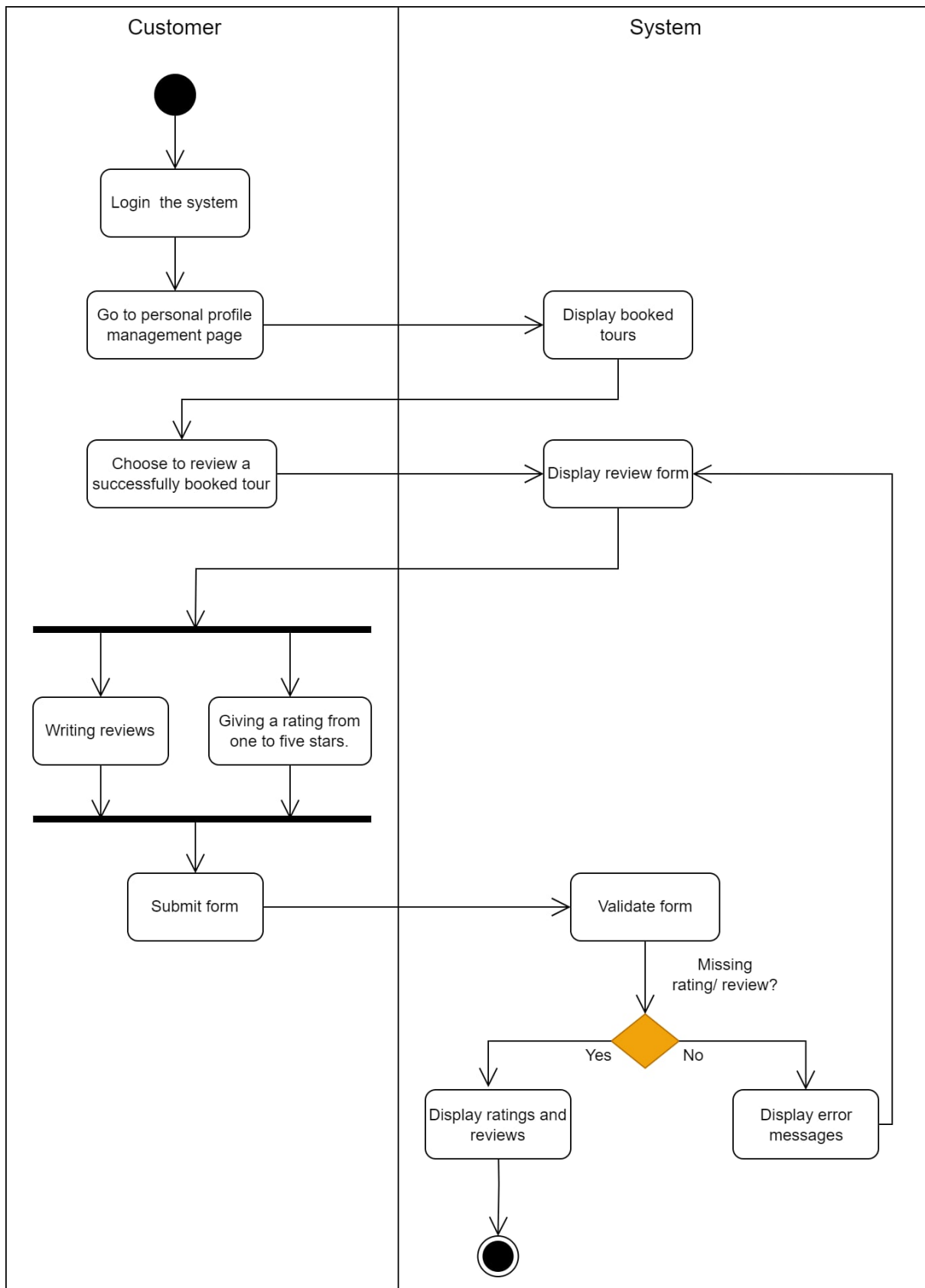


Figure 21. Activity diagram for customer ratings and reviews function

3.2.3. Staff Activity Diagrams

Login function

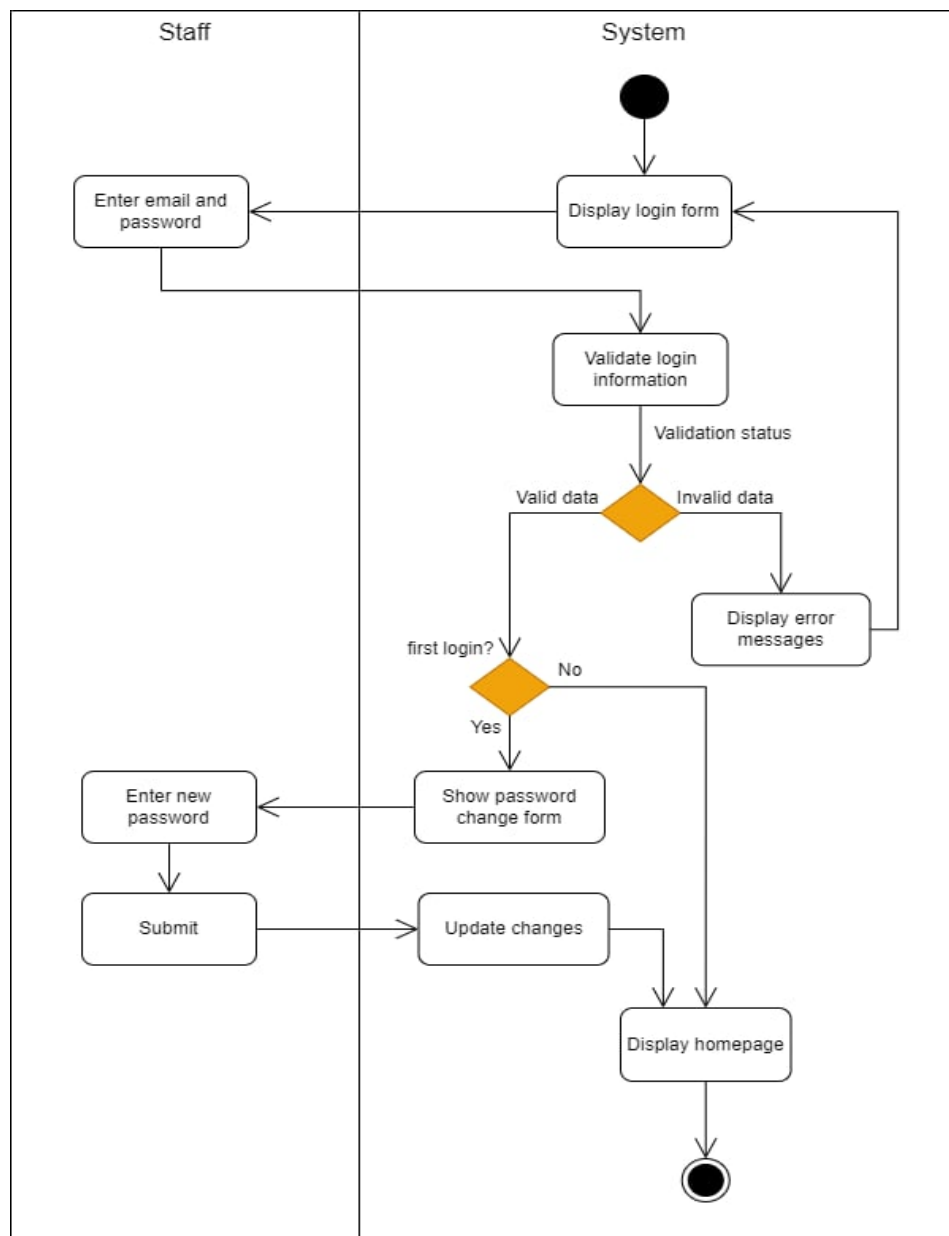


Figure 22. Activity diagram for staff login function

The diagram in Figure 22 models the activity flows of staff login function. Firstly, staff goes to the login page and enters their email and password. If the information is valid, the system then checks if it is the first login or not. If not, the system redirects to the staff homepage. However, if it is the first time the staff logs in, the system will show a form for staff to change password. Staff now enters a new password and submits the form. System updates changes and redirect staff to homepage.

Tour management functions

The diagram in Figure 23 models the activity flows of staff assigned tour management functions. Firstly, staff have to log in the system and go to the staff homepage. This page displays a list of assigned tours of a staff. The staff can choose a tour and view it in detail. The staff can request the manager to accept their creation, edition or cancelation for the tour:

- If the staff wants to edit the tour, they click the “Create new” button. The system then displays a new form and the staff will fill in all the necessary information and submit.
- If the staff wants to edit the tour, they have to click the “Edit” icon to request editing the tour and the system displays a draft with the current tour information. Customer then edits the tour information and submit the request.
- If the staff requests to cancel the tour, they have to click the “Cancel” icon and the system will display a form to enter the reasons. Customer then enters the reasons and submit the request.

The request will be saved and displayed in the staff request management page and sent to the manager.

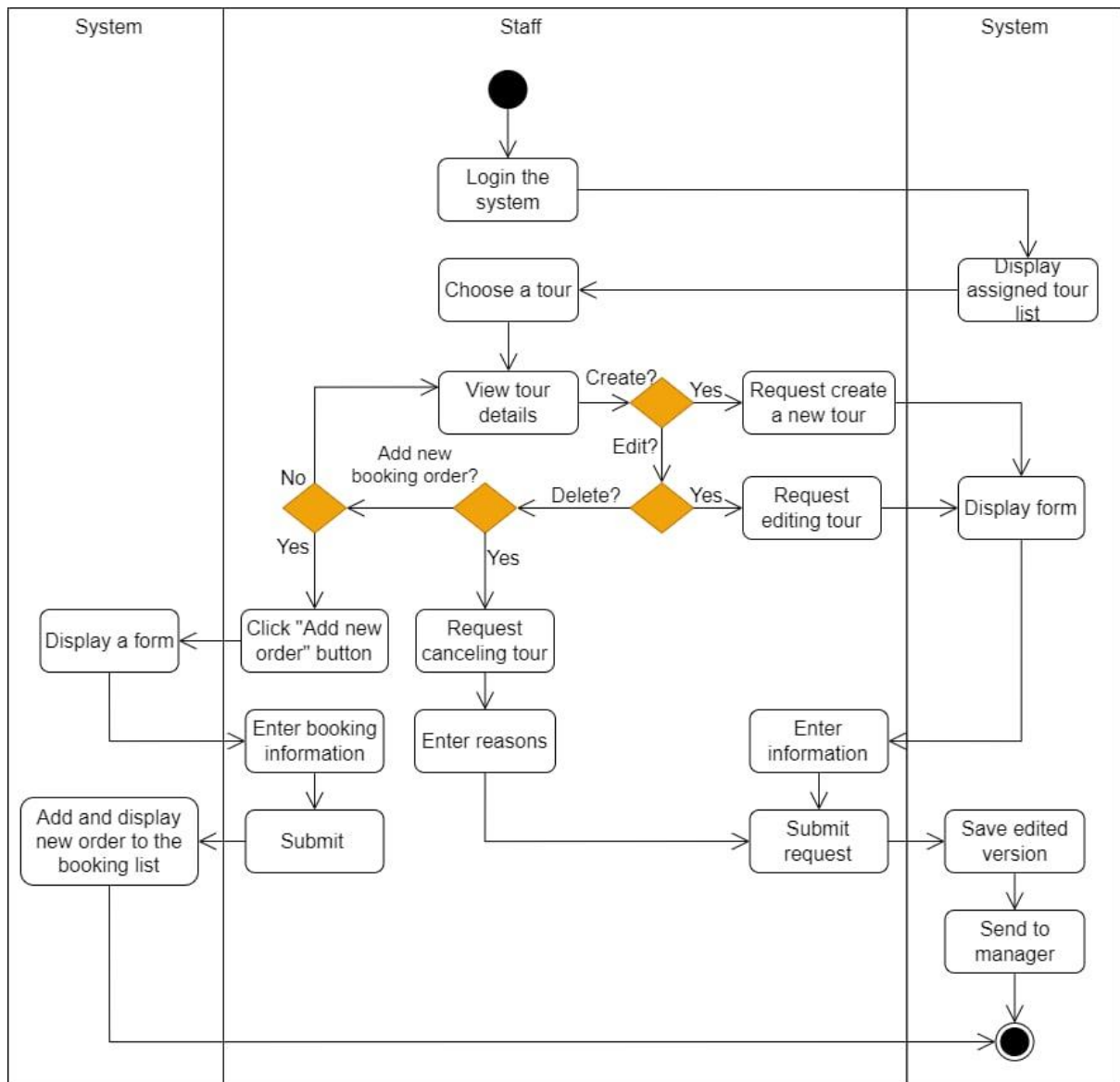


Figure 23. Activity diagram for staff tour management function

3.2.4. Manager Activity Diagrams

Staff account deactivation function

The diagram in Figure 24 models the activity flows of manager account deactivation function. Manager first has to log in and go to the staff management page to view the staff list. Manager then chooses a staff member and clicks the “Deactivate” button. The system then checks if this staff member has any assignment left. If yes, the manager has to assign all the tours to other staff and click the “Deactivate” button again. If no assignment is left, then the manager can deactivate the staff successfully. System will remove this staff out of the manager’s list of working staff.

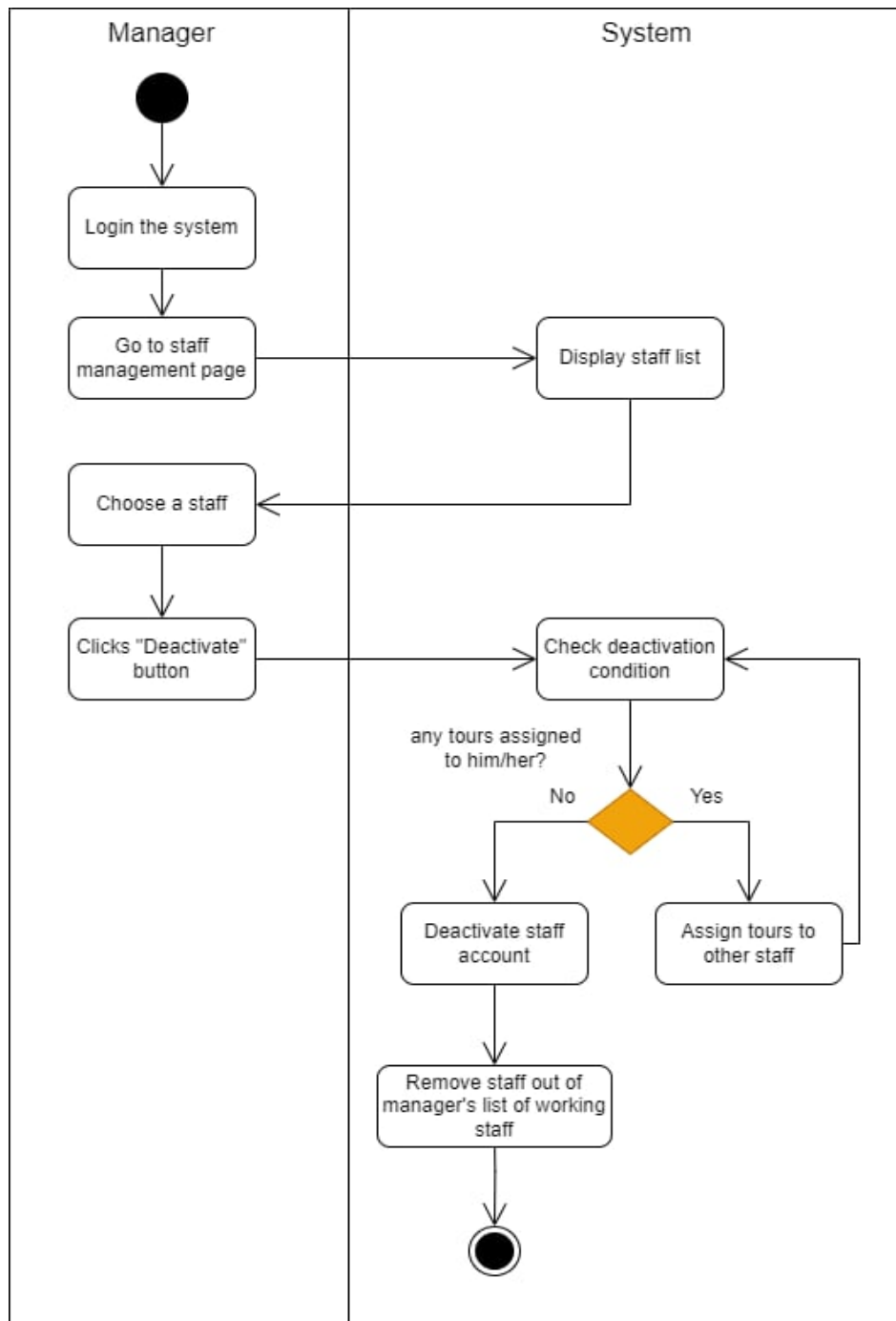


Figure 24. Activity diagram for staff account deactivation function

Tour management

The diagram in Figure 25 models the activity flows of the tour management. Manager first has to log in and go to the tour management page to view the list of tours. Manager can choose a tour to view its details, edit, cancel the tour or change the responsible staff.

- To edit the tour, the manager will click “View details” and click the “Edit” icon. Manager only needs to edit the desired information without having to fill in other

information again. When the draft is submitted, the system will update the changes and display the tour with new information.

- To cancel the tour, the manager will click “View details” and click the “Cancel” button. When they click the button, the system will update the tour status to “Canceled”.
- To change the responsible staff, the manager will select the staff option. The system will display a list of available staff and the manager will choose one and submit the option. When the manager finishes, the new responsible staff will be updated.

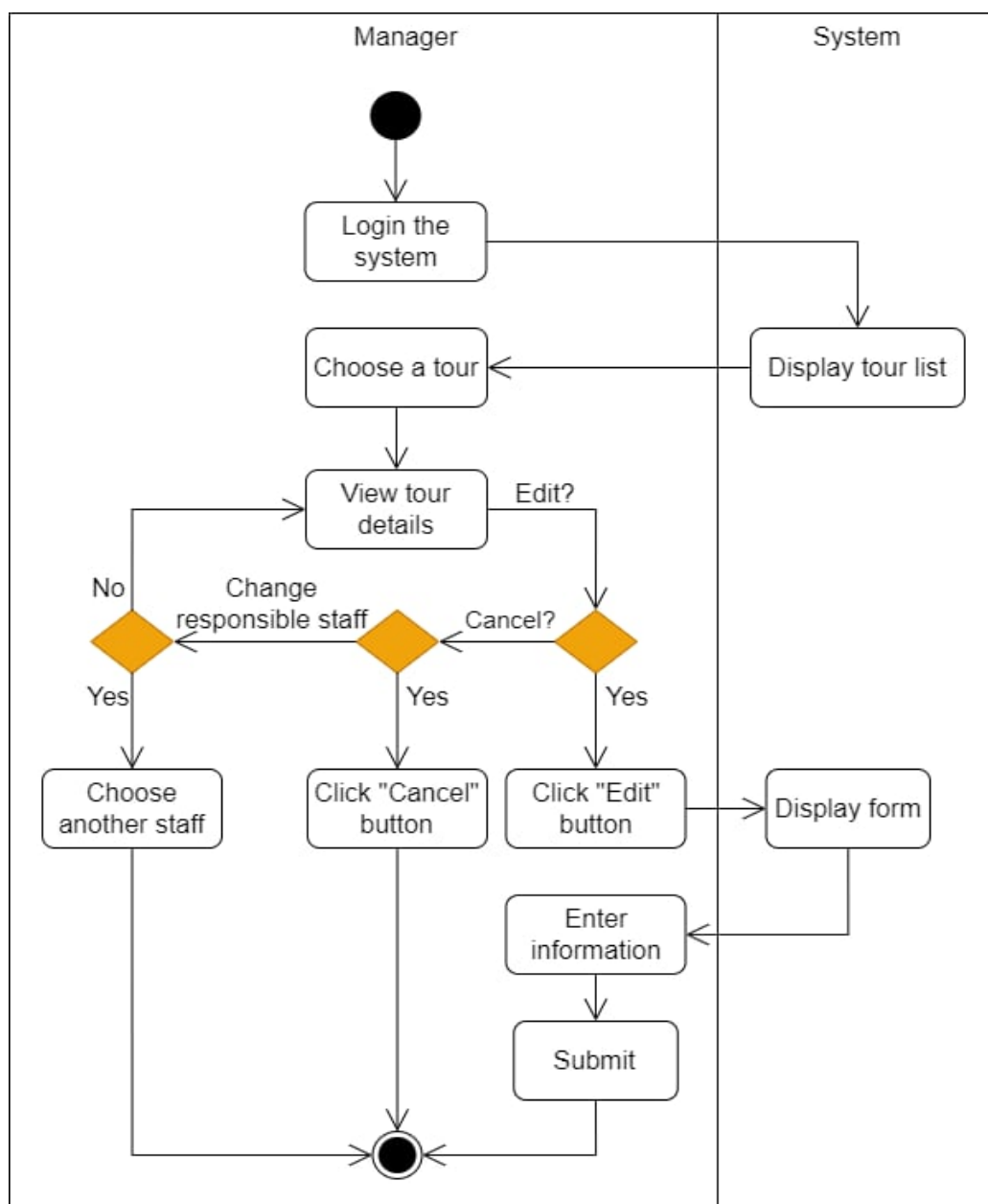


Figure 25. Activity diagram for manager tour management function

3.3. Sequence Diagrams

3.3.1. Customer Sequence Diagrams

Log in and Sign Up

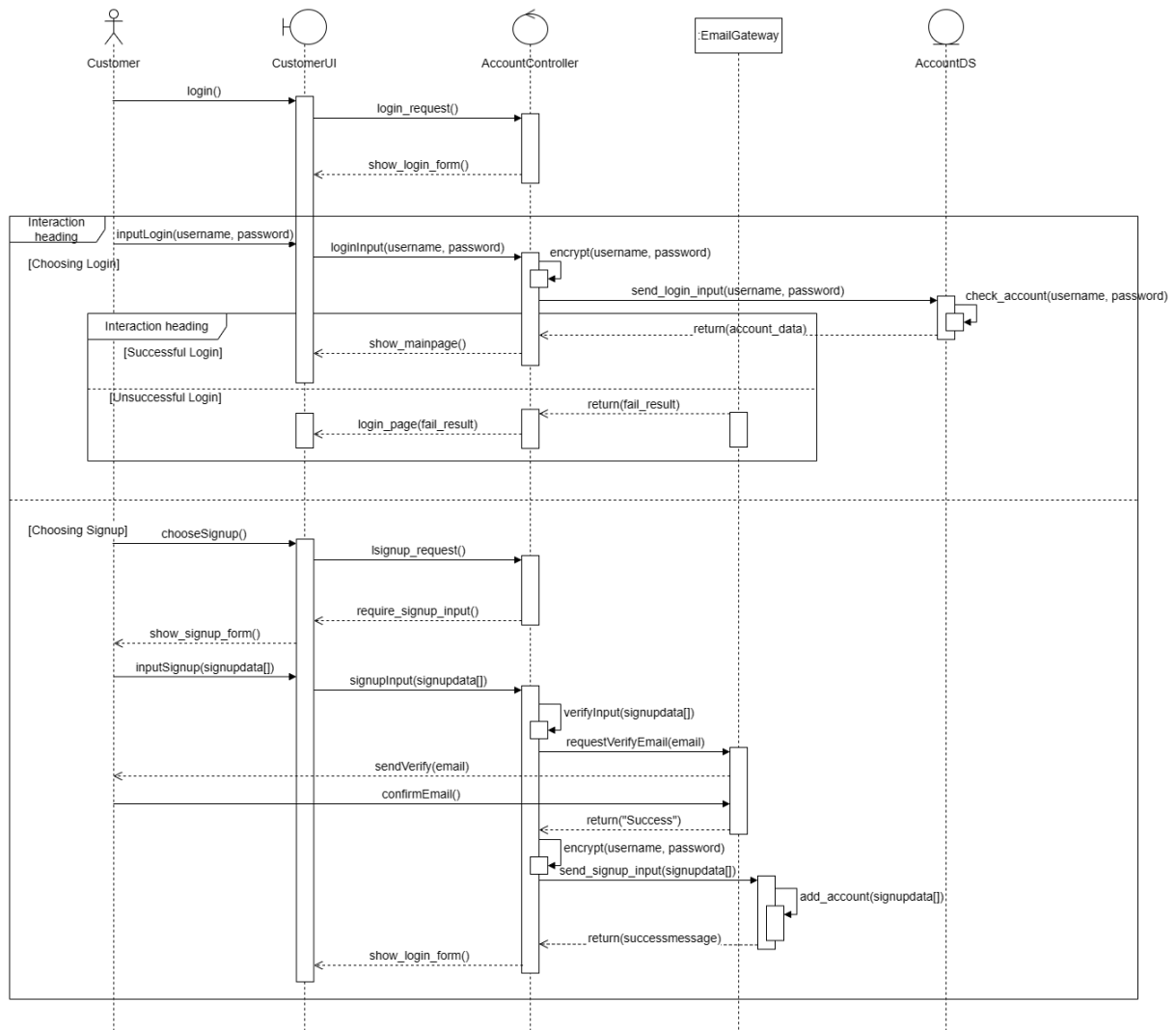


Figure 26. Sequence Diagram for Customer Login and Signup

The Figure 26 shows clearly how logging in and signing up activity are processed. To login, customers are required to input username and password. The system then uses the provided inputs to authenticate. If the authentication is successful, then the system will show the customer the main page view in customer's session.

On the other hand, to sign up, customers need to provide all required data, including a valid email. The system will use an Email Gateway to verify the email provided by the customer. In our system, the data of customers are encrypted to higher the data security.

View Tour Detailed Inforamtion

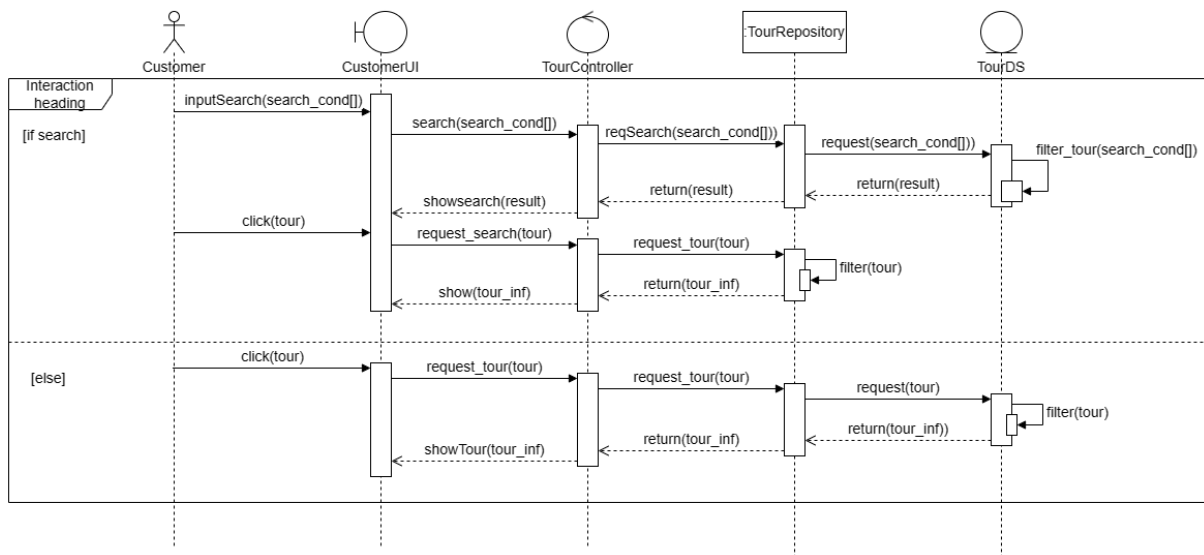


Figure 27. Sequence Diagram for Customer Viewing Detailed Information of a Tour

The Figure 27 shows that there are two main ways for customers to view detailed information of a tour. Once the customer is in the main page, he/she can directly click on a tour showing in the recommendations for further information. Another way is to search his/her desired tour with searching bar. Customer only needs to provide conditions to the searching form. The system will filter tours based on the provided conditions and shows the list of related tours for customer. Finally, customer can click on the tour shown in the list for detailed information.

Book and pay for a Tour

The Figure 28 illustrates step by step how booking and paying process is done. When customer choose to book a tour, a booking form will be shown. Customer is required to input all needed fields. Once the system receives all needed information, it will send an email to the customer to verify. If the verification is successful, customer can then move to the paying page. Here, all provided paying inputs form customer will be send directly to the payment gateway of the bank or online payment company. When the payment is successful, an email will be sent to the customer's provided email for confirmation.

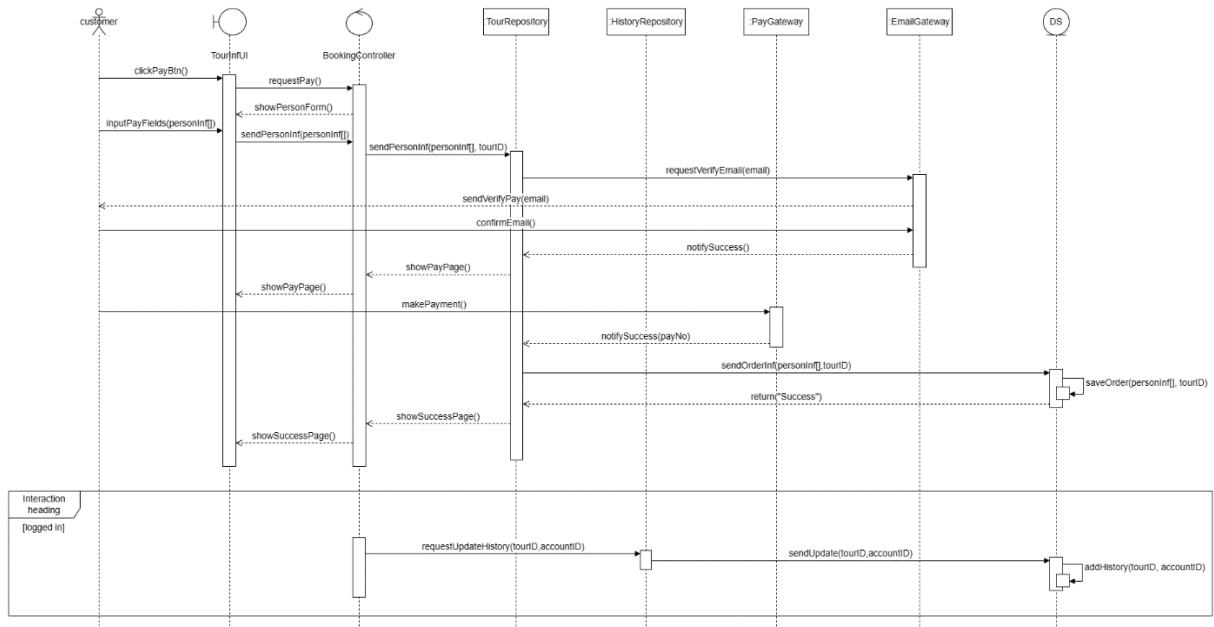


Figure 28. Sequence Diagram for Customer Booking and Paying for a Tour

View News and Tours in a Place

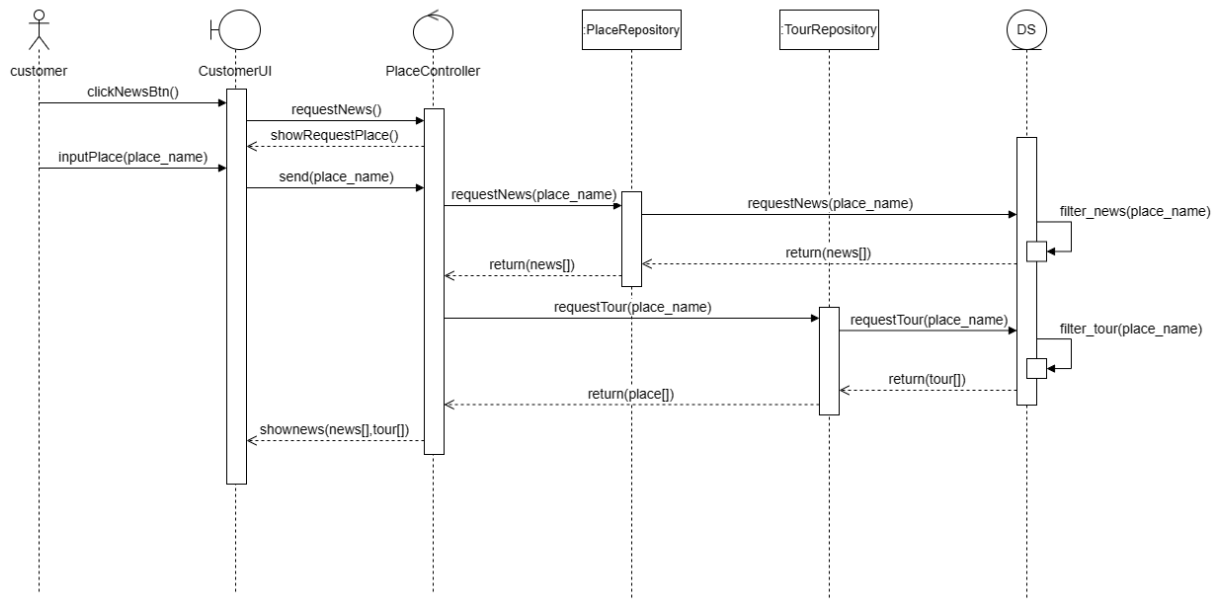


Figure 29. Sequence Diagram for Customer View News and Tours Suggestions for a Place

The Figure 29 indicates that our system also provides customers another way of finding a desirable tour. Customers can obtain news of a specific place via searching name of the place or an area or province. The system then shows many suggestions for related tours.

3.3.2. Staff Sequence Diagram

Login

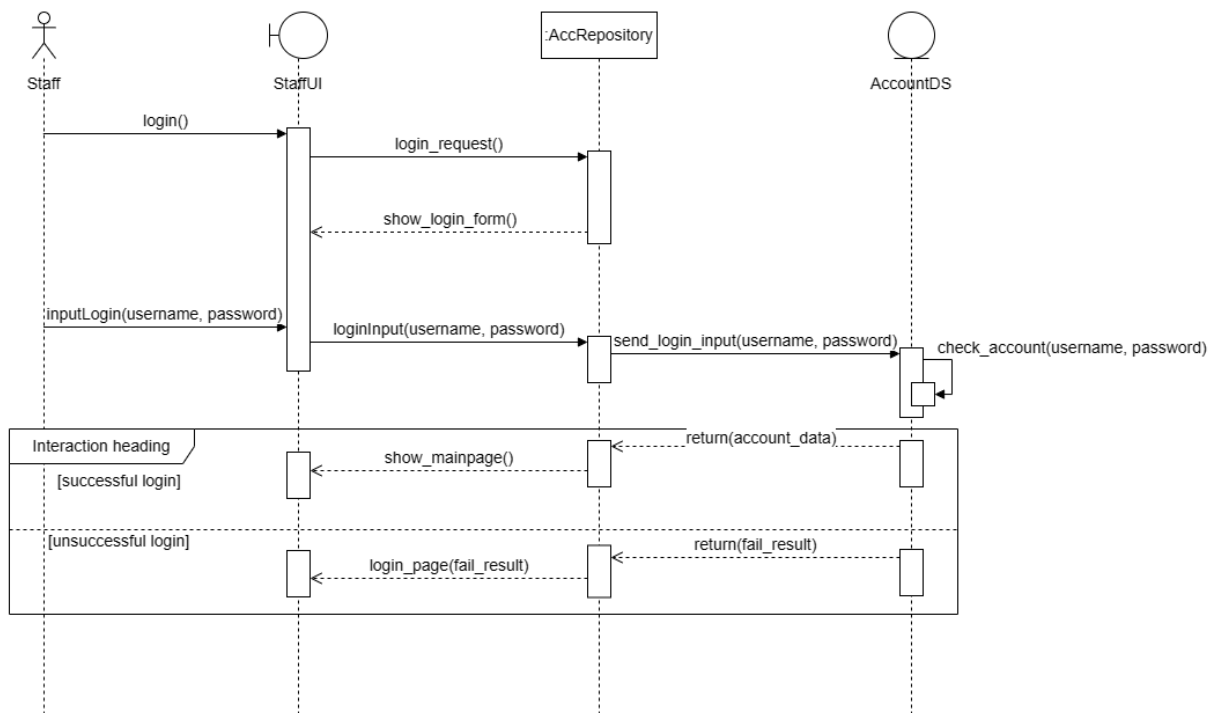


Figure 30. Sequence Diagram for Staff Login

The Figure 30 shows overall how a staff login to the system. All accounts has been created and then provided to staff. To login to the system as a staff, staff can choose staff mode in the login page and then uses the provided account to login to the system. In the first login, staff is required to change password.

Make a Request for a Tour

The Figure 31 shows that staff can make three kinds of requests, including editing, deleting, and adding a tour. To edit a tour, a staff need to have access to the tour, then he/she can choose the edit mode. All modification will be saved as a draft on the database, waiting for determination. Also, having access to the tour, a staff can also request deleting tour only if the deleting conditions are met. On the other hand, adding a tour does not need any access, a staff only needs to inputs all required data to the system and wait for the tour being confirmed.

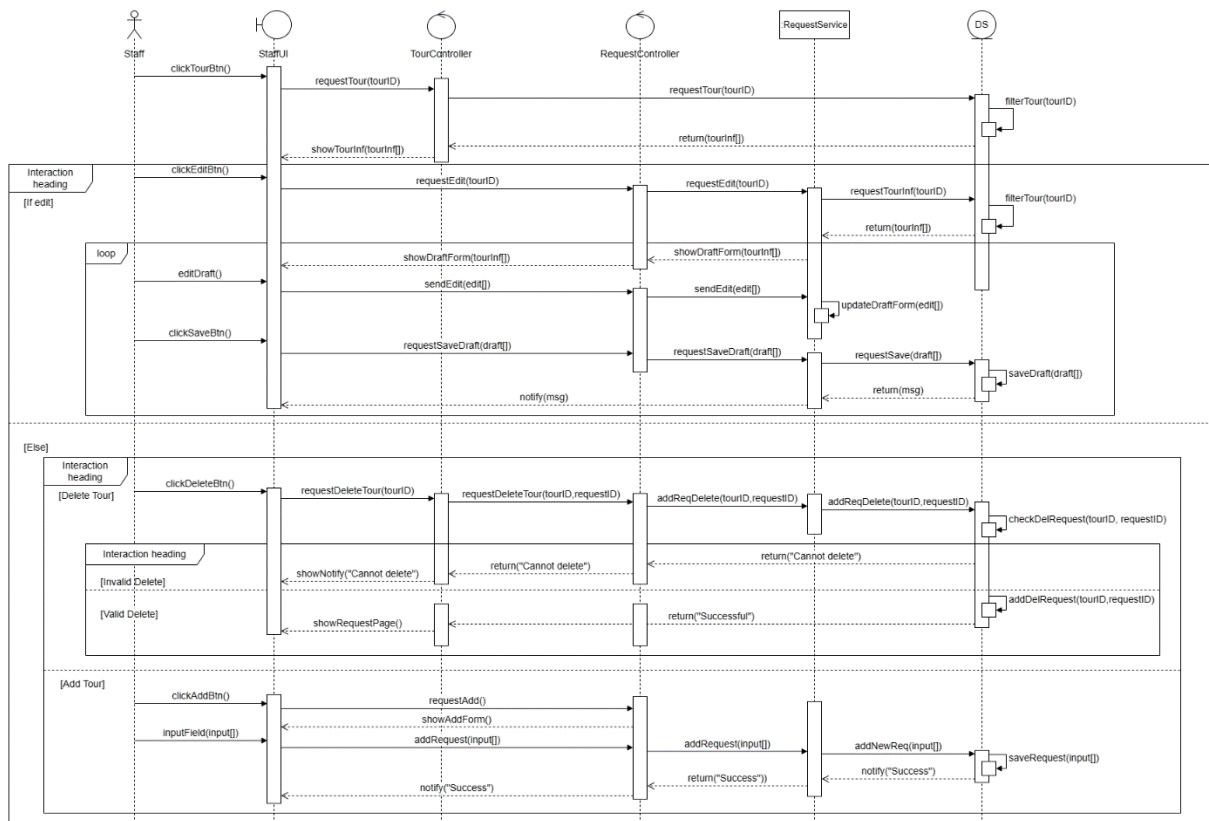


Figure 31. Sequence Diagram for Staff Making a Request

Add a Customer into a Tour

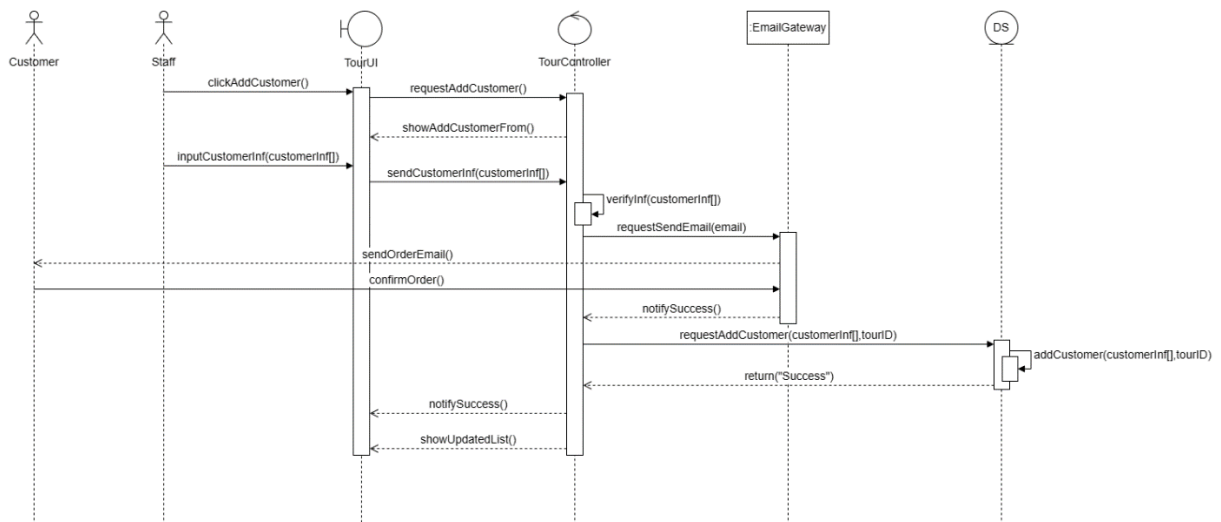


Figure 32. Sequence Diagram for Staff Adding a Customer into a Tour

The Figure 32 shows that we also consider if the customer wants to come to the office and pays for the tour by cash, then a staff can also add a customer manually with steps showing in the Figure 32 above. Once the adding is successful, an confirmation email is sent to the customer.

3.3.3. Manager Sequence Diagram

Login the system

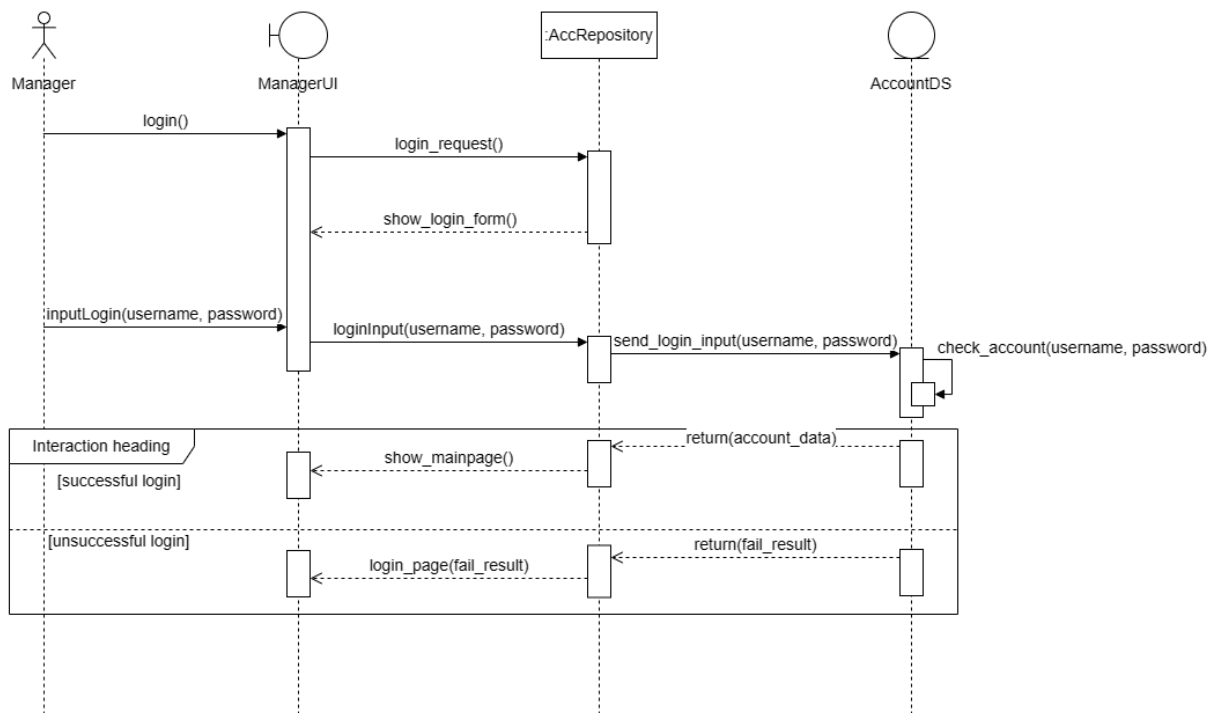


Figure 33. Sequence Diagram for Manager Logging in the system

The Figure 33 shows overall how a manager login to the system. This account is modified in the database. To login to the system as a manager, manager can choose manager mode in the login page and then uses the configured account to login to the system.

View and Decide on a Request

The Figure 34 shows that manager can determine the status of all requests from the staff. If the manager accept a request, the data is then updated with the accepted one. The system will automatically deleted the accepted requests and give announcement for the staff making the request. If the manager decline, the staff will also receive a notification and the comment on the declined requests.

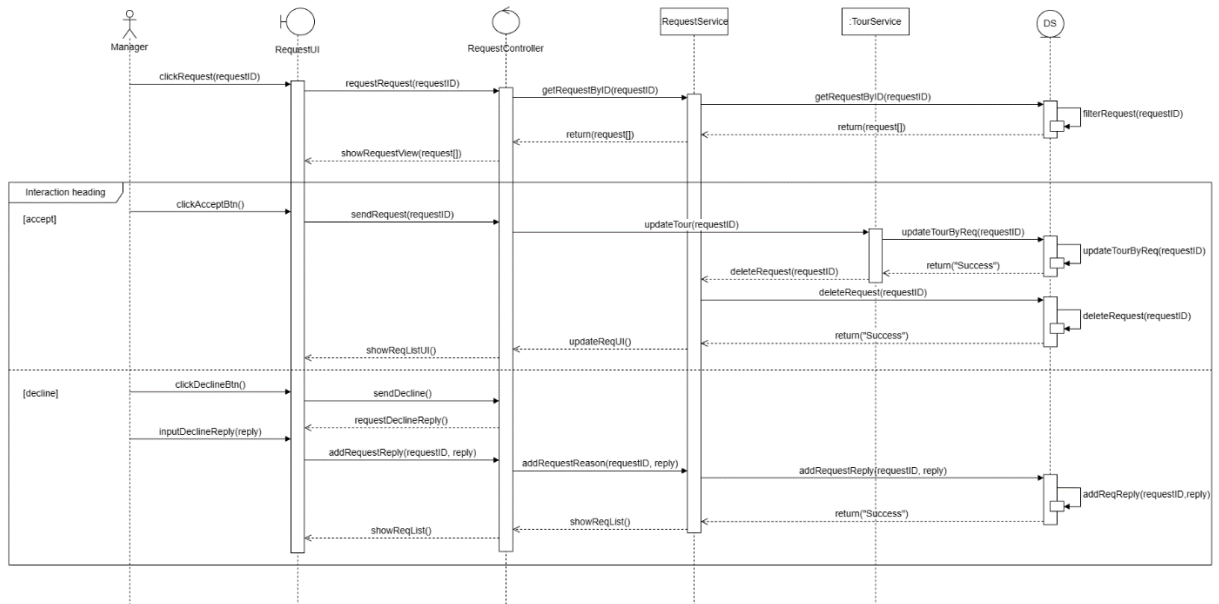


Figure 34. Sequence Diagram for Manager Viewing and Determining a Request

Deactivate a Staff Account

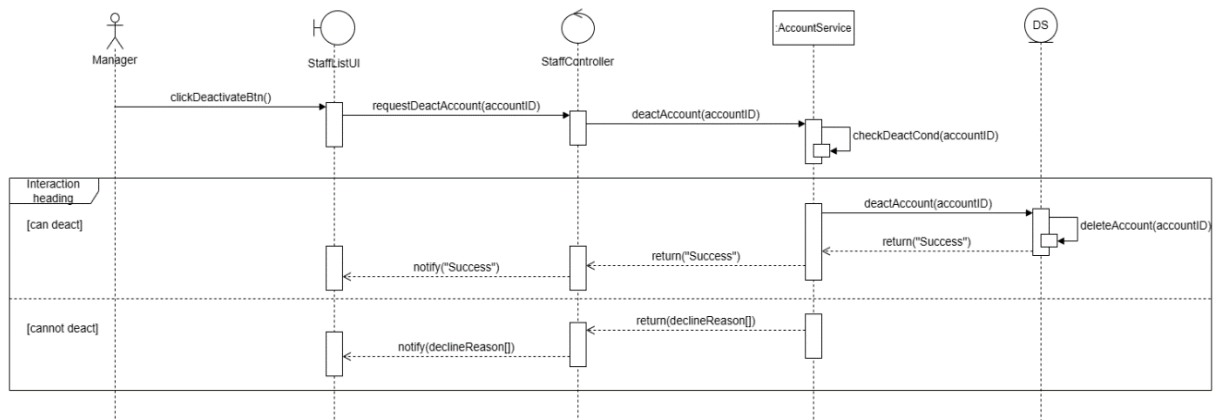


Figure 35. Sequence Diagram for Manager Deactivating an Account

The Figure 35 clearly illustrates that manager can also deactivate staff accounts if the deactivating conditions are met. If these conditions do not meet, the the system will notify the manager with the needed conditions.

3.4. Database design

3.4.1. Enhanced Entity Relationship Diagram (EERD)

Our Enhanced Entity Relationship Diagram is shown in Figure 36. The Entities in our database include:

- **Customer:** which contains information of a customer. This entity includes Id, username, phone number, email, encrypted password for security reasons.
- **Order:** which contains information of a booking order. The entity includes Id, payment method and total bill.
- **Tour:** which contains information of a tour. The entity includes Id, name, price, vehicle, seatNum, departure, starting date, duration, many schedules, many services, note, booking deadline, average ratings and isActive.
- **Feedback:** which contains information of a customer's feedback for a tour. The entity includes Id, reviews, ratings and datetime.
- **Place:** which contains information of a tourist attraction where the company will organize the tours. The entity includes Id, name, province where the place belongs to, many images and the description.
- **Staff:** which contains information of a staff member. The entity has Id, name including firstname and lastname, gender, date of birth, phone number, email and encrypted password for security reasons.
- **Request:** which contains information of a request that a staff sends to a manager. The entity includes Id, status, reply and type. The entity includes Edit Request, Add Request and Cancel Request depending on the discriminative attribute "type".
- **Manager:** which contains information of a manager. The entity includes Id, email and password.

Relationships in our database design include:

- **Customer views Place:** which contains information about the articles of tourist attractions that customers can view. Many customers can view an article and an

article can be viewed by many customers, so the relationship will be many to many.

- **Tour has Place:** which contains information about a place that a tour includes as a destination. Many tours can include a place and many places can be included in a tour, so the relationship will be many to many.
- **Customer adds Feedback:** which contains information about the feedback that a customer gives to a tour. A customer can give a lot of feedback but feedback is given by only one customer, so the relationship will be one to many.
- **Feedback belongs to Tour:** which contains information about the feedback that a customer gives to a tour. The feedback belongs to only one tour while a tour can have many feedback, so the relationship will be many to one.
- **Order belongs to Tour:** which contains information about a booking order of a tour. An order belongs to one tour while a tour can have many orders, so the relationship will be many to one.
- **Order of Customer is added by Staff:** which contains information about an order of a customer which is added manually by staff. A staff can add many orders for many customers, so the relationship will be many - many - one.
- **Staff is managed by Manager:** which contains information about a staff managed by a manager. A staff is managed by a manager but a manager can manage many staff, so the relationship will be many to one.
- **Request is reviewed by Manager:** which contains information about a request reviewed by a manager. A request is reviewed by a manager and a manager can review many requests, so the relationship will be many to one.

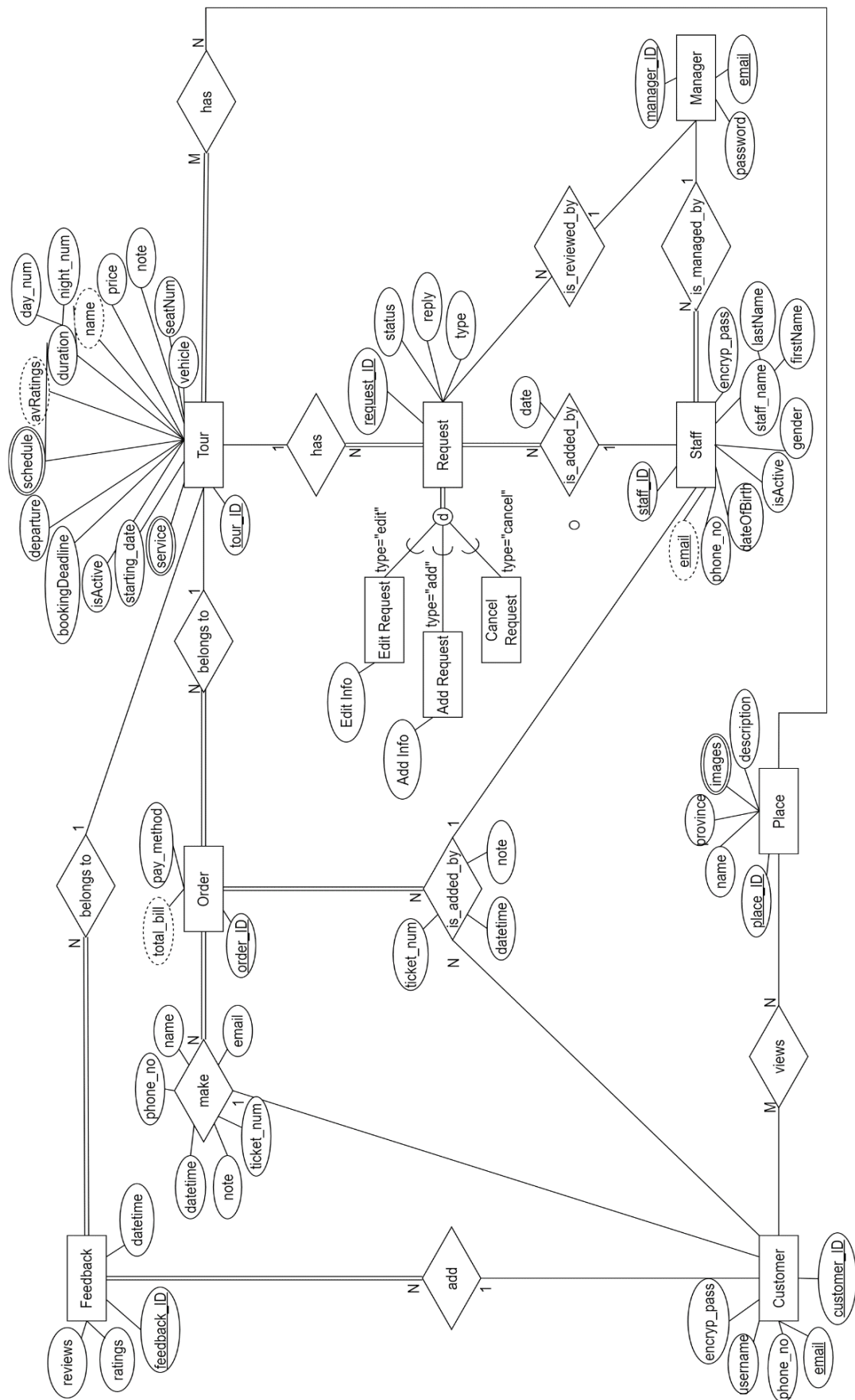


Figure 36. Enhanced Entity Relationship Diagram (EERD)

3.4.2. Relational mapping

From conceptual design, we perform mapping techniques for logical database design. In the Figure 37 below, the bolded words are key attributes and the arrows represent the foreign keys of one entity reference to the primary key of another entity.

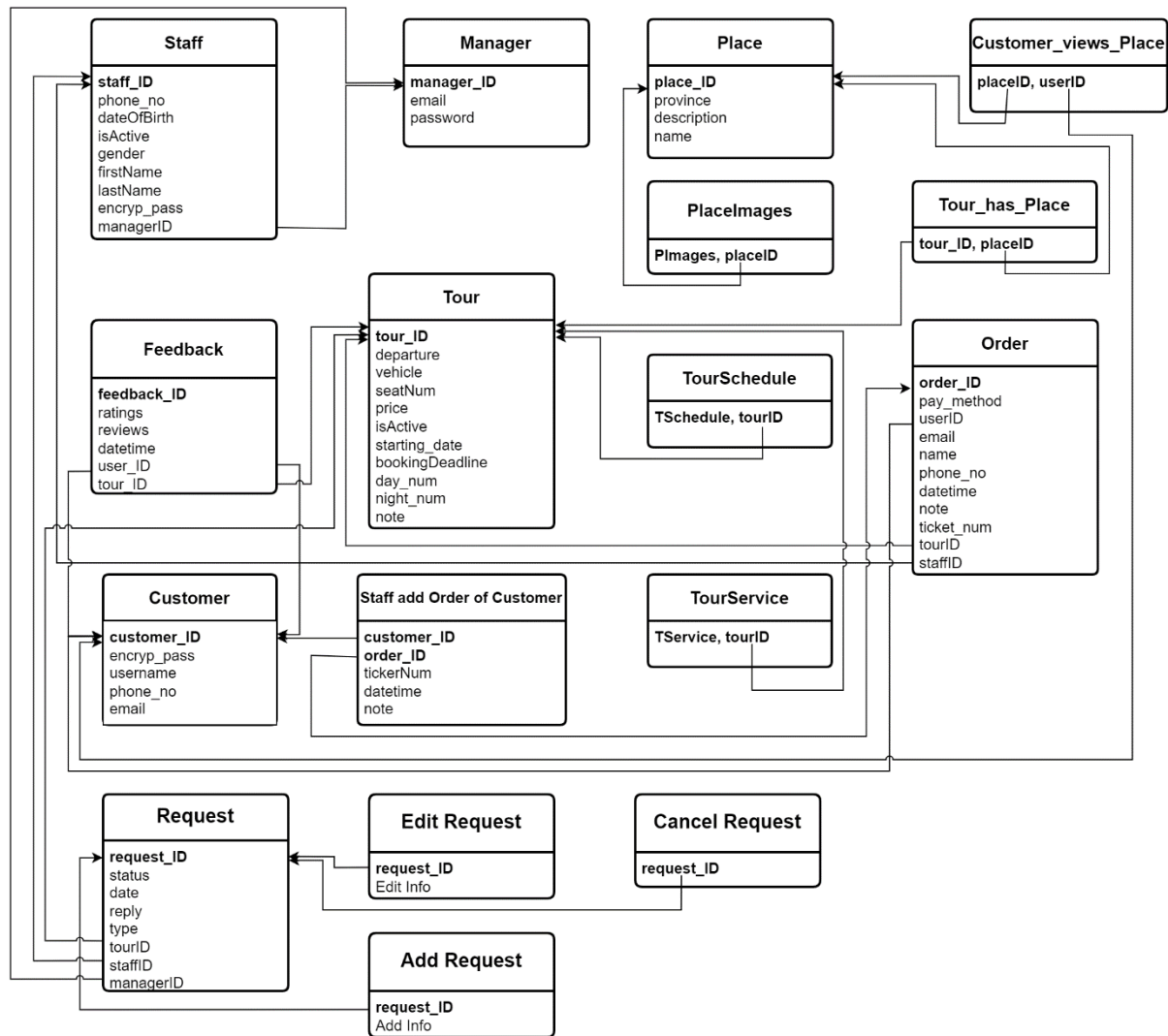


Figure 37. Relational Mapping of EERD

3.5. Architecture design

3.5.1. Architectural pattern

In this project, we will use the MVCS model. The standard architecture of MVCS is shown in Figure 38:

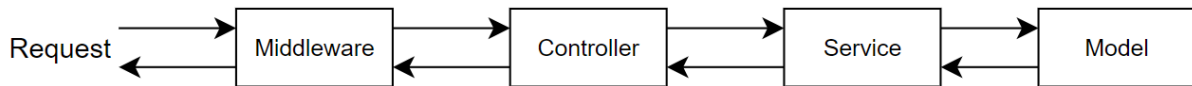


Figure 38. MVCS pattern

MVCS stands for Model-View-Controller-Service. It allows a coder to predefine the interaction between each piece of code and have easy modifications by dividing code into layers according to its functionality. This architecture doesn't belong to any programming language or framework. This pattern separates Database or networking interaction from the Controller. Each part plays an important role in application development. [3]

- View Layer: Represents how the data will be rendered to the view
- Controller layer: Responsible only for receiving the request and displaying the result of the operation. Controller is used for communicating between the Model, Service and View.
- Service Layer: Also known as the Business Layer, contains the functionality that compounds the core of the application, thus becoming highly reusable for controllers and services.
- Model Layer: Responsible for defining how the data model is defined.

3.5.2. Architecture diagram

The Figure 39 illustrates overall how we planned to implement our system. For frontend, all designs have been done with Figma will be implemented by ReactJs and the stucture will be organized as shown in the figure. Frontend is connected to the Django – Backend via http respest. The main models in the backend include “User”, “Places”, “Order”, “Tour Info”, “Feeback”, “Image”, “Staff & Manager Account”, and “Request” model. These models connect directly to the database – MySQL.

To controll code and version, our group use Github and from that we plan to implement the website through AWS Cloud using Elastic Beanstalk, AWS pipeline, and Amazon RDS, all of which are provided by Amazon.

For testing, Postman and AutoCannon will be used to check and evaluate API. Jira Software is also be considered to be used for bug tracking and project managing purpose.

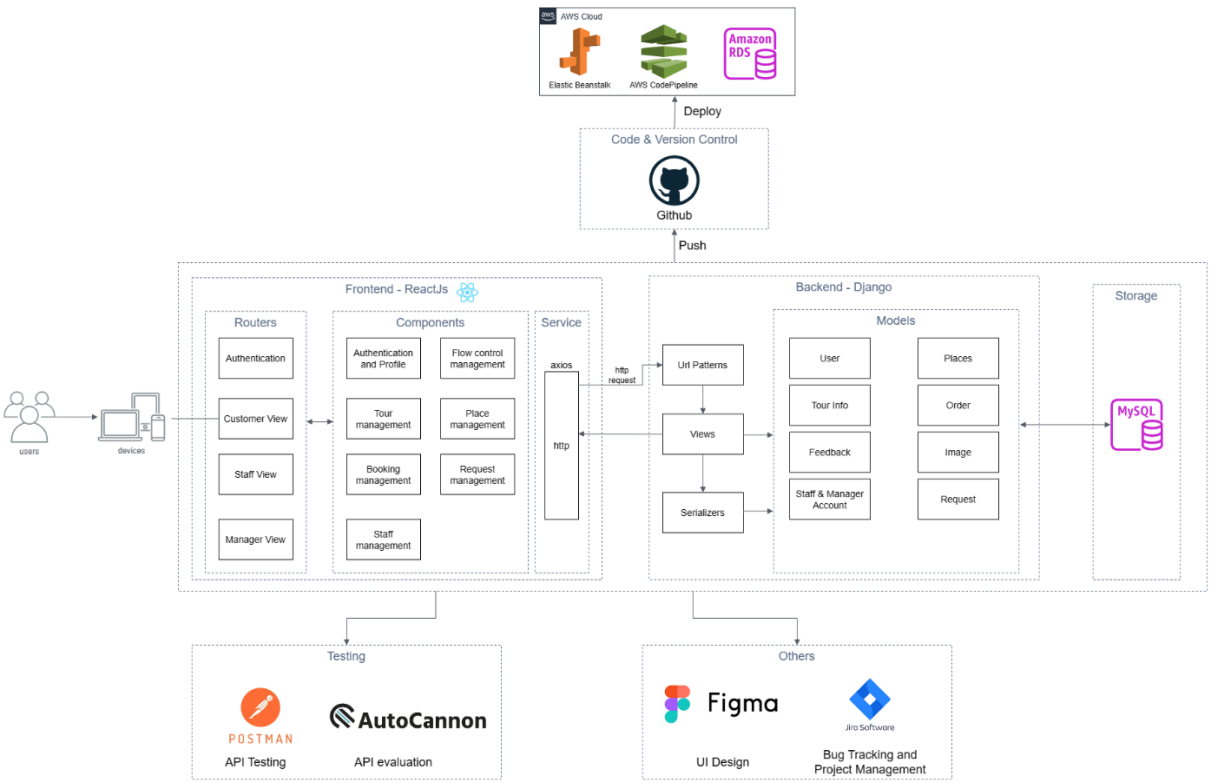


Figure 39. Architecture Diagram for the system

3.6. Component diagram

Customer Component Diagram

The Figure 40 shows the architecture structure of the system that handle the activities of customers. Customer can get views of login page, history of orders (this also including given feedbacks and status of each order), booking page, news and suggested tours of a place, tour lists, and detailed information of a particular tour.

All interactions between customers and the model are coordinated by the controller. All controllers in the system, including authentication, order controller, order processor, email gateway, place controller, tour controller, and tour processor, controls the application logic and acts as the coordinator between the View and the Model for the customers' activities.

For model, as in the order history page, customers can view both ordered tour list and their feedback, then the model “Order & Feedback” needs to be the composition of “Order” model and “Feedback” model.

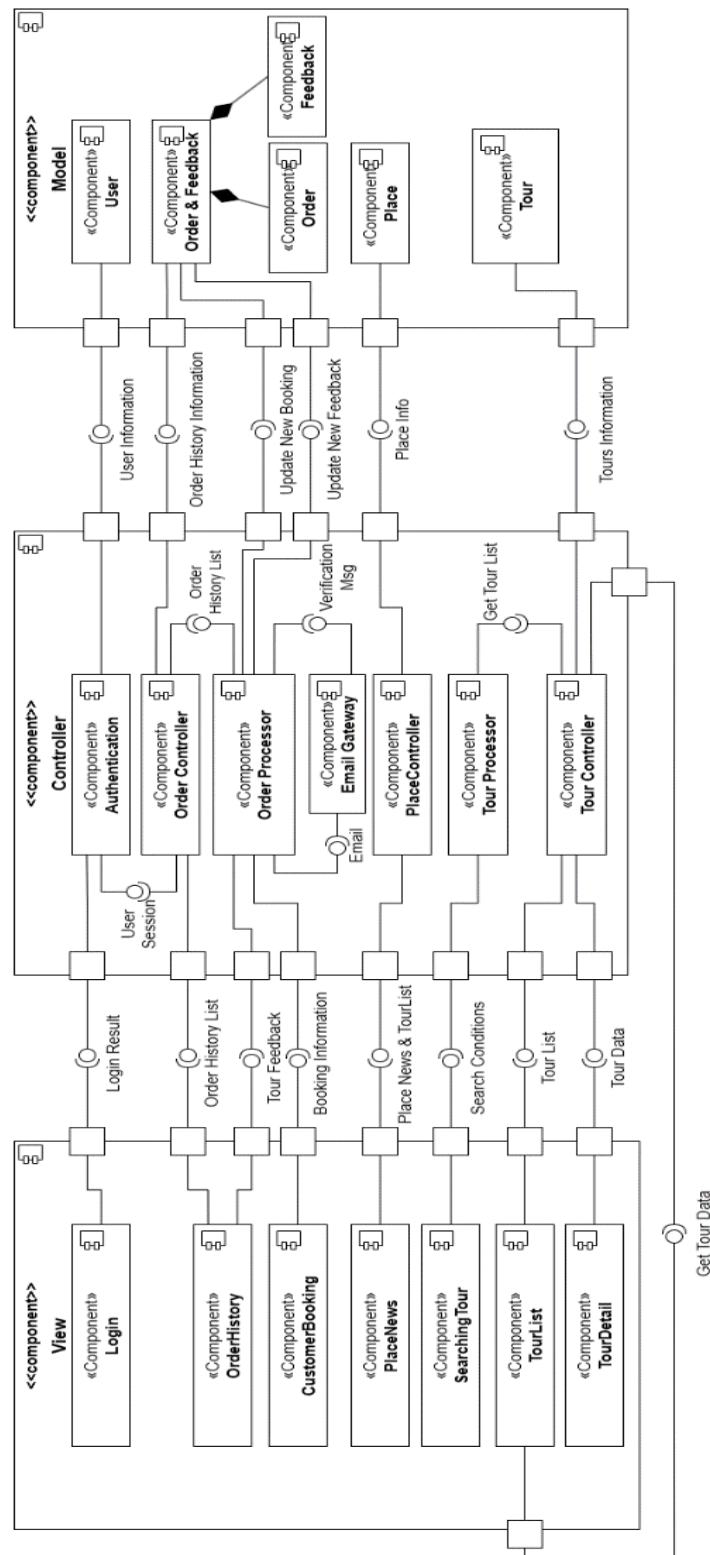


Figure 40. Component Diagram for Customer

Staff and Manager Component Diagram

The Figure 41 shows that the component for Staff and Manager is complex due to the interaction of both stakeholders with each other.

Staff can view list of tour that they have created or assigned, and all requests they have saved. On the other hand, manager can view all tours, all staff, and all requests. All activity of the manager on to the system can affect extensively on the staff's view. The effect are mostly related with the request component, which is displayed on the Figure 41 below.

A notable point on the model is that the "Tour Info" model is a composition of 4 other models: "TourSchedule", "TourService", "Tour", and "Order".

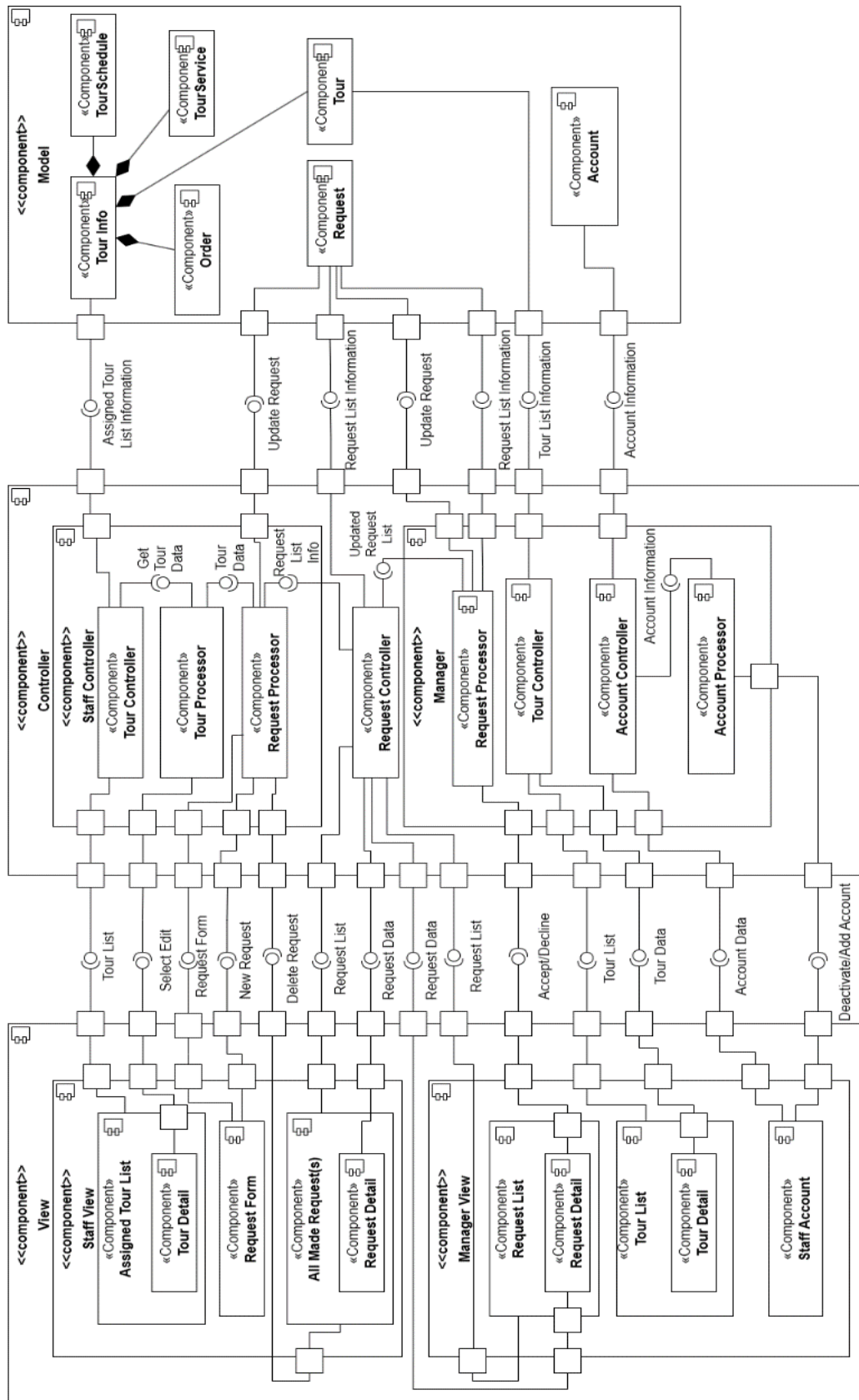


Figure 41. Componet diagram for Staff and Manager

3.7. Class diagram

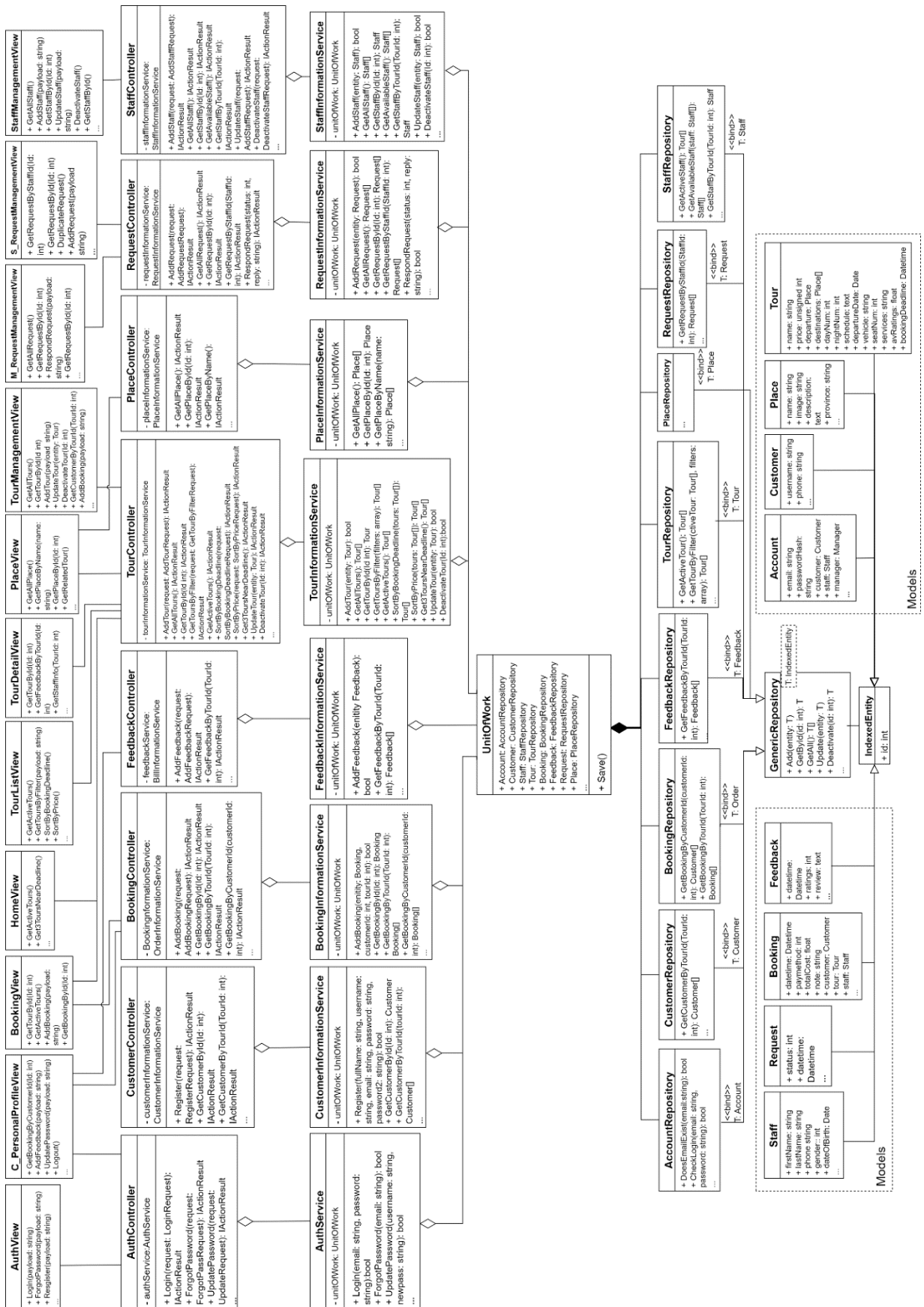


Figure 42. Class diagram

The Figure 42 illustrates our class diagram. Our architecture follows the MVCS model:

3.7.1. View layer

The View layer contains all the available requests a user can make. The request is then sent to the Controller. There are 11 main views as shown in Figure 43.

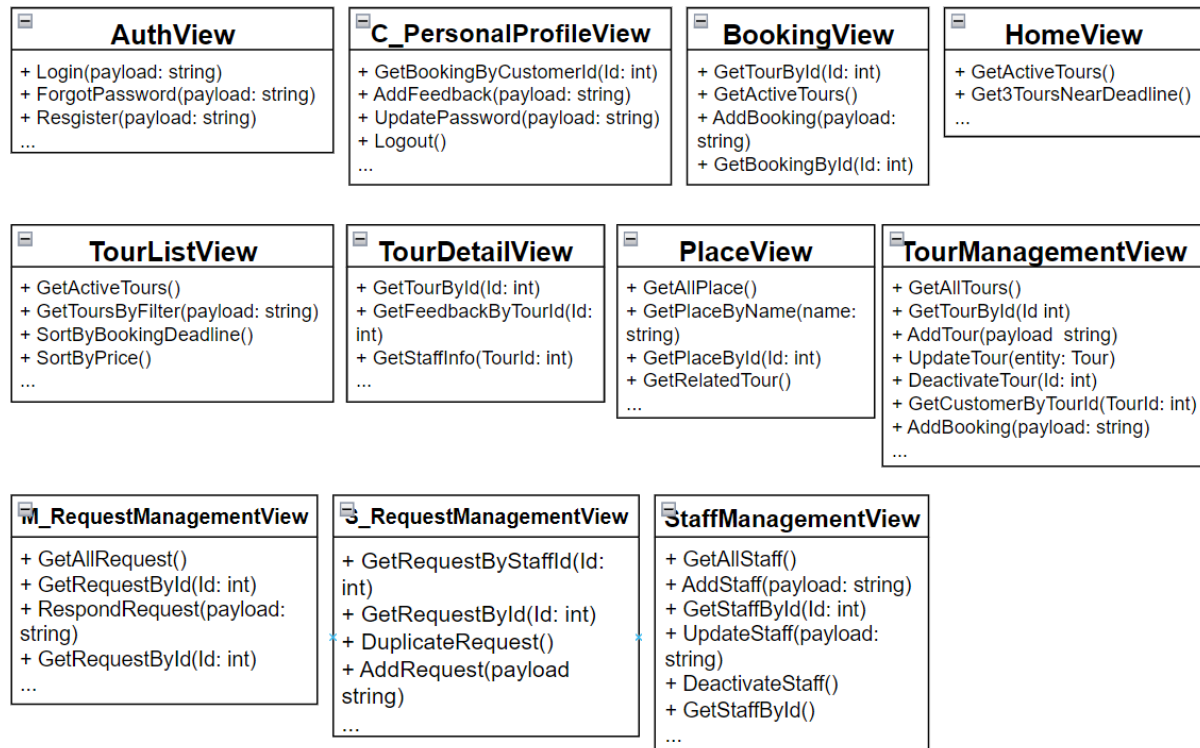


Figure 43. Class diagram in Repository Layer

3.7.2. Controller layer

The Controller layer receives requests from the View layer, calls a Service and returns an interface action result. When a user clicks onto some particular button onto the front-end application, our system would receive the request and return a response by displaying an announcement on the screen. The Controller classes are displayed in Figure 44

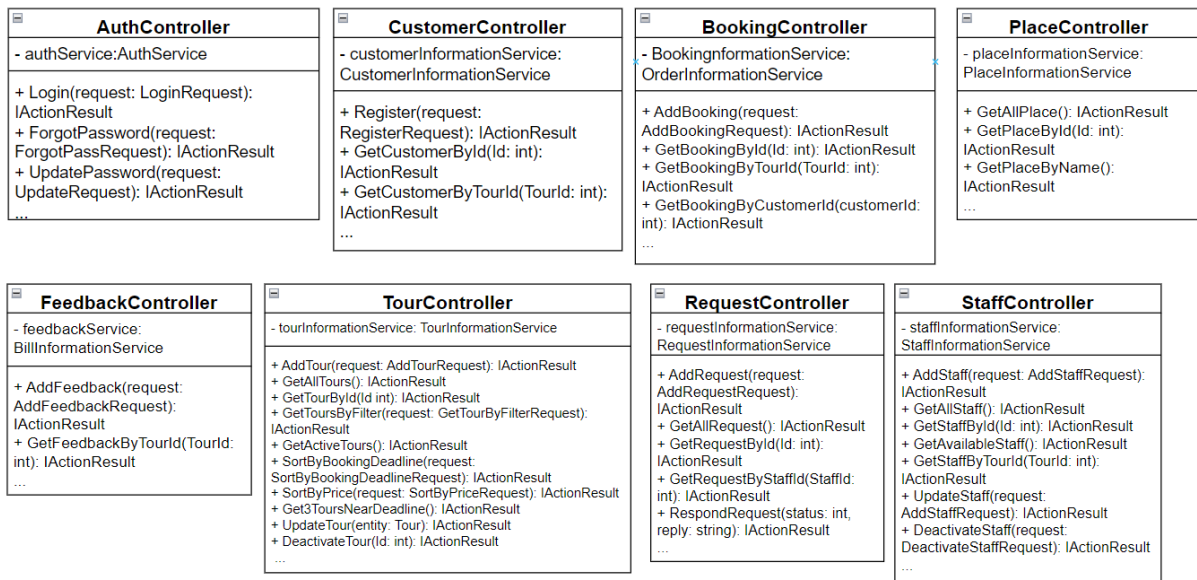


Figure 44. Class diagram in Controller Layer

3.7.3. Service layer

The Service layer encapsulates all the business logic that could be in the controller. That is, the Controller will call this layer's objects to get or update Models, or other requests. The Service layer would potentially utilize the fundamental methods defined in repositories. The Service classes are displayed in Figure 45.

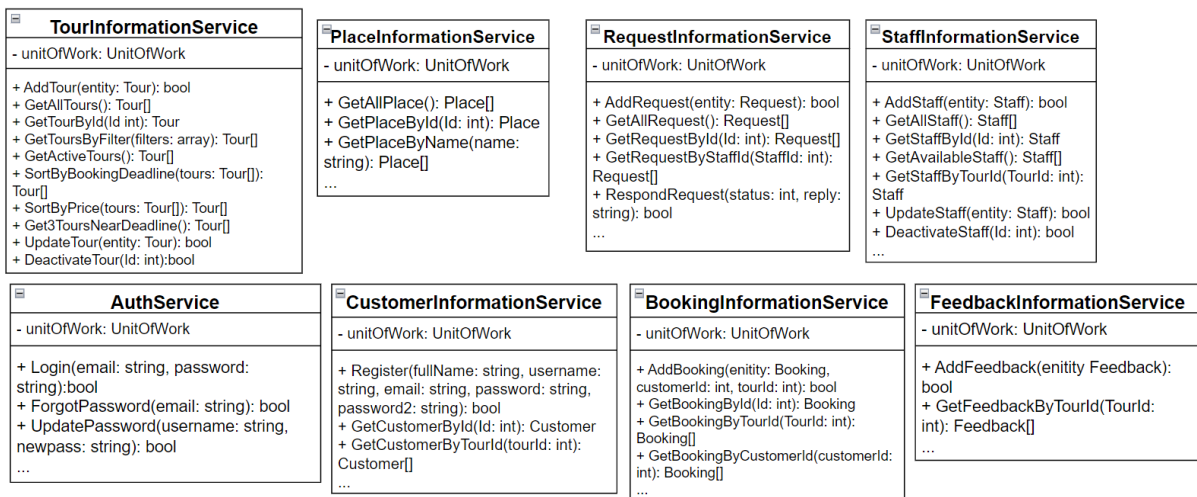


Figure 45. Class diagram in Service Layer

3.7.4. Repository layer

Besides, our architecture will contain the Data Access Objects following the researched patterns: The Service layer will call these abstract classes to get or update the data it needs.

Repository pattern

The Repository pattern is a Domain-Driven Design pattern intended to keep persistence concerns outside of the system's domain model by creating an abstraction layer between the data access layer and the business logic layer of an application

A concrete repository gives access to an entity, an entity consisting of one or more related tables. A repository encapsulates all CRUD operations, in addition the repository also acts as an interface between domain and data model. This means that the domain and data model are only loosely coupled via well-defined interfaces. [4]

Generic Repository

Creating a repository class for each entity type could result in a lot of redundant code, and it could result in partial updates. One way to minimize redundant code is to use a generic repository. The generic repository will handle typical CRUD requirements. When a particular entity type has special requirements, such as more complex filtering or ordering, we create a derived class that has additional methods for that type.

Unit of Work pattern

This is a pattern that is commonly used with the repository pattern. All database operations should be handled in a single database transaction. This means that the repositories do not commit their respective transactions themselves, the database session of the repositories is managed by the unit of work instance, therefore the unit of work is also responsible for committing or rolling back the transaction. [4]

Suppose you have to update two different entity types as part of the same transaction. If each uses a separate database context instance, one might succeed and the other might fail. Therefore, this is especially important when multiple repositories are involved: the

unit of work instance ensures that either all or none of the database operations are committed, keeping our data consistent.

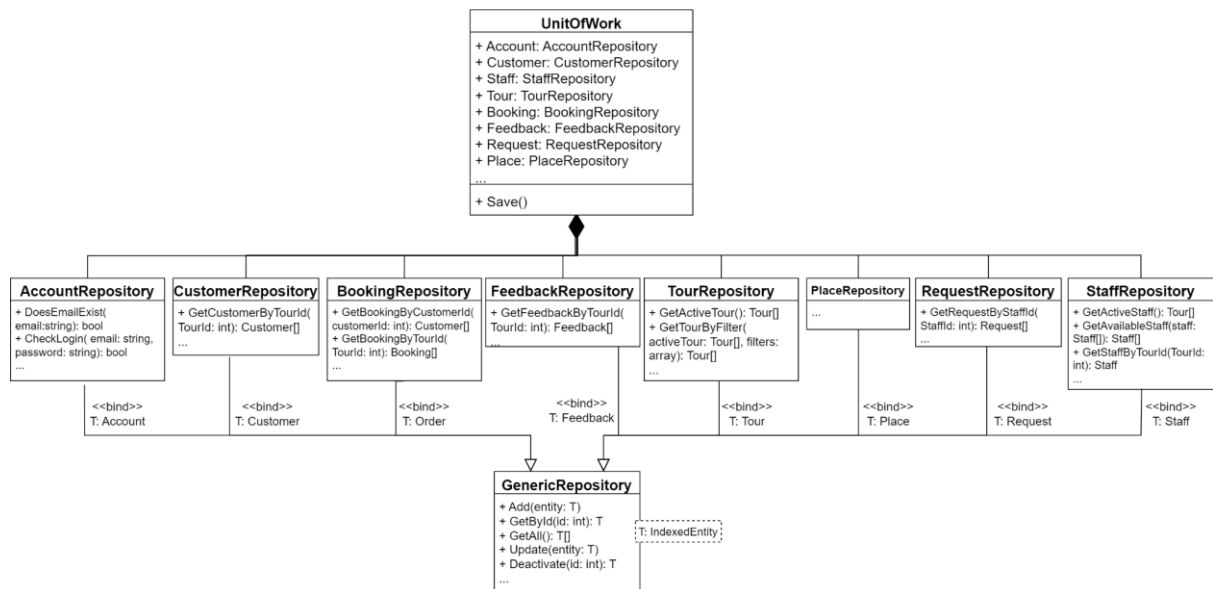


Figure 46. Class diagram in Repository Layer

The Figure 46 shows all the classes in the Repository layer.

- **GenericRepository class:** This is the abstract class of a repository, which defines general repository methods such as: Add, GetById, GetAll, Update, Deactivate, etc to complement concrete repositories.
- **<Entity>Repository class:** these repositories are concrete repositories which inherit from the GenericRepository. The entity specific methods are implemented in these repositories.
- **UnitOfWork class:** UnitOfWork defines the basic functions of the Unit of Work interface. It is used to wrap and manage the database session.

3.7.5. Model layer

The Model layer shown in Figure 47 contains all the tables in the Database. For the sake of simplicity, we design this diagram with only some main entities: Customer, Booking, Tour, Feedback, Place, Staff, Request. The Account entity represents the account information of all users. This entity is only drawn to separate the methods of Account Repository, therefore, it is not drawn in the EERD. Since all of the entities in our database are designed as strong entities which means every table will have the key

attributes Id. We generalize the concept by adding the IndexedEntity class and utilizing inheritance to reduce code duplication.

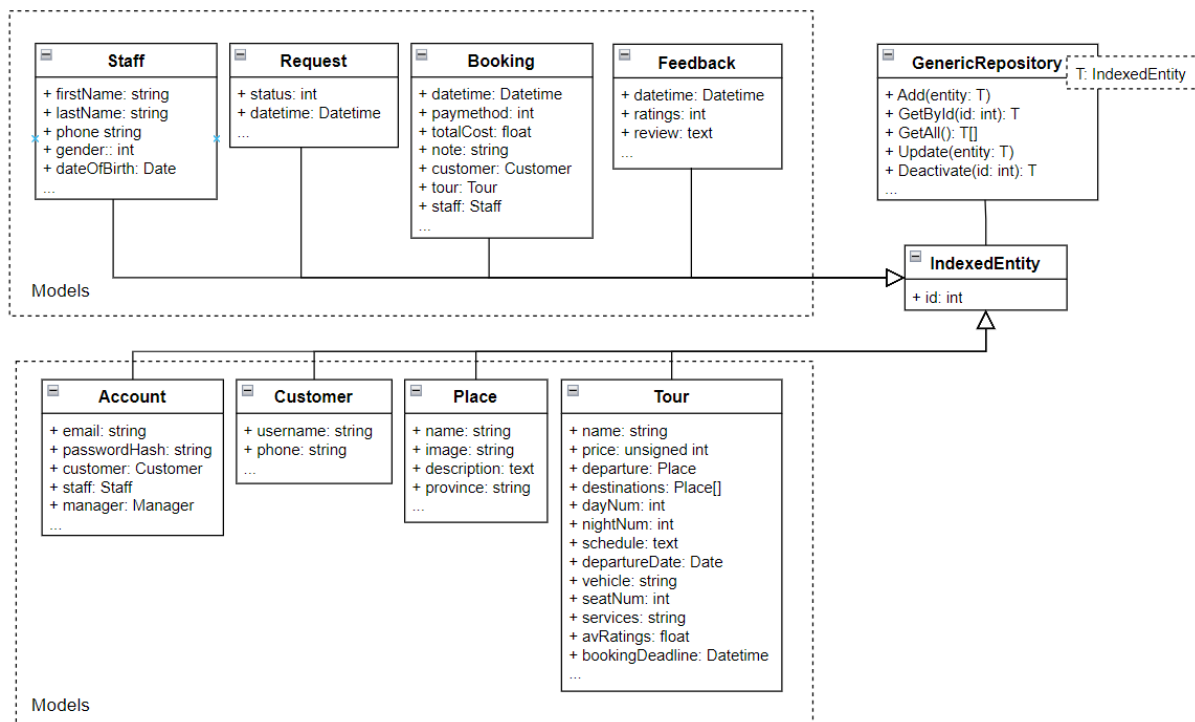


Figure 47. Class diagram in Model Layer

Chapter 4

Technology research

4.1. Front-end development research

The front-end can be seen as client-side development and refers to the component of the program that customers can view and interact with. The responsibility of a front-end developer is not restricted to the implementation of visual components alone. It also involves designing and managing interactive functions, such as navigation, buttons, and anything else that affects how the application works. [5]

Front-end frameworks are software development tools used in custom web application development for building user interfaces. They provide a collection of pre-written code that can be reused by following specific standards, which typically include conventions for organizing different parts of the code. Front-end frameworks help developers understand and work with code written by others more easily, making them especially useful when different dedicated developers are working on the same codebase. This also provides developers with a single set of rules, documentation, and best practices to write code. [6]

With a well-designed front-end, our website can impress and attract more customers as well as help the staff and managers work more effectively. Understanding the importance of front-end development, we have done research on which front-end framework to use.

4.1.1. AngularJS

AngularJS is a structural framework for dynamic web apps. It lets you use HTML as your template language and lets you extend HTML's syntax to express your application's components clearly and briefly. AngularJS's data binding and dependency injection eliminate much of the code you would otherwise have to write. And it all happens within the browser which is ideal for any server technology. [7]

AngularJS is what HTML would have been, had it been designed for applications. HTML is a great declarative language for static documents. The impedance mismatch between dynamic applications and static documents is often solved with: [7]

- **Library** - a collection of functions which are useful when writing web apps. Your code is in charge and it calls into the library when it sees fit. E.g., jQuery.
- **Frameworks** - a particular implementation of a web application, where your code fills in the details. The framework is in charge and it calls into your code when it needs something app specific. E.g., durandal, ember, etc.

Benefits of AngularJS [8]

- **MVC support:** AngularJS offers a facility that stitches these separated codes dynamically. As a result, it saves a lot of time for developers.
- **Code reuse:** AngularJS allows them to use the predefined templates and modify them as required.
- **Data binding feature:** AngularJS supports one-way and two-way data binding features. It allows developers to manipulate data in the model layer and update the view layer simultaneously.
- **Large community support:** AngularJS is managed by Google so the features will receive regular updates and support. It also garners support from the open community as it is an open source framework.
- **Real-time testing:** AngularJS supports both end-to-end testing and unit testing.
- **Quick application prototyping:** Less code, time, effort, and resources is needed for interactive application implementation.

Challenges of AngularJS [8]

- **Javascript support mandatory:** Older computers and hardware that lack Javascript support won't have the necessary hardware and software to reach the AngularJS page
- **Performance issues:** Applications developed using AngularJS find it difficult to handle heavy traffic. They sometimes lag, resulting in below-par app performances.

- **Difficult to learn:** AngularJS lacks the necessary documentation for knowledge sharing and learning. Additionally, some directives and features are quite complex for traditional developers to grasp.
- **Memory leakage:** Memory leakage occurs when a memory unit that is not used or necessary for an application is not returned to the OS. If such leaks affect the applications developed through AngularJS, then it can give rise to a series of problems such as application slowdown or crash, latency issues, and can even affect the performance of other applications.

4.1.2. Vue

Vue is a JavaScript framework for building user interfaces. It builds on top of standard HTML, CSS, and JavaScript and provides a declarative and component-based programming model that helps developers to create interactive UI for both simple and complex applications. [9]

Benefits of Vue.js [9]

- **Simplicity:** Vue.js consists of single-file components that include all kinds of the codes such as HTML, CSS, and JavaScript.
- **Easy to learn:** Developers only need basic knowledge with the basics of HTML, CSS and JavaScript, which is not the case when it comes to Angular or React.
- **Simple integration:** It is easy to integrate Vue.js into other frameworks.
- **Virtual DOM performance and rendering:** Vue.js uses the virtual DOM, a copy of the website's original DOM is created so that an adequate amount of changes can be made without rendering the entire DOM and this minimizes the number of DOM manipulations needed to be handled by Vue.
- **Two-Way binding:** Vue inherited this feature from Angular. This feature makes the framework ideal for applications with real-time updates.
- **Readability and Single-File Components:** Vue uses different components to build your web pages. An architecture with traits like this also means a couple of benefits: Reusable Components, Great Unit-testing, Code Reading.

Challenges of Vue.js [9]

- **Language constraint:** VueJS was initially developed by the developers in China and is immensely popular there. therefore, for any forum conversations, plugin descriptions, or any other type of instructional manuals, they are developed in Chinese language, which means you need to get translations for them to proceed ahead in work. This can sometimes act as a language barrier.
- **No Streamlined Approach:** VueJS is a community that is funded and maintained by developers therefore it lacks significant funding or is not strongly supported by big size companies. The technology isn't reliable or well-supported enough to deliver the fast solutions for bugs that is expected by big shot firms and while on other hand could be provided by React or Angular support. Furthermore, for all the tools that are being built, VueJS still has a long way to go in comparison to the sheer number of custom tool/s that React and Angular offer.
- **Lack of Experienced Users:** There is a lack of experienced VueJS developers which makes it less preferable for developers to use their services.

4.1.3. React

ReactJS is the highly used open-source JavaScript Library. It helps in creating impressive web apps that require minimal effort and coding. The main objective of ReactJS is to develop User Interfaces (UI) that improves the speed of the apps. There are important pros and cons of ReactJS given as following:

Benefits of ReactJS [10]

- **Easy to Learn and Use:** ReactJS comes with a good supply of documentation, tutorials, and training resources.
- **Creating Dynamic Web Applications Becomes Easier:** It provides less coding and gives more functionality. It makes use of the JSX (JavaScript Extension), which is a particular syntax letting HTML quotes and HTML tag syntax to render particular subcomponents.
- **Effortless Maintenance:** ReactJS provides a modular and flexible architecture, leading to modernizing each component independently without affecting others.

In addition, objects defined for a particular element can be reassigned to other components to reduce the developer's effort.

- **Component Reusability:** React JS provides the architecture and ecosystem to build business software based on components and reuse each component multiple times.
- **Fast Rendering:** React JS comprises the in-built functionality of virtual DOM, making the rendering faster and enhancing the overall user experience.
- **SEO Friendly:** The Virtual DOM feature of ReactJS renders the application more accurate and returns the web app as a typical web page. Due to this, search bots are able to crawl through the React JS website and effortlessly analyze the relevance and rank the website on the search engine.
- **Ample Community Support:** Compared to other open-source JavaScript libraries, ReactJS community is immensely growing. Therefore, it will be easier to resolve the difficulties during development.

Challenges of ReactJS [11]

- **The high pace of development:** The high pace of development has an advantage and disadvantage both. In case of disadvantage, since the environment continually changes so fast, some of the developers do not feel comfortable to relearn the new ways of doing things regularly.
- **Poor Documentation:** React technologies updating and accelerating so fast that there is no time to make proper documentation.
- **View Part:** ReactJS Covers only the UI Layers of the app and nothing else. So you still need to choose some other technologies to get a complete tooling set for development in the project.

Considering the advantages and disadvantages of ReactJS, we see that ReactJS' strengths outweigh its weaknesses and that Reactjs is a better fit for our project, so we decided to use ReactJS for this project. Some ReactJS components are presented below.

4.1.3.1. Component

A Component is one of the core building blocks of React. Components let you split the UI into independent, reusable pieces, and think about each piece in isolation. Components make the task of building UIs much easier. Conceptually, components are like JavaScript functions. They accept arbitrary inputs (called “props”) and return React elements describing what should appear on the screen [12]. You can see a UI broken down into multiple individual pieces called components and work on them independently and merge them all in a parent component which will be your final UI.

Functional Components [13]

Functional components are simply javascript functions. We can create a functional component in React by writing a javascript function. These functions may or may not receive data as parameters. We will use functional components when we are sure that our component does not require interacting or working with any other component. That is, these components do not require data from other components however we can compose multiple functional components under a single functional component.

Class Components [13]

The class components are a little more complex than the functional components. The functional components are not aware of the other components in your program whereas the class components can work with each other. We can pass data from one class component to another class component. We can use JavaScript ES6 classes to create class-based components in React. We can also use class-based components when our component requires interacting or working with other components. but it is not recommended as using class-based components without need will make your application in-efficient.

4.1.3.2. React Router Dom

In traditional websites, the browser requests a document from a web server, downloads and evaluates CSS and JavaScript assets, and renders the HTML sent from the server. When the user clicks a link, it starts the process all over again for a new page.

React Router DOM is an npm package that enables you to implement dynamic routing in a web app. That is, React Router enables “client side routing” which allows your app to update the URL from a link click without making another request for another document from the server. Instead, your app can immediately render some new UI and make data requests with “fetch” to update the page with new information. [14]

This enables faster user experiences because the browser doesn't need to request an entirely new document or re-evaluate CSS and JavaScript assets for the next page. It also enables more dynamic user experiences with things like animation. [15]

Nested Routes

Nested Routing is the general idea of coupling segments of the URL to component hierarchy and data. In nearly every case, segments of the URL determine: [15]

- The layouts to render on the page
- The data dependencies of those layouts

React Router embraces this convention with APIs for creating nested layouts coupled to URL segments and data.

4.1.3.3. React Memo

React memo lets you skip re-rendering a component when its props are unchanged. We can wrap a component in memo to get a memoized version of that component. This memoized version of your component will usually not be re-rendered when its parent component is re-rendered as long as its props have not changed. But React may still re-render it: memoization is a performance optimization, not a guarantee. [16]

4.1.3.4. React Redux

Redux is an open source and client-side State management library, creating a global store in which State can be shared between components. Redux can be used with any JS library or framework like React, Angular, or Vue. It helps you write applications that behave consistently and are easy to test.

State is a JavaScript object used to store a component's data. When a component's state changes, React will automatically re-render the component to reflect these changes on the user interface.

The three components of Redux are: Action, Reducer and Store. [17]

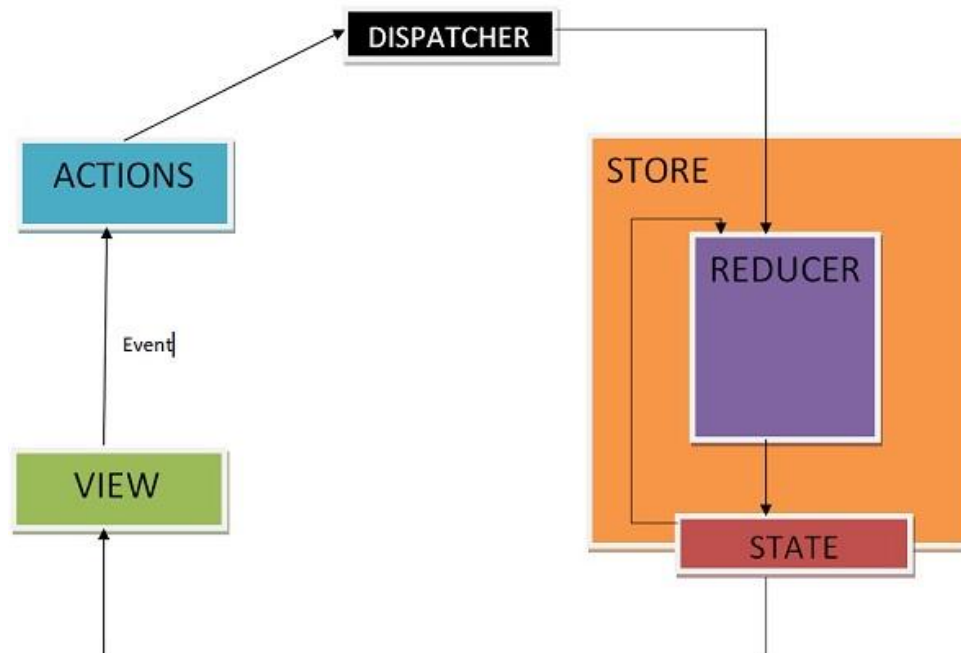


Figure 48. Redux architecture

- **Action:** Action is a simple function that returns an object and is responsible for sending data to the store through dispatch and reducer functions. It has two properties: payload and type. The action is called “type” and the value applied to change the state of the application is called “payload”.
- **Reducer:** “Reducer” is a function used to change the state of the application in a way that depends on specific types of actions. The function takes two arguments: current state and action, and returns a new state. The reducer's main function is to create a new state for the store.
- **Store:** Store is a JavaScript object and a central repository for state management, accessible from the entire application. The only method to access the store is through sending an action.

Redux Persist: With the Redux Persist library, developers can save the Redux store in persistent storage, for example, the local storage. Therefore, even after refreshing the browser, the site state will still be preserved. Redux Persist also includes methods that allow us to customize the state that gets persisted and rehydrated, all with an easily understandable API.

Redux Thunk Middleware: It allows us to return functions instead of objects from redux actions. Plain redux doesn't allow complex logic inside action functions, you can only perform simple synchronous updates by dispatching actions. This middleware extends its ability and lets you write complex logic that interacts with the store. Thunk doesn't interfere with the action until it returns a function. Thunk allows us to dispatch actions manually, which gives us the power to incorporate some logic or run some asynchronous code before dispatching an action. [14]

4.1.3.5. Axios

Axios, which is a popular library, is mainly used to send asynchronous HTTP requests to REST endpoints. This library is very useful to perform CRUD operations. [18]

- This popular library is used to communicate with the backend. Axios supports the Promise API, native to JS ES6.
- Using Axios we make API requests in our application. Once the request is made we get the data in Return, and then we use this data in our project.
- This library is very popular among developers.

4.1.4. CSS framework

We decided to use Tailwind CSS for the project. Tailwind CSS is a utility-first CSS framework having built-in classes. Tailwind CSS has many classes that cover different and important CSS properties, we can easily extend and create new classes with its own classes. [19]

Tailwind CSS offers several advantages over Bootstrap – one of the most popular open-source CSS frameworks:

- **Utility-First Approach:** Tailwind provides a vast set of pre-defined utility classes that can be directly applied to HTML elements. This approach allows for rapid development, as you can style elements by composing classes, reducing the need for custom CSS.
- **Customizable:** It is easy to extend or modify the default theme to match the project's specific design requirements.
- **Smaller File Size:** Unlike Bootstrap that come with a lot of pre-designed components and styles, Tailwind CSS generates a smaller file size since you only include the classes you use. This leads to more efficient and faster-loading web pages.
- **Flexibility:** Tailwind provides low-level utility classes, giving the flexibility to build unique and custom designs without being constrained by predefined components.
- **Developer Productivity:** The utility-first approach and the availability of numerous ready-to-use classes help developers focus on building functionality rather than spending time writing custom CSS for styling.
- **Responsive Design:** It is easy to create responsive designs. It provides responsive utility classes for quickly adapting your layout and elements to different screen sizes and devices.
- **Community and Ecosystem:** Tailwind CSS has gained popularity, leading to a thriving community and ecosystem. You can find a wide range of plugins, tools, and resources to enhance your development workflow and add additional functionality.

In conclusion, we will use ReactJS as the Front-end framework and Tailwind CSS as the CSS framework to develop the Front-end for this project..

4.2. Server development research

Choosing a suitable framework for building the system is also one of the essential steps. After experiencing some framework and searching for widely used frameworks. We have figured out three frameworks including NodeJS (ExpressJS), Ruby on Rails, and

Django. To choose the most appropriate framework, we have listed out some notable benefits and challenges.

4.2.1. Node.js (Express.js)

Express is a lightweight and flexible web application framework for Node.js that provides a set of necessary features for building web and mobile applications. It also provides HTTP utility methods and middleware for API development.

Express library has four main components including Application, Request, Response, and Router. Together, these components are composited and then create a lightweight backend framework that provides routing and middleware functionality.

Benefits

- ExpressJS supports JavaScript, a widely used programming language that is simple to learn and has numerous instructing documents or support.
- ExpressJS is a minimal and lightweight web framework, which makes it fast and suitable for building both small and large applications.
- ExpressJS has a robust middleware system that allows developers to enhance and extend the functionality of applications.
- Being built on top of Node.js, ExpressJS benefits from the extensive Node.js ecosystem and npm packages.

Challenges

- ExpressJS does not include many advanced features out of the box. We see that we will have to use additional modules or build custom functionality to implement some functionalities for our system.
- ExpressJS is not immune to security vulnerabilities.
- ExpressJS does not have lots of features or tools that provide help for managing the complexity and challenges of such applications that we expect: Data modeling and validation, Database integration and migration, Authentication and authorization, etc.

Overall, we consider that ExpressJS is a lightweight and easy to learn framework. However, it may be shortened for our needed applications and tools, and it may not be a suitable choice for customer's data security.

4.2.2. Ruby on Rails

Ruby on Rails, often referred as Rails, is a web application framework written in Ruby. Ruby on Rails follows the Model-View-Controller (MVC) architectural pattern and emphasizes convention over configuration.

Benefits

- Ruby on Rails provides an easy-to-use Object-Relational Mapping (ORM) system for database interaction.
- Ruby on Rails has a large ecosystem that allows excellent solutions for developers. The community's gem hosting services allows access to a number of 'gems' that could be add-ons, libraries, snippets and so on.

Challenges

- Ruby on Rails has slow runtime speed and performance. It is slower compared to other frameworks such as Django and NodeJS.
- Ruby on Rails has a limitation on good documentation, particularly for the less popular gems and for libraries which make heavy use of mixins (which is most of Rails). This may cost us time in figuring needed libraries.

Overall, Ruby on Rails is suitable for our project's MVC model as it follows MVC architecture. However, its the runtime speed and performance may make the latency time higher. Also, lacking of good documentation is a weak point of this framework.

4.2.3. Django

Django is a high-level web framework in Python that follows the Model-View-Controller (MVC) architectural pattern. It encourages rapid development and clean, pragmatic design.

Benefits

- Django includes many features by default, some of them maybe needed in our system such as an admin interface, authentication, and an ORM, reducing the need for external libraries.
- Due to being python written, Django provides more amount of dynamism and flexibility to developers.
- Django incorporates security best practices by default, helping developers build secure applications.
- Django is widely used and has large support community. Also, there are several clear instructing documentations.
- Django also provides an elegant and powerful way to interact with the database in an Object-Oriented fashion.

Challenges

- Django might make developers face learning curve due to the richness of features and the Django way of doing things.
- Having experienced Django, we see that the ORM system lacks features. This may cause difficulties in querying data.

Overall, like Ruby on Rails, Django is suitable for our project's MVC model as it also follows MVC architecture. Moreover, Django also contains the richness of tutorials and instructing documentations, providing lots of useful features, especially for security. On the other hand, the challenge of using Django is that, learning curve, lack of features on ORM.

Conclusions: After considering both advantages and disadvantages of three popular frameworks namely ExpressJS, Ruby on Rails, and Django, our group has come to the decision that the Django framework will be used to implement the system.

Although there are still some challenges in using Django, it is still the most suitable framework due to extensive features, architecture pattern, and widely support from community and documentations.

4.3. Database management system research

For Database management, we decide using SQL-based database management systems, including Postgres and MySQL. To choose the most suitable one, we have listed out some important characteristics to compare.

4.3.1. Postgres

Postgres is an object-relational database management system. It uses SQL (Structured Query Language) as its primary query language.

Benefits

- Postgres is an open-source database management system. Therefore, it is free to use, modify, and implement.
- PostgreSQL has robust support for working with JSON data, including a JSONB data type that provides efficient storage and querying for JSON.

Challenges

- The default configuration of Postgres may not be optimized for specific use cases, requiring users to fine-tune settings for optimal performance.
- The performance of Postgres can be slower as compared to MySQL.
- Further development or debugging in Postgres may consume time.

4.3.2. MySQL

MySQL is an open-source relational database management system (RDBMS) that is widely used for building and managing databases.

Benefits

- MySQL often has high evaluation for its performance in read-heavy operations, making it suitable for applications with a significant number of SELECT queries.
- MySQL is also easy to use and learn. Therefore, further development or modification will consume less amount of time.

- MySQL has a large and active community, and it is widely used in the web development community. There are many third-party tools and libraries available for MySQL.
- MySQL does support stored procedures and triggers, providing some level of procedural programming.

Challenges

- MySQL is limited in features, compared with Postgres.
- MySQL is dual-licensed under the GPL and a commercial license, which may have implications depending on the type of project.

Conclusions: After comparing and considering with system features, we see that MySQL is more appropriate than Postgres. This is due to its high performance in read-heavy operations, suitability for development and modifications, and the wide support community with lots of good tutorials and documentations.

4.4. Testing technology research

4.4.1. API testing

We have researched and selected the 3 most popular and recommended API testing tools to compare and consider to find the most suitable tool for our group's project.

4.4.1.1. REST Assured

REST Assured is a popular testing framework for RESTful APIs that caters to developers familiar with the Java programming language.

It simplifies API testing by providing a convenient syntax, allowing testers to create requests without the need to write code from scratch. [20]

Key Features of REST Assured [20]

- **No Need to Understand HTTP:** Users of REST Assured do not require an in-depth understanding of the HTTP protocol, as the framework abstracts the complexity and provides a user-friendly interface for API testing.
- **Java Compatibility:** REST Assured is designed for Java developers and can be easily integrated into Java projects. Users need Java version 8.0 or higher to utilize the tool effectively.
- **Integration Testing:** REST Assured supports integration testing, enabling users to test API endpoints and interactions with other components or systems in a holistic manner.

Advantages of REST Assured [20]

REST Assured offers several advantages that make it a popular choice for API testing, particularly for developers familiar with Java:

- **Simplified API Testing:** REST Assured provides a user-friendly and straightforward syntax, making it easy for developers to write API tests without the need to deal with complex HTTP details. This simplicity enhances the efficiency of API testing.
- **Integration with Java Projects:** Being a Java-based framework, REST Assured seamlessly integrates with Java projects, enabling developers to incorporate API tests directly into their existing codebase.
- **Readable Test Code:** REST Assured expressive syntax allows developers to write API test code in a clear and readable manner. This readability makes it easier to understand test scenarios, promoting better collaboration among team members.

Limitations of REST Assured [20]

While REST Assured is a powerful and popular tool for API testing, it also has some limitations that users should be aware of:

- **Java Dependency:** REST Assured is designed for Java developers, which means users without Java programming skills may find it challenging to use or understand the framework.

- **Learning Curve:** For testers who are new to REST Assured or API testing in general, there may be a learning curve, particularly if they are not already proficient in Java.
- **Limited Language Support:** REST Assured primarily focuses on Java, which means teams with developers proficient in other programming languages may not be able to fully utilize the tool's capabilities.

4.4.1.2. Google Apigee

Apigee API is a powerful and reliable tool that provides a comprehensive set of features for designing, monitoring, testing, and securing APIs.

Built with multi-step JavaScript capabilities, it offers users the flexibility to manage their API features efficiently. Apigee API allows tracking and analyzing API traffic, response behavior, and error rates, ensuring optimal performance and reliability. [20]

Key Features of Apigee API [20]

- **API Design and Management:** Apigee API facilitates the design and management of APIs, allowing users to define endpoints, request/response formats, and policies for API behavior.
- **API Monitoring:** The tool offers robust monitoring capabilities, enabling users to track API traffic, usage patterns, performance metrics, and error rates. This helps in identifying and resolving potential issues proactively.
- **API Testing:** Apigee API supports API testing, allowing users to verify the functionality and correctness of their APIs through various test scenarios.

Advantages of Apigee API [20]

Apigee API offers several advantages that make it a leading API management platform in the industry:

- **Comprehensive API Management:** Apigee API provides a complete set of tools and features for designing, monitoring, testing, and securing APIs. It offers a

unified platform for end-to-end API management, simplifying the entire API lifecycle.

- **Scalability and Performance:** Apigee API is designed to handle high traffic volumes and large-scale API deployments. Its architecture ensures scalability and maximum performance, making it suitable for enterprise-level applications.
- **Flexibility with JavaScript Policies:** The multi-step JavaScript policy framework in Apigee API allows users to implement custom logic and workflows in their APIs, providing the flexibility to meet specific business requirements.

Limitations of Apigee API [20]

While Apigee API offers many advantages, it also has some limitations that users should consider:

- **Cost:** Apigee API is a feature-rich API management platform, but it can be relatively expensive, especially for small businesses or projects with limited budgets. Organizations need to carefully evaluate the cost versus benefits for their specific use case.
- **Learning Curve:** Implementing and managing Apigee API may have a learning curve, especially for teams and developers who are new to the platform and API management concepts. Training and onboarding may be required for effective usage.
- **Overhead for Small Projects:** For small-scale projects or APIs with limited complexity, the extensive features of Apigee API may introduce unnecessary overhead, making it more suitable for larger and more complex projects.

4.4.1.3. Postman

Postman is a leading API testing tool that is compatible with Linux, MacOS, and Windows operating systems. It offers a wide range of features, allowing users to create functional, integration, and regression tests efficiently.

Users can choose between using the Command Line Interface (CLI) or a Graphical User Interface (GUI) that is easy to navigate. [20]

Key Features of Postman [20]

- **User-Friendly Interface:** Postman provides an interface that caters to users of all levels. Experienced testers can leverage its scripting capabilities, while beginners can achieve good results without much complexity using the GUI.
- **Automated Testing:** Postman supports automated testing, enabling users to monitor APIs and execute various requests for testing and debugging purposes. This automation streamlines the testing process and improves efficiency.
- **Runtime Support for Popular Patterns and Libraries:** Postman's runtime supports widely used patterns and libraries, facilitating quick testing and debugging of APIs with familiar tools.

Advantages of Postman [20]

- **User-Friendly Interface:** Postman's intuitive and user-friendly interface makes it accessible to users of all levels, including beginners and experienced testers.
- **Cross-Platform Compatibility:** Postman is compatible with Linux, MacOS, and Windows, allowing users to access and utilize the tool on different operating systems.
- **Versatile Testing Capabilities:** Postman supports various testing types, including functional, integration, and regression testing, enabling comprehensive API evaluation.

Limitations of Postman [20]

- **Dependency on Cloud Features:** Some advanced features in Postman, such as API monitoring, require a cloud subscription, which may involve additional costs for organizations.
- **Limited Offline Capabilities:** While Postman offers some offline functionality, certain features may require an internet connection, which can be restrictive in certain testing environments.
- **Learning Curve for Advanced Features:** Although Postman's basic functionalities are easy to grasp, leveraging more advanced features and scripting may have a steeper learning curve for less experienced users.

Conclusion: From all the information we just mentioned above, our group agrees to choose the third tool - Postman, due to some main reasons:

- In comparison with the cost of Google Apigee, the additional cost of Post can be affordable - not as expensive as Google Apigee.
- Our group's project is small and Google Apigee is more suitable for larger and more complex projects.
- At present, we do not have enough ability to work with Google Apigee because training and onboarding may be required for effective usage.
- We are not actually experts in Java which challenges us to take full advantage of the REST Assured tool's capabilities.

4.4.2. Performance testing

4.4.2.1. Artillery

Artillery is a modern, powerful & easy-to-use performance testing toolkit. Use it to ship scalable applications that stay performant & resilient under high load.

Artillery prioritizes developer productivity and happiness, and follows the "batteries-included" philosophy. [21]

Artillery was designed with best practices for modern production-engineering in mind. Test at cloud-scale, test any stack, and go from zero to production-grade testing fast.

Table 26. Some features of Artillery testing tool

Feature	Description
Test at cloud-scale	Distributed load testing built in. Run distributed multi-region load tests using AWS Lambda or AWS Fargate without managing any infrastructure
Test any stack	Test HTTP APIs, WebSocket and Socket.io services, or complex web apps with real browsers
Go from zero to production-grade fast	Designed to be easy to start with, with extensions and plugins for common tasks and integrations

	with monitoring, o11y and CI/CD
Extensible	Extend Artillery and build custom integrations with Node.js
Scalable and cost-efficient	Run large-scale load tests from your own AWS account with AWS Lambda or serverless AWS Fargate cluster
Permissive open source	Licensed under MPL-2.0 to make building on top of Artillery easy for platform and SQA teams

4.4.2.2. Apache Jmeter

JMeter Performance Testing is Testing method performed using Apache JMeter to test the performance of a web application. JMeter for performance testing helps to test both static and dynamic resources, helps to discover concurrent users on website and provides a variety of graphical analysis for performance testing.

Apache JMeter testing tool offers following benefit in Performance Testing [22]

- Meter can be used to test the performance of both static resources such as JavaScript and HTML, as well as dynamic resources, such as JSP, Servlets, and AJAX.
- JMeter can discover maximum number of concurrent users that your website can handle
- JMeter provides a variety of graphical analyses of performance reports.

4.4.2.3. Autocannon

Autocannon is a fast HTTP/1.1 benchmarking tool written in Node.js that can simulate high traffic to your application. It supports HTTPS and HTTP pipelining, and can be used programmatically through an API or as a command-line tool.

You can use Autocannon to measure the performance of your Node.js APIs, such as how many requests per second they can handle, how long they take to respond, and how they behave under load.

Autocannon also provides a graphical user interface (GUI) called autocannon-ui that allows you to visually analyze your endpoints and apply the optimal changes you need to obtain a reliable, fast and optimal service.

Autocannon offers various flexibilities that enable you to specify the number of connections, connection rate, requests, duration, port, maximum connection, timeout, and several other options. [23]

Conclusion: From all the information we just mentioned above, our group agrees to choose the third tool - Autocannon, due to some main reasons:

- Autocannon can simulate thousands of requests per second and access Node.js worker threads, which makes it even faster than some other tools.
- Can easily install it just by running: `npm install -g autocannon`.

Chapter 5

System implementation

5.1. Wireframe design

5.1.1. Wireframe for Customer View

5.1.1.1. UI Flow Diagram

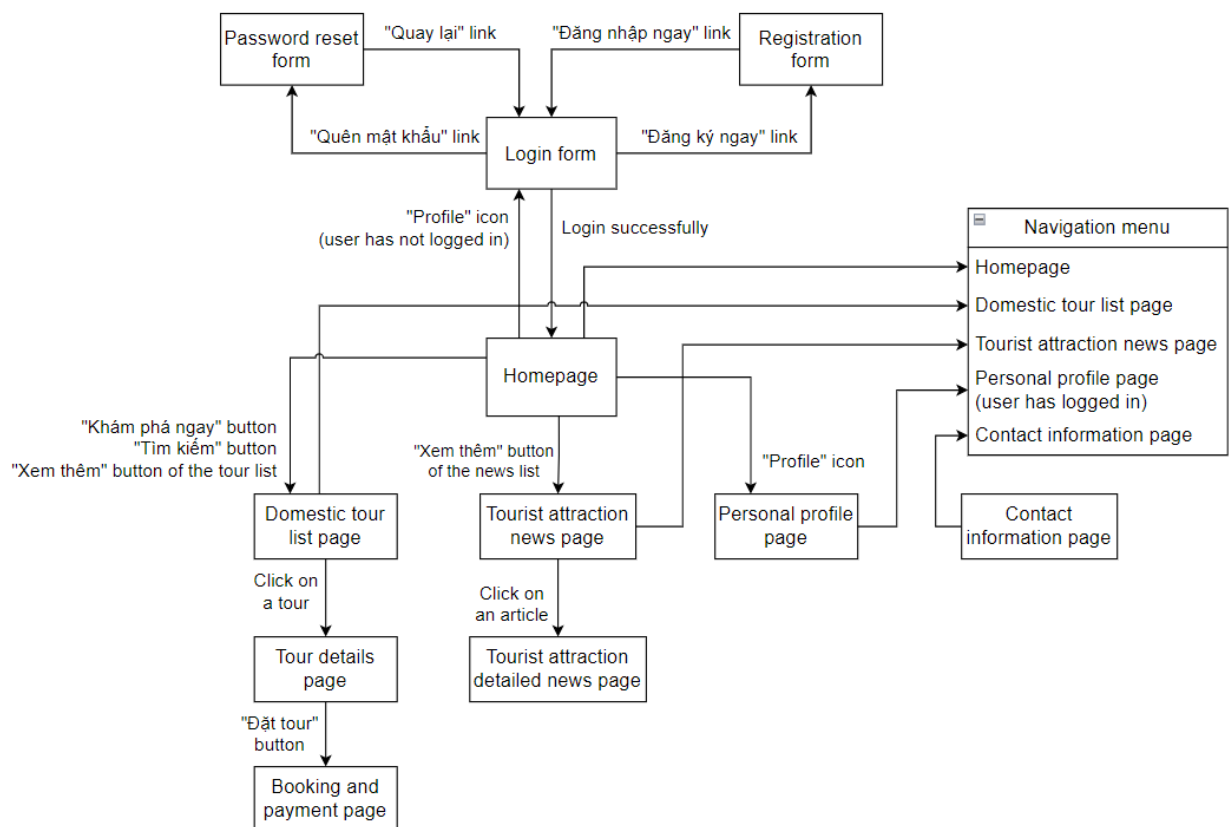


Figure 49. UI Flow Diagram for Customer View

The Figure 49 shows an overview of the guest and customer interface in our application and models the interactions that users have with our software. Users will have a navigation system - which contains links to appropriate pages in the website - to help them in traversing our website. Guests can access the homepage, domestic tour list page, tour details page, booking and payment page, login form, registration form and password reset form. In addition to the above pages, Guests who have logged in (Customers) can access the personal profile page.

5.1.1.2. Detailed UI design

A website structure consists of 3 main parts: header, main, footer:

- **Header:** This represents introductory content, is generally found at the top of a document, a section, or an article and usually contains the main heading and some navigation and search tools.
- **Main:** This is the main content that will be displayed on the website to support the messages and purposes that the website wants to bring to users. The main content will usually be: Who we are, what services we provide, what you will get when choosing us, etc.
- **Footer:** This is the bottom part of the page, typically containing navigation menus, copyright information, contact details, author information, and more.

Header

First impressions are vital in any product design, and headers are often the first thing a new user sees. Header contains important information and functionality for users, including the site's branding information and usually primary navigation links. Our header contains the elements described below:

- **Logo:** Logo position is left-aligned within the header. This accomplishes 3 goals:
 - It reinforces our brand.
 - It reminds users of where they are and what they're viewing.
 - It allows quick navigation to the homepage through a single click.
- **Navigation bar:** includes a list of links allow visitors to have easy access to all pages:
 - **News link:** This link navigates users to the tourist attractions news page.
 - **Tourism dropdown menu:** The dropdown menu includes a list of various choices that a customer can select such as: Domestic tours, Most popular tours, Tours during the month, Economic (Low-cost) tours.
 - **Contact link:** This link navigates users to the company's contact information page.
 - **Person icon:**

- When the user is not logged in, this icon will navigate them to the login page.
- Once the customer has logged in, this icon will redirect them to their personal profile page.

The header design is shown in Figure 50.



Figure 50. Design for header

Footer

A footer contains information that improves a website's overall usability. The footer design is shown in Figure 51. Our header contains the elements described below:

- Company information section: includes logo, brand name, email, branch addresses, and hotline of each branch
- Sitemaps: contains a list of all the pages on our site, like an outline, helping to organize content and create intuitive navigation. Our main pages listed are: Tours page, Tourist attractions news page and Contact page.
- Recommendations: suggest customers some convenient ways to use the website. For example, customers can log in to save their tour booking history and give feedback for their trips.
- Copyright: includes a copyright symbol, or word, the year of publication, the organization's name and a statement of rights.



Figure 51. Design for footer

Registration form

The registration form contains a list of fields that a user will input their information and submit to create an account on our website. The design for our registration form is shown in Figure 52. Our registration form has the elements which are described below:

- The username field requires the user to input their name.
- The phone number field is used to collect the user's phone number.
- The email field requires the website visitors to enter an email address.
- The password field lets users fill in their password. For the password field, password masking is used to hide passwords in the form of bullets when the user enters the password. Password masking is an attempt to protect the user against shoulder surfers who look at the screen. Moreover, the interface provides a check box (the eye) that the users can select or unselect if they wish to see or hide the password. Users can choose to see the password for their convenience if they are alone or they need to confirm if they have entered the correct password they were aiming for.
- The password confirmation field requires the user to type their password again for confirmation.
- "Register" button: User clicks this button when they want to submit the form.
- "Login now" link: This link redirects a user to the login form.
- Error messages: If the user enters incorrect input requirements, error messages will be shown under each field. As usual, the red color is used for error message text.

The image shows a registration form titled "ĐĂNG KÝ TÀI KHOẢN" (Register Account) on a blue background with the "DU LICH" logo at the top. The form is a white card with the following fields: "Tên người dùng" (Username) with input "Nhập tên người dùng", "Số điện thoại" (Phone number) with input "Nhập số điện thoại", "Email" with input "Nhập email", "Mật khẩu" (Password) with input "Nhập mật khẩu" and an eye icon, and "Xác nhận mật khẩu" (Confirm password) with input "Nhập lại mật khẩu" and an eye icon. A blue "Đăng ký" (Register) button is at the bottom, followed by a link "Bạn đã có tài khoản? Đăng nhập ngay" (Do you have an account? Log in now).

Figure 52. Design for registration form

Login form

A login form utilizes the credentials of a user, in order to authenticate their access. In the Figure 53, our login form consists of the elements described below:

- The email field requires the website visitors enter an email address.
- The password field lets users fill in their password. For the password field, password masking is also used.
- "Register now" link: User clicks "Register now" link if they want to create a new account.
- "Submit" button: User will click "Submit" button when they want to submit the form.
- "Forgot password" link: The form provides a link called "Forgot password" above the password field to help users reset their password if they forgot it.
- Error messages: If user enters an incorrect email or password, error messages will be shown under each field. As usual, the red color is used for error message text.

The image shows a login form titled "ĐĂNG NHẬP THÀNH VIÊN" (Member Login) on a blue background with the "DU LICH" logo at the top. The form is white and contains two input fields: "Email" with a placeholder "Nhập email" and "Mật khẩu" (Password) with a placeholder "Nhập mật khẩu" and an eye icon for toggling visibility. Below the password field is a link "Quên mật khẩu" (Forgot password). A blue button labeled "Đăng nhập" (Login) is positioned below the inputs. At the bottom, a text link says "Bạn chưa có tài khoản? Đăng ký ngay" (Don't have an account? Register now).

Figure 53. Design for login form

Password reset form

When the user forgot their password, they can choose the “Quên mật khẩu” option to reset the password. When they click on the link, they will be navigated to a form with an email field as in Figure 54. After they submit their email, a link is sent to the email address for password reset.

The image shows a forgot password form titled "QUÊN MẬT KHẨU" (Forgot Password) on a blue background with the "DU LICH" logo at the top. The form is white and contains a single input field for "Email" with a placeholder "Nhập email". Below the input field is a blue button labeled "Gửi" (Send). At the bottom, a text link says "Quay lại" (Go back).

Figure 54. Design for forgot password form

The figure below shows the steps after a user clicks on the link sent to their email. They will enter the new password into the form and submit to successfully change the password.

Reset your password

for freelance.rohit7@gmail.com

New password

..... I

SAVE

Password changed

You can now sign in with your new password

Figure 55. Design for password reset form

Homepage for customers and guests

This is the homepage for the customers and guests visiting our website. The homepage includes 3 parts. The first part is the top of the page including: a banner, search tool and last minute tours. The second part is the main part of the website which shows the newest tours. If the user wants to see more, they will click on the “Xem thêm” button. In the third part, we suggest customers with the most popular tours. Finally, we show some articles of attractions. If the user has not yet decided on the destination for their trip, they can research some attractions provided to find a desired place and tours related to that place are also suggested.

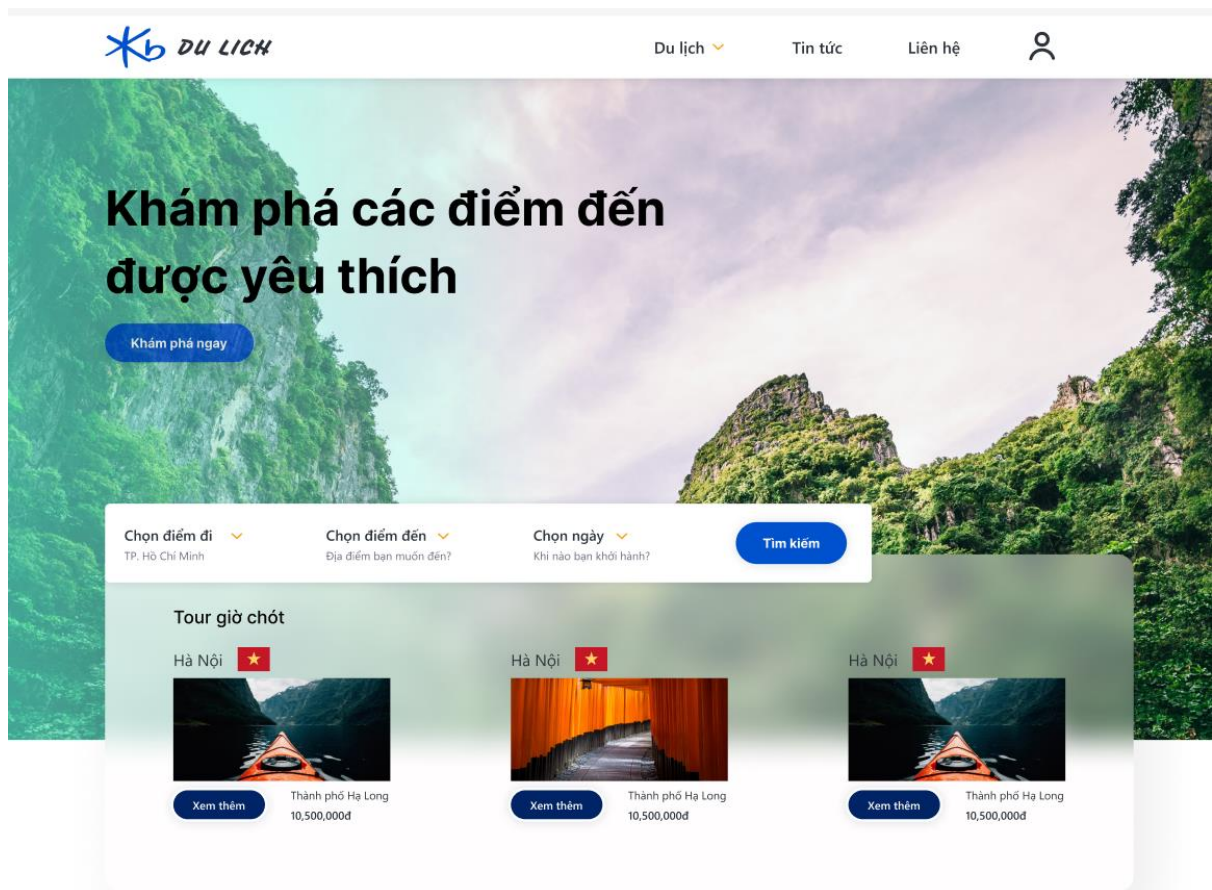


Figure 56. Design for Guest and Customer homepage (1)

Figure 56 illustrates the first part of this page - search bar and last minute tours:

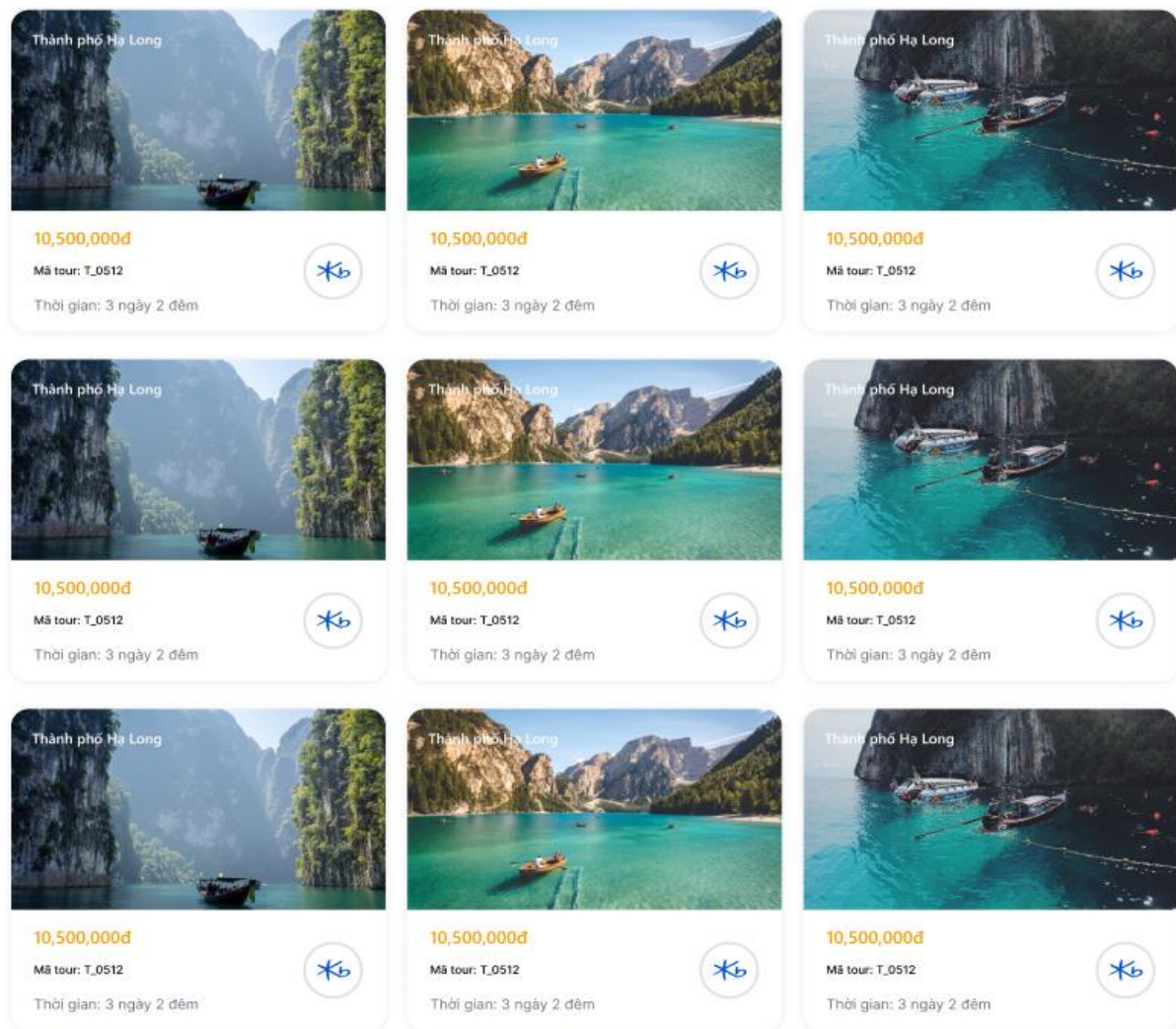
- “Khám phá ngay” button will navigate users to the “Domestic tour list page” to see all tours.
- For the search bar, users will fill in the search inputs and click “Tìm kiếm” button to go to the “Domestic tour list page” and see tours that match their search conditions.
- “Xem thêm” button on each tour will lead customers to the detailed information page of that tour.

Figure 57 illustrates the second part of this page which are the newest tours available:

- Users can click on the tour to see its detailed information.
- If users want to see more tours, they can click on the “Xem thêm” button.

Tour mới nhất

Khám phá ngay các tour du lịch vô cùng hấp dẫn!



Xem thêm

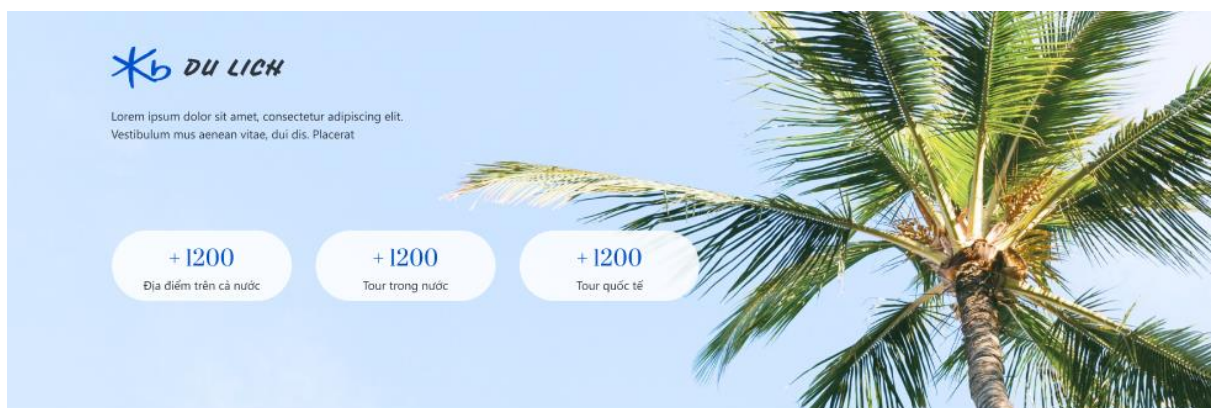
Figure 57. Design for Guest and Customer homepage (2)

Figure 57 illustrates the third part of this page - favorite destinations:

- Users can click on the tour to see its detailed information.
- If users want to see more tours, they can click on the “Xem thêm” button.

Figure 58 illustrates the final part of this page - the news section:

- Users can click on the article to see its detailed information.
- If users want to see more articles, they can click on the “Xem thêm” button.



Điểm đến được yêu thích nhất

Sự lựa chọn hoàn hảo và được yêu thích nhất dựa trên đánh giá từ du khách.

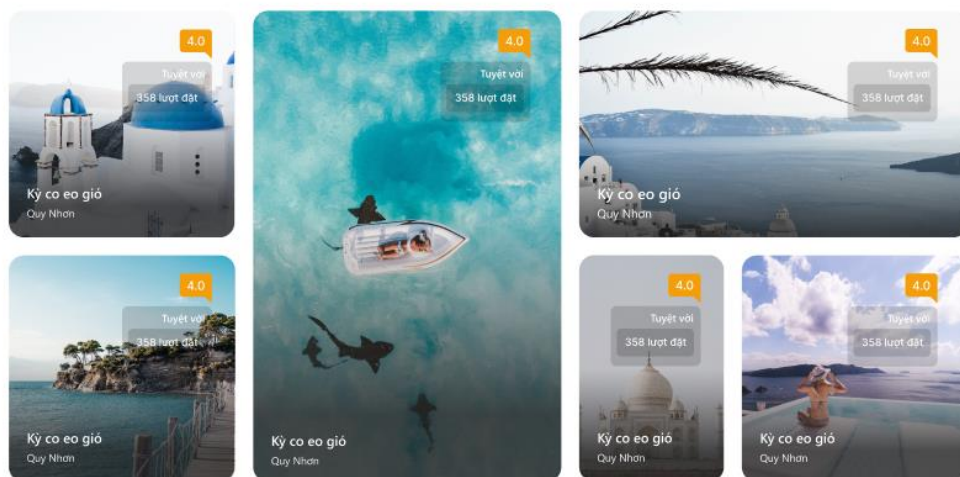


Figure 58. Design for Guest and Customer homepage (3)

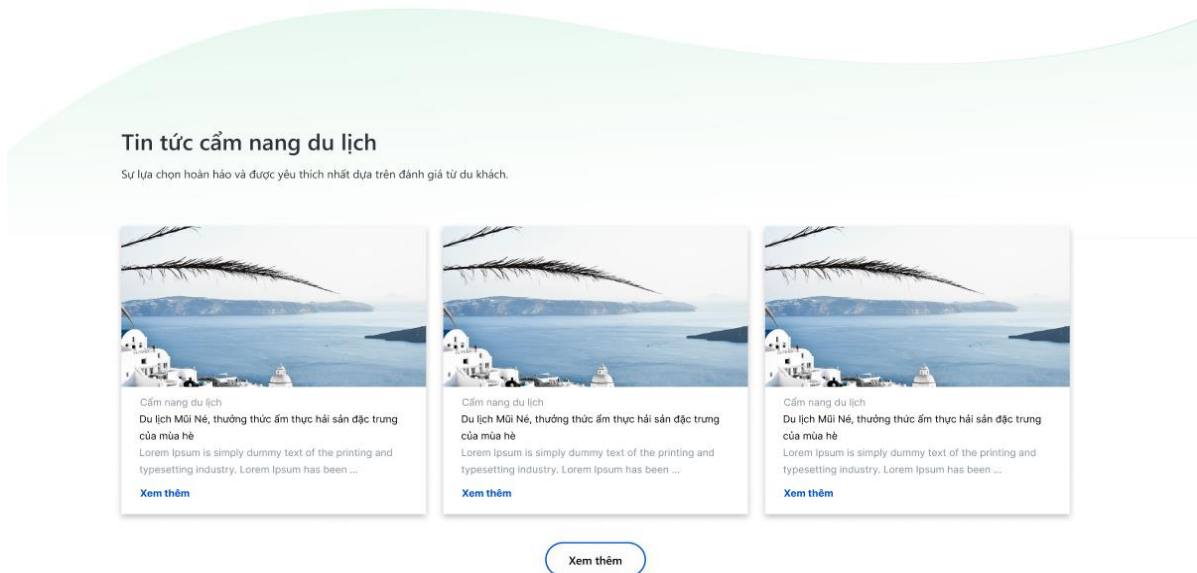


Figure 59. Design for Guest and Customer homepage(4)

Domestic tour list page

You can see the design for this page in Figure 60. This page displays a list of available domestic tours. Users can find the tour easier by using the filter bar on the left hand side. In this filter bar, users can choose: a desired departure, one or many destinations, number of days, departure date, number of guests, tour type, price range and transportation. User then clicks on the “Tìm kiếm” button to filter the tours. They can also choose to sort by price or date.

Du lịch > Du lịch trong nước

Lọc kết quả

Điểm đi:
TP. Hồ Chí Minh

Điểm đến:
--- Chọn điểm đến ---

Số ngày:
1-3 ngày 4-7 ngày
8-14 ngày Trên 14 ngày

Ngày đi:
18/08/2022

Số lượng người:
1 người 2 người
3-5 người 5+ người

Dòng Tour:
Dưới 5 người 5 - 7 người
8 - 16 người Trên 16 người

Bộ lọc tìm kiếm
TỪ 600.000đ - 50.000.000đ Filter

Thông tin vận chuyển
Máy bay Ô tô
Xe khách

Tìm kiếm

Du lịch trong nước

Du lịch trong nước luôn là lựa chọn tuyệt vời. Đường bờ biển dài hơn 3260km, những khu bảo tồn thiên nhiên tuyệt vời, những thành phố nhộn nhịp, những di tích lịch sử hào hùng, nền văn hóa độc đáo và hấp dẫn, cùng một danh sách dài những món ăn ngon nhất thế giới, Việt Nam có tất cả những điều đó. Với lịch trình dày, khởi hành đúng thời gian cam kết, Vietravel là công ty lữ hành uy tín nhất hiện nay tại Việt Nam, luôn sẵn sàng phục vụ du khách mọi lúc, mọi nơi, đảm bảo tính chuyên nghiệp và chất lượng dịch vụ tốt nhất thị trường.

Sắp xếp theo: Tất cả

Thành phố Hạ Long



10,500,000đ

Mã tour: T_0512

Thời gian: 3 ngày 2 đêm

Thành phố Hạ Long



10,500,000đ

Mã tour: T_0512

Thời gian: 3 ngày 2 đêm

Thành phố Hạ Long



10,500,000đ

Mã tour: T_0512

Thời gian: 3 ngày 2 đêm

Thành phố Hạ Long



10,500,000đ

Mã tour: T_0512

Thời gian: 3 ngày 2 đêm

Thành phố Hạ Long



10,500,000đ

Mã tour: T_0512

Thời gian: 3 ngày 2 đêm

Thành phố Hạ Long



10,500,000đ

Mã tour: T_0512

Thời gian: 3 ngày 2 đêm

Figure 60. Design for domestic tour list page

Tour details page

This is the page showing details of a tour. All information for the tour is displayed. User can click on the “Đặt tour” button to book the tour.

Du lịch > Du lịch trong nước > Đà Nẵng - Bà Nà - Cầu Vàng - Sơn Trà - Biển Mỹ Khê - Hội An - Đà Nẵng

Đà Nẵng - Bà Nà - Cầu Vàng - Sơn Trà - Biển Mỹ Khê - Hội An - Đà Nẵng



Điểm nhấn hành trình

Hành trình	Đà Nẵng - Bà Nà - Cầu Vàng - Sơn Trà - Biển Mỹ Khê - Hội An - Đà Nẵng (Khách sạn 4* trọn tour)
Lịch trình	3 ngày 2 đêm
Khởi hành	18/08 - 20/08/2022
Vận chuyển	Xe du lịch đời mới, máy bay
Số chỗ	Còn 5 chỗ

Được mệnh danh là “thành phố đáng đến” với dòng sông Hàn thơ mộng với cây cầu Rồng biểu tượng của Thành phố biển du lịch Đà Nẵng - nơi mà quý khách có thể cảm nhận được sự pha trộn giữa khí hậu miền Bắc, miền Nam. Ngoài ra Đà Nẵng còn sở hữu nhiều danh lam thắng cảnh làm say lòng người như: Bà Nà Hills, Bán Đảo Sơn Trà, Ngũ Hành Sơn, Đà Nẵng, phố cổ Hội An.... Tour du lịch Đà Nẵng sẽ đưa quý khách khám phá bãi biển được Forbes lựa chọn là bãi biển quyến rũ nhất hành tinh với bờ biển dài, làn nước trong xanh, không khí mát mẻ.... Tham khảo kinh nghiệm du lịch Đà Nẵng và Đặt ngay tour Đà Nẵng của Du Lịch Việt để khám phá Đà Nẵng có gì mà lại luôn là điểm đến hấp dẫn như vậy.

Đà Nẵng - Bà Nà - Cầu Vàng - Sơn Trà - Biển Mỹ Khê - Hội An - Đà Nẵng

Mã tour: T_0512

Thời gian: 3 ngày 2 đêm

Khởi hành: 18/08 - 20/08/2023

Vận chuyển: Máy bay

Điểm xuất phát: TP. Hồ Chí Minh

8,500,000đ

ĐẶT TOUR

Điểm nhấn hành trình

Lịch trình

Dịch vụ bao gồm và không bao gồm

Ghi chú

Ngày khởi hành khác

Figure 61. Design for tour details page (1)

Lịch trình

Ngày 1: TP. Hồ Chí Minh - Đà Nẵng - Khu du lịch Bà Nà (Ăn chiều)

- Quý khách tập trung tại điểm hẹn, Ga đi trong nước, sân bay Tân Sơn Nhất. Hướng dẫn viên Vietravel hỗ trợ làm thủ tục cho đoàn đáp chuyến bay đi Đà Nẵng. Tại sân bay Đà Nẵng xe và HDV Vietravel đón đoàn đi tham quan:
 - Bán đảo Sơn Trà và viếng Chùa Linh Ứng: Nơi đây có tượng Phật Quan Thế Âm cao nhất Việt Nam...
 - Ngũ Hành Sơn: Động Tàng Chơn, Động Hoa Nghiêm, Chùa Non Nước,...
 - Làng Đá Non Nước Nguyễn Hùng: mua sắm sản phẩm đá mỹ nghệ...

Ngày 2: Đà Nẵng - Bán đảo Sơn Trà (Ăn sáng, chiều, tối tự túc)

- Dùng bữa sáng tại khách sạn. Xe đưa Quý khách đi tham quan:
 - Khu du lịch Bà Nà (chỉ phí cáp treo & Ăn trưa tự túc): tận hưởng không khí se lạnh của Đà Lạt tại miền Trung...
 - Bãi biển Mỹ Khê: Một trong những bãi biển quyến rũ nhất hành tinh. Quý khách tự do dạo biển, chụp hình....
 - Buổi tối Quý khách tự túc đi dạo phố, thưởng thức đặc sản địa phương...Nghỉ đêm tại Đà Nẵng.

Ngày 3: Đà Nẵng - TP. Hồ Chí Minh (Ăn sáng)

- Dùng bữa sáng tại khách sạn. Xe tiễn Quý khách ra sân bay Đà Nẵng đón chuyến bay trở về Tp.Hồ Chí Minh.
- Chia tay Quý khách và kết thúc chương trình du lịch tại sân bay Tân Sơn Nhất

Ghi chú

- Hành trình có thể thay đổi thứ tự điểm đến tùy vào điều kiện thực tế.
- Khách Sạn có thể ở xa trung tâm thành phố vào các mùa cao điểm.

Dịch vụ

- ✓ Bảo hiểm
- ✓ Hướng dẫn viên
- ✓ Xe đưa đón
- ✓ Bữa ăn

Ngày khởi hành khác

STT	Ngày khởi hành	Phương tiện	Giá	Số chỗ	Book tour	
1	20/08/2022	Xe khách	6,500,000đ	Còn 10 chỗ	Book	Chi tiết
2	21/08/2022	Máy bay	8,500,000đ	Còn 10 chỗ	Book	Chi tiết

Figure 62. Design for tour details page (2)

The average ratings and reviews of the previous customers are also displayed in the comment section. The comment section is displayed in the Figure 63.

Bình luận

4.0 ★★★★★☆ (18 đánh giá)

Quỳnh Trân ★★★★★

18:20, 18/09/2019

Khởi hành: 05/12/2022

Phương tiện: Máy bay

Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s,

Quỳnh Trân ★★★★★

18:20, 18/09/2019

Khởi hành: 05/12/2022

Phương tiện: Máy bay

Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s,

Quỳnh Trân ★★★★★

18:20, 18/09/2019

Khởi hành: 12/08/2022

Phương tiện: Xe khách

Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s,

Quỳnh Trân ★★★★★

18:20, 18/09/2019

Khởi hành: 13/02/2022

Phương tiện: Máy bay

Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s,

Quỳnh Trân ★★★★★

18:20, 18/09/2019

Khởi hành: 13/02/2022

Phương tiện: Máy bay

Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s,



Figure 63. Design for average ratings and reviews section

At the end of the page, as in Figure 64, we suggest customers with some related tours so that they can have a variety of choices for their trips.

Tour liên quan

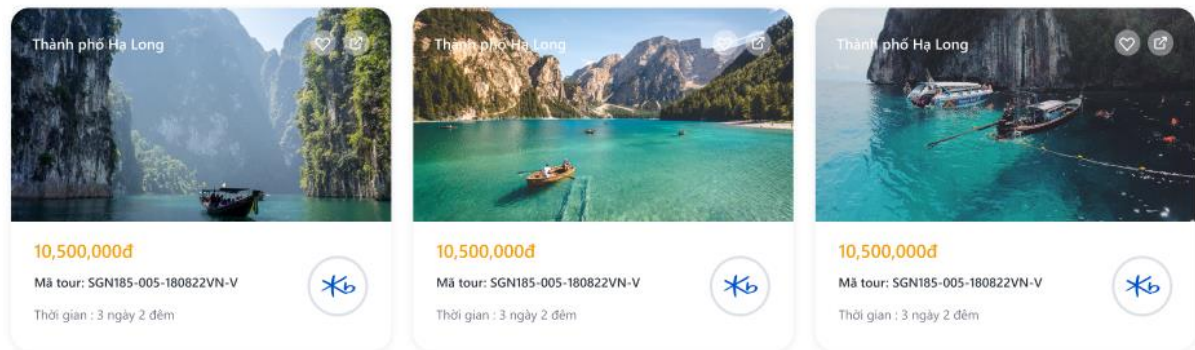


Figure 64. Design for related tour suggestion section

Booking and payment page

This is the page for booking the tour. The information of the tour is displayed on the right hand side of the page. User needs to go through 3 steps to successfully book the tour.

The first step is shown in Figure 65. In step 1, they have to fill in their contact details including: full name, email address, phone number and notes (if any). After that, they click “Tiếp tục” button to continue.

The second step is shown in Figure 66. In step 2, users choose a payment method and click “Xác nhận”. When the user confirms the payment method, the system will automatically make payment.

1

Điền thông tin

2

Thanh toán

3

Xem lại hóa đơn

Điền thông tin liên hệ

Họ và tên *

Email *

Số điện thoại *

+

Thêm ghi chú

TIẾP TỤC

Thông tin đặt tour



Đà Nẵng - Bà Nà - Cầu Vàng - Sơn Trà - Biển Mỹ Khê - Hội An - Đà Nẵng

Mã tour

T_0512

Ngày khởi hành

31/08/2023

Số khách

1 khách

Giá 1 khách

8,500,000 VND

Tổng tiền

8,500,000 VND

Figure 65. Design for booking step 1 - Filling in booking information

> Du lịch > Du lịch trong nước > Đà Nẵng - Bà Nà - Cầu Vàng - Sơn Trà - Biển Mỹ Khê - Hội An - Đà Nẵng > Thanh toán

✓

Điền thông tin

2

Thanh toán

3

Xem lại hóa đơn

Lựa chọn ngân hàng bạn sẽ thanh toán

☒ Ngân hàng Vietcombank



☐ Ngân hàng Techcombank



☐ Ngân hàng Mbbank



XÁC NHẬN

Thông tin đặt tour



Bà Nà - Cầu Vàng - Sơn Trà - Biển Mỹ Khê - Hội An

Mã tour

T_0512

Ngày khởi hành

31/08/2023

Số khách

1 khách

Giá 1 khách

8,500,000 VND

Tổng tiền

8,500,000 VND

Figure 66. Design for booking step 2 - Make payment

In the final step, the system shows a bill to the user to review their booking. If user has logged in, their transaction will be saved and displayed in their personal profile. However, for users that do not log in the website, the system will send an email of the transaction information to the email address they gave.

> Du lịch > Du lịch trong nước > Đà Nẵng - Bà Nà - Cầu Vàng - Sơn Trà - Biển Mỹ Khê - Hội An - Đà Nẵng > Thanh toán



KB DULICH xin cảm ơn Quý Khách !



Đặt tour thành công

Chi tiết đơn hàng



Bà Nà - Cầu Vàng - Sơn Trà - Biển Mỹ Khê - Hội An

Mã đơn hàng	1523
Ngày khởi hành	31/08/2023
Số khách	1 khách
Tổng tiền	22,690,000 VND

Thông tin liên hệ

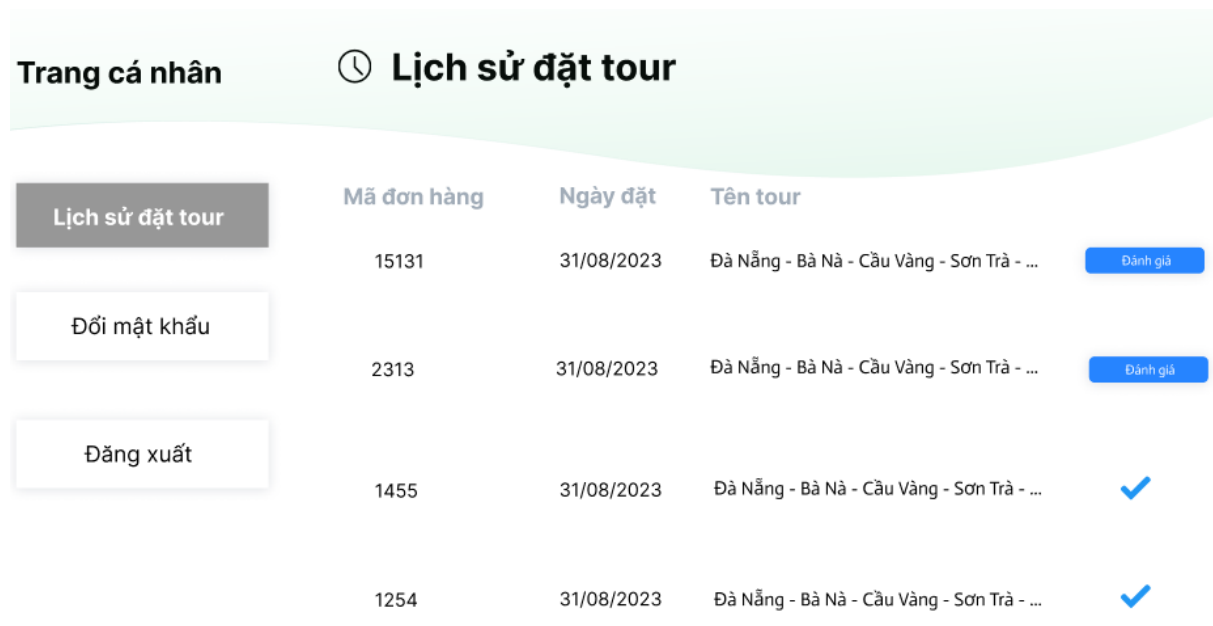
Họ và tên	Phạm Quỳnh Trân
Email	abc@gmail.com
Số điện thoại	029644846151
Ghi chú	Phòng tầng cao

VỀ TRANG CHỦ

Figure 67. Design for booking step 3 - Review booking

Personal profile page

The personal profile page is for users who have logged in our system. They can view all of the tours they have booked here. They can also give feedback for the tours that have finished.



Mã đơn hàng	Ngày đặt	Tên tour	Đánh giá
15131	31/08/2023	Đà Nẵng - Bà Nà - Cầu Vàng - Sơn Trà - ...	Đánh giá
2313	31/08/2023	Đà Nẵng - Bà Nà - Cầu Vàng - Sơn Trà - ...	Đánh giá
1455	31/08/2023	Đà Nẵng - Bà Nà - Cầu Vàng - Sơn Trà - ...	✓
1254	31/08/2023	Đà Nẵng - Bà Nà - Cầu Vàng - Sơn Trà - ...	✓

Figure 68. Design for booking history section

The form in Figure 69 is the feedback form. Customers can give feedback by rating from 1 to 5 stars and giving a review or comment for the tour. Customer then clicks “Gửi bình luận” to submit the tour feedback.



Đà Nẵng - Bà Nà - Cầu Vàng - Sơn Trà - Biển Mỹ Khê - Hội An - Đà Nẵng

Đánh giá tour

★ ★ ★ ★ ★
Rất tệ Tệ Thường Tốt **Rất tốt**

Bình luận về tour

Bình luận của bạn

☐ Ẩn tên trên bình luận

[Gửi bình luận](#)

Figure 69. Design for feedback form

Users can also change their password by entering the current password, the new password and re-typing the new password. Customer then clicks “Lưu thay đổi” and the system will check the password if it satisfies the requirements before the password is successfully changed.

The image shows a web interface for changing a password. On the left is a sidebar with three buttons: "Lịch sử đặt tour", "Đổi mật khẩu" (highlighted in dark grey), and "Đăng xuất". The main content area has a light green header with the title "Đổi mật khẩu" and a key icon. Below the header, there are three input fields with labels: "Mật khẩu hiện tại" (with sub-label "Nhập mật khẩu hiện tại"), "Mật khẩu mới" (with sub-label "Nhập mật khẩu mới"), and "Nhập lại mật khẩu mới" (with sub-label "Nhập lại mật khẩu mới"). At the bottom right of the form is a blue button labeled "Lưu thay đổi".

Figure 70. Design for password change section

Contact information page

Contact information page is shown in Figure 71. This is the page that contains contact information of the company. If users want to contact directly, they can see the company addresses, their locations on a map and the working time. If users want to contact online, they can contact through the email addresses or phone numbers.

Trang chủ > Liên hệ

Thông tin liên hệ



09:00 - 21:00

KB Dulich - Công ty TNHH Giao thông vận tải và du lịch Sài Gòn

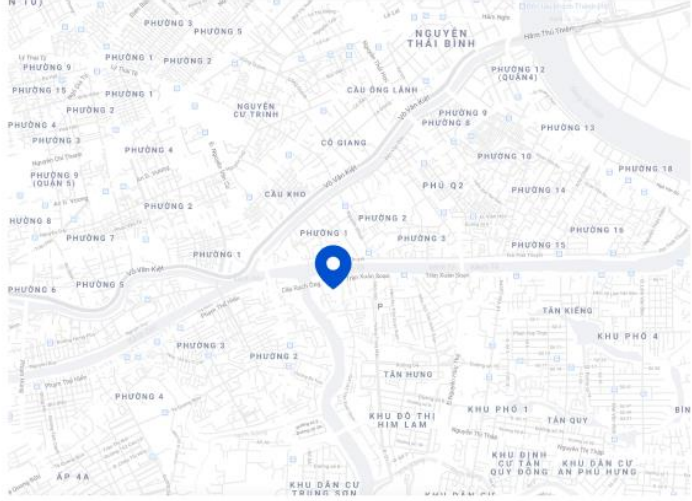
Chi nhánh Quận Thanh Xuân - Hà Nội



09:00 - 21:00

KB Dulich - Công ty TNHH Giao thông vận tải và du lịch Sài Gòn

Chi nhánh Quận Tân Phú - TP.HCM



ADDRESS

TP. Hồ Chí Minh: Số 219 Võ Văn Tần, Phường 5, Quận 3

Hà Nội: Tầng 6, tòa nhà Zentower, số 12 Khuất Duy Tiến, P. Thanh Xuân Trung, Q. Thanh Xuân

PHONE

TP. Hồ Chí Minh: 1900 5862

Hà Nội: 1900 0871

EMAIL

Support@gmail.com

Figure 71. Design for contact information page

131

Tourist attraction news page

Tourist attraction news page is shown in Figure 72. This page has articles about tourist attractions information. Users can access this page and search for their desired place by entering its name into the search box. Users will click on the article that they want to view in detail.

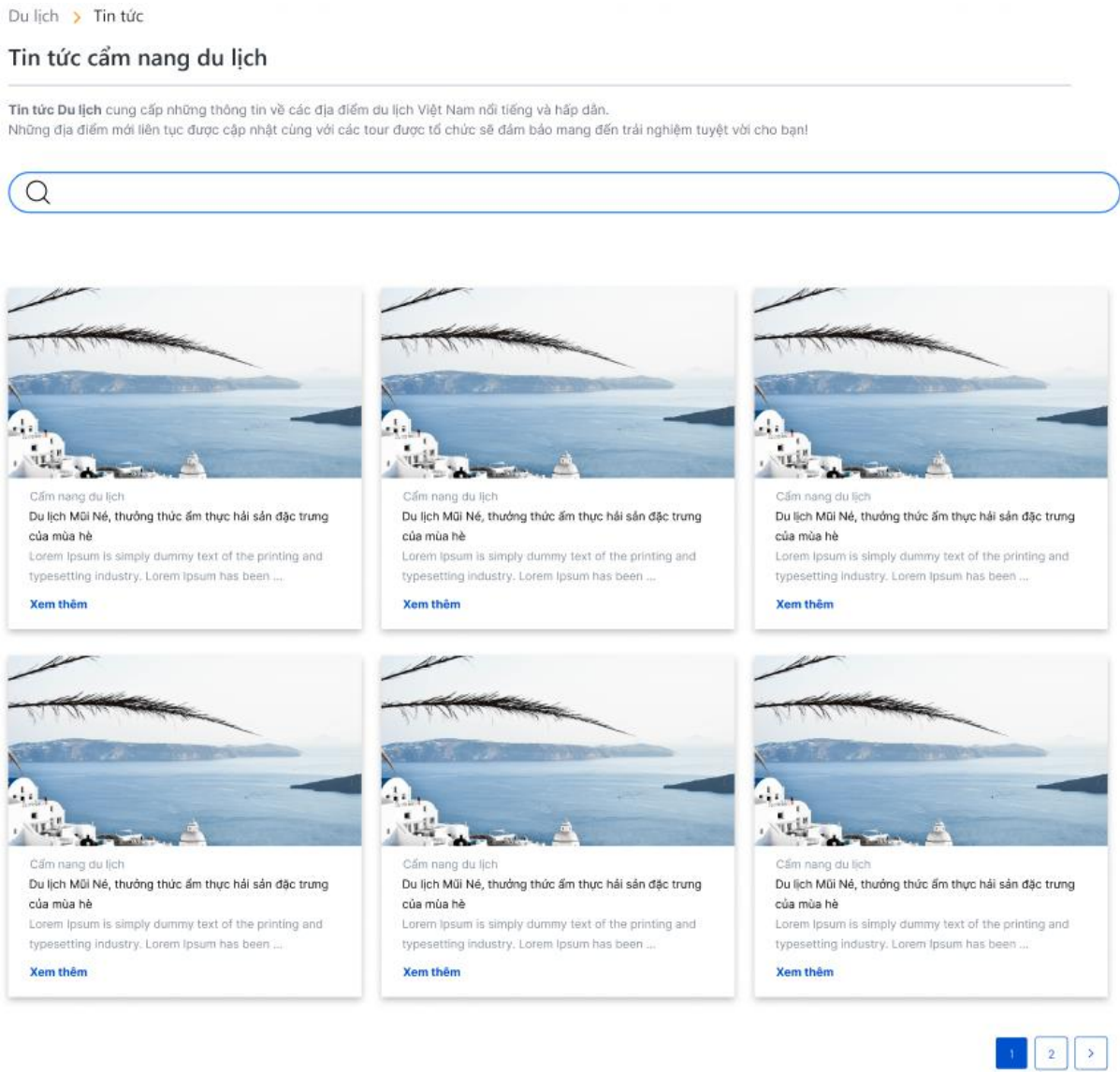
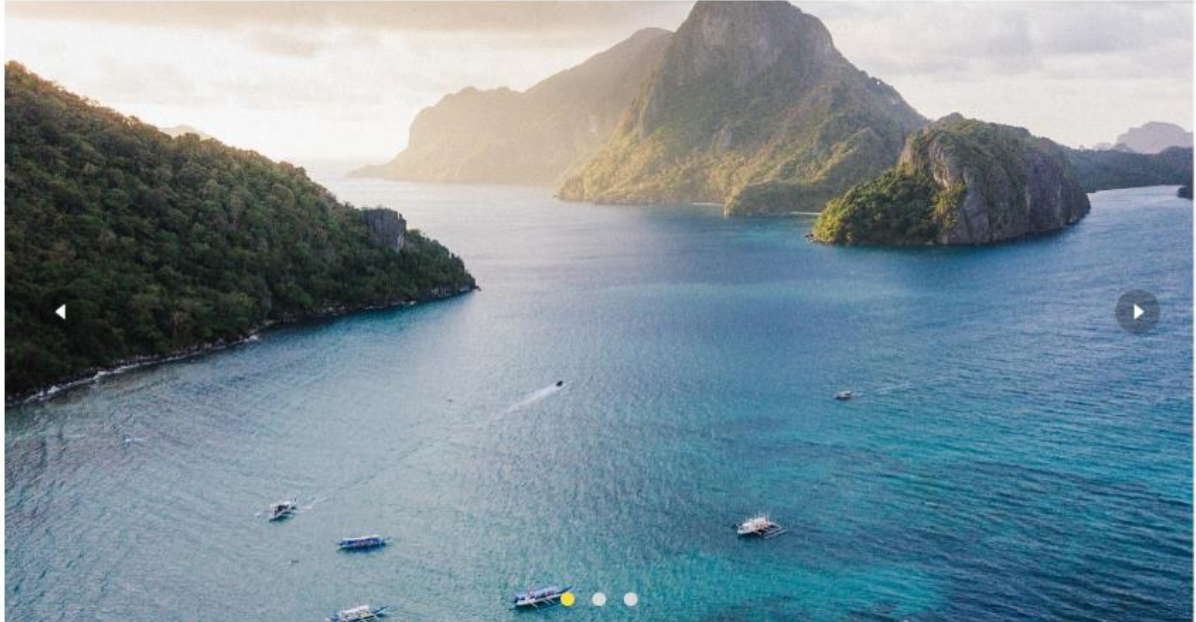


Figure 72. Design for tourist attraction news page

Tourist attraction detailed news page

This is the detailed information of the destination. Some images, descriptions and entertainment activities will be shown to users as in Figure 73.

Du lịch Mũi Né, thưởng thức ẩm thực hải sản đặc trưng của mùa hè



Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aliquam placerat, augue a volutpat hendrerit, sapien tortor faucibus augue, a maximus elit ex vitae libero. Sed quis mauris eget arcu facilisis consequat sed eu felis. Nunc sed porta augue. augue a volutpat hendrerit, sapien tortor faucibus bugd augue. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aliquam placerat, augue a volutpat hendrerit, sapien tortor faucibus augue, a maximus elit ex vitae libero. Sed quis mauris eget arcu facilisis consequat sed eu felis. Nunc sed porta augue. augue a volutpat hendrerit, sapien tortor faucibus bugd augue.

Activities

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aliquam placerat, augue a volutpat hendrerit, sapien tortor faucibus augue, a maximus elit ex vitae libero. Sed quis mauris eget arcu facilisis consequat sed eu felis. Nunc sed porta augue. augue a volutpat hendrerit, sapien tortor faucibus bugd augue.

- Lorem ipsum dolor sit amet, consectetur adipiscing elit.
- Aliquam placerat, augue a volutpat hendrerit, sapien tortor faucibus augue, a maximus elit ex vitae libero.
- Sed quis mauris eget arcu facilisis consequat sed eu felis. Nunc sed porta augue.
- augue a volutpat hendrerit, sapien tortor faucibus bugd augue.

Figure 73. Design for detailed article section

Some tours related to this place will be recommended for customers. The design is the same as the Figure 64 in the section “Tour details page”.

5.1.2. Wireframe for Staff View

5.1.2.1. UI Flow Diagram

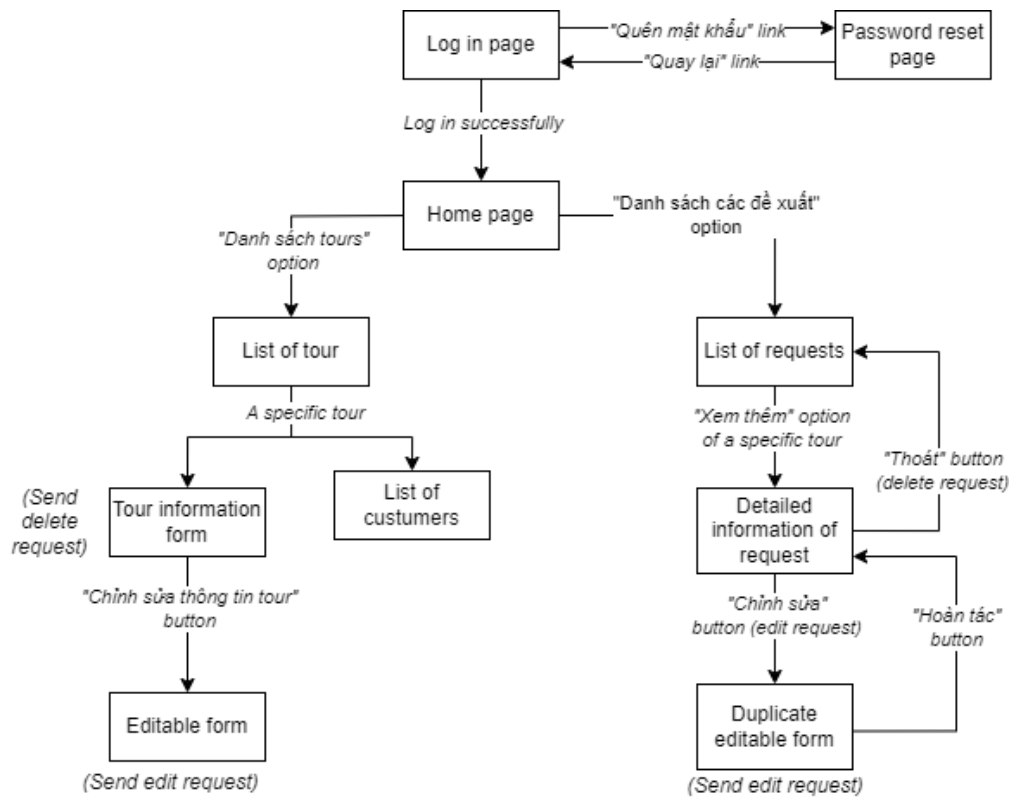


Figure 74. UI Flow Diagram for Staff View

The Figure 74 above illustrates how the staff can interact with the system. When the staff login successfully, they are brought to the main page view. Each of them will have three main options - “Danh sách tours”, and “Danh sách các đề xuất”. For each different option, there will be different operations with the system. To understand more specifically how those operations are performed, we continue to analyze in the *Detailed UI design* part below.

5.1.2.2. Detailed UI design

There are two main parts for Staff UI: Header and Main.

Header



Figure 75. Design for Header in staff view

In Figure 75, from the left-hand side to the right-hand side, we have:

- Home icon: We use this icon to show that a staff is at the main page.
- Bell icon: We use this icon to show received notifications.
- Account icon: The staff can click on this icon and choose to log out of the system.

Main: All information as well as buttons are shown in this part corresponding to each option. The staff can easily view and operate.

- UI designs for Staff Authentication

Login form

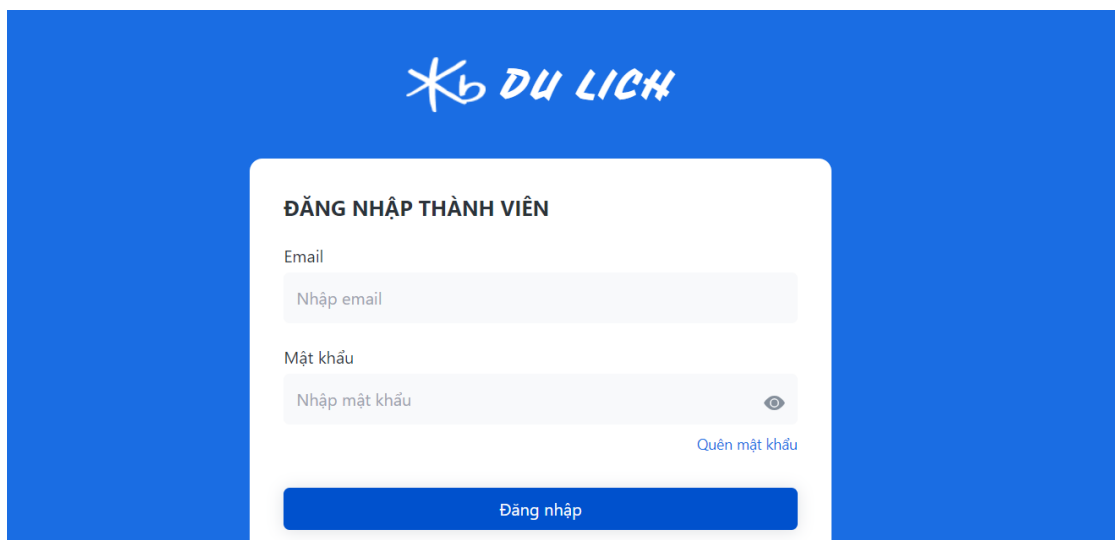
The image shows a login form titled "ĐĂNG NHẬP THÀNH VIÊN" (Member Login) centered on a blue background. At the top of the form is a logo with a stylized star and the text "DU LỊCH". Below the title, there are two input fields: "Email" with a placeholder "Nhập email" and "Mật khẩu" (Password) with a placeholder "Nhập mật khẩu" and a toggle icon. To the right of the password field is a link "Quên mật khẩu" (Forgot password). At the bottom of the form is a blue button labeled "Đăng nhập" (Login).

Figure 76. Design for staff and manager login form

The Figure 76 illustrates the login interface when staff or manager access the web page. When a staff accesses the web page, the system will show the Login form. A staff member is required to login successfully in order to use the system. If a staff forgets his/her password, he/she can click on “Quên mật khẩu” option to reset the password.

Reset password form

When a staff clicks on the “Quên mật khẩu” option, the system will show a reset password form. The reset password form for staff is similar to the form for customers shown in the Figure 54. A staff can fill his/her email address in the blank cell named “Email”. After finishing and clicking on the “Gửi” button. He/She will receive a link to

reset his/her password via email address that he/she just sent before. If a staff recalls his/her password and does not want to change the password anymore, at the reset password form, he/she can click on the “Quay lại” option to go back.

Change password form for the first login



Figure 77. Change password form for the first login

The Figure 74 illustrates the interface when a staff logs in the system for the first time. The system will show a form and require the staff to input his/her new password. After finishing, the staff clicks on the “Xác nhận” button to update the new password.

- Main page designs for Staff (Login successfully):

Reset password page

The Figure 78 illustrates the interface when a staff chooses the “Đổi mật khẩu” option of the account icon on the header bar. The system will show a password change form. A staff can change his/her password by entering the current password, the new password and re-typing the new password. A staff then clicks on the “Lưu thay đổi” button and the system will check the password if it satisfies the requirements before the password is successfully changed.

Đổi mật khẩu

Mật khẩu hiện tại

Nhập mật khẩu hiện tại

Mật khẩu mới

Nhập mật khẩu mới

Nhập lại mật khẩu mới

Nhập lại mật khẩu mới

Lưu thay đổi

Figure 78. Design for password change

Tour management pages

Trang chủ

Danh sách tours

Danh sách các đề xuất

Danh sách tours

Danh sách tours

Tìm kiếm

Mã tour	Tên tour	SL khách đặt tour	Ngày khởi hành
T1	Du lịch Bà Nà Hills - Cầu Rồng - Bán Đảo ...	30	01/01/2024
T2	Tour Bán Đảo Sơn Trà - Cầu Quay Sông H..	50	09/09/2024

Tổng số: 2

+ Thêm tour mới

Figure 79. Design for tour list management page

The Figure 79 illustrates the interface when clicking on the “Danh sách tours” option, the system will show a table of created tours. A staff can click on a specific tour to see more details. A staff can find any tours he/she wants by using a search bar. Furthermore, in this option, a staff can click on the “Thêm tour mới” button to create a new tour and send the request to the manager.

Trang chủ

Danh sách tours

Danh sách các đề xuất

Phiếu thông tin tour

Tên chương trình:

Gía chương trình:

VND

Hành trình:

Xuất phát:

Điểm đến:

Thêm điểm dừng

Phương tiện:

☐ Xe 4 chỗ

☐ Xe 7 chỗ

☐ Xe khách

☐ Máy bay

Lịch trình:

Số ngày:

Ngày khởi hành:

Mô tả chi tiết:

Thêm mô tả chi tiết

Ghi chú:

Dịch vụ bao gồm:

☐ Bảo hiểm

☐ Bữa ăn

☐ Hướng dẫn viên

☐ Xe đưa đón

☐ Vé tham quan

Nhân viên đảm nhận:

Nguyễn Văn A

Have a nice trip ^^

Hoàn tất và gửi đề xuất

Hủy bỏ

Figure 80. Design for adding a new tour function

The Figure 80 illustrates the interface when clicking on the “Thêm tour mới” button, the system will show a new form for a staff to fill in. After finishing, a staff can click on “Hoàn tất và gửi đề xuất” to send the request to the manager. If a staff does not want to add a new tour anymore, he/she can click on the “Hủy bỏ” button to cancel the new form.

Trang chủ

Danh sách tours

Danh sách các đề xuất

Phiếu thông tin tour

Tên chương trình: Du lịch Bà Nà Hills - Cầu Rồng - Bán Đảo Sơn Trà

Giá chương trình: 5.000.000 VND

Hành trình:

Xuất phát: Thành phố Hồ Chí Minh

Điểm đến 1: Bà Nà Hills Điểm đến 2: Cầu Rồng Điểm đến 3: Bán Đảo Sơn Trà

Phương tiện:

☒ Xe 4 chỗ

☐ Xe 7 chỗ

☐ Xe khách

☐ Máy bay

Lịch trình:

Số ngày: 5 Ngày khởi hành: 1/1/2024

Mô tả chi tiết:

Ghi chú: Không có

Dịch vụ bao gồm:

☒ Bảo hiểm

☒ Hướng dẫn viên

☒ Xe đưa đón

☒ Vé tham quan

☒ Bữa ăn

Nhân viên đảm nhận: Nguyễn Văn A

Have a nice trip ^^

Chỉnh sửa thông tin tour

Đề xuất xóa tour

Danh sách khách hàng

Tìm kiếm

Họ	Tên	SĐT	Email	SL	Ghi chú	Đánh giá	Bình luận
Nguyễn	Ngọc Anh	0123456789	ngocanhnguyen@gmail.com	03	Ghế hàng đầu	★★★★☆	Khá hài lòng
Ngô	Kiến Văn	0193696780	kvngo@gmail.com	01		★★★★★	Dịch vụ tốt

Tổng số: 2

Thêm khách hàng

Figure 81. Design for view of tour details

The Figure 81 illustrates the interface when a staff chooses a specific tour, the system will show a detailed information form of that tour along with a table of customers' information. Similar to the table of tours, a staff can also search for any specific customer by using a search bar. A staff can operate with this interface by clicking on the “Chỉnh sửa thông tin tour”, “Đề xuất xóa tour” or “Thêm khách hàng” button.

139

Trang chủ

Danh sách tours

Danh sách các đề xuất

Phiếu thông tin khách hàng

Họ:

Tên:

SL:

SDT:

Email:

Ghi chú (nếu có):

Xác nhận thêm

Hủy bỏ

Figure 82. Design for add-new-customer form

The Figure 78 illustrates the interface when clicking on the “Thêm khách hàng” button. The system will show a form with blank cells, the staff will fill in the customer’s information. After finishing, the staff can click on the “Xác nhận thêm” button to add the customer to the list. If the staff does not want to add anymore, he/she can click on the “Hủy bỏ” button to cancel and back to the tour details page.

Trang chủ

Danh sách tours

Danh sách các đề xuất

Phiếu thông tin tour

Tên chương trình: Du lịch Bà Nà Hills - Cầu Rồng - Bán Đảo Sơn Trà

Giá chương trình: 5.000.000 VND

Hành trình:

Xuất phát: Thành phố Hồ Chí Minh

Điểm đến 1: Bà Nà Hills Điểm đến 2: Cầu Rồng Điểm đến 3: Bán Đảo Sơn Trà

Phương tiện:

☒ Xe 4 chỗ

☐ Xe 7 chỗ

☐ Xe khách

☐ Máy bay

Lịch trình:

Số ngày: 5 Ngày khởi hành: 1/1/2024

Mô tả chi tiết:

Ghi chú: Không có

Dịch vụ bao gồm:

☒ Bảo hiểm

☒ Hướng dẫn viên

☒ Xe đưa đón

☒ Vé tham quan

☒ Bữa ăn

Nhân viên đảm nhận: Nguyễn Văn A

Have a nice trip ^^

*Lý do đề xuất hủy

Hoàn tất và gửi đề xuất

Hủy bỏ

Figure 83. Design for tour cancellation request view

The Figure 79 illustrates the interface when clicking on the “Đề xuất xóa tour” button, the system will show a compulsory blank cell for the staff to fill in the reason. After finishing, a staff can click on the “Hoàn tất và gửi đề xuất” button to send the request to the manager. If a staff does not want to delete anymore, he/she can click on the “Hủy bỏ” button to cancel the request.

141

Trang chủ

Danh sách tours

Danh sách các đề xuất

Phiếu thông tin tour

Tên chương trình: Du lịch Bà Nà Hills - Cầu Rồng - Bán Đảo Sơn Trà

Gía chương trình: 5.000.000 VND

Hành trình:

Xuất phát: Thành phố Hồ Chí Minh

Điểm đến 1: Bà Nà Hills

Điểm đến 2: Cầu Rồng

Điểm đến 3: Bán Đảo Sơn Trà

Thêm điểm dừng

Phương tiện:

Xe 4 chỗ

Xe 7 chỗ

Xe khách

Máy bay

Lịch trình:

Số ngày: 5

Ngày khởi hành: 1/1/2024

Mô tả chi tiết:

Thêm mô tả chi tiết

Ghi chú: Không có

Dịch vụ bao gồm:

Bảo hiểm

Bữa ăn

Hướng dẫn viên

Xe đưa đón

Vé tham quan

Nhân viên đảm nhận: Nguyễn Văn A

Have a nice trip ^^

Hoàn tất và gửi đề xuất

Hoàn tác

Figure 84. Design for tour edit request view

The Figure 80 illustrates the interface when clicking on the “Chỉnh sửa thông tin tour” button, the system will show an editable detailed information tour form. A staff member can change anything he/she wants. After finishing, a staff can click on the “Hoàn tất và gửi đề xuất” button to send the request to the manager. If a staff member does not edit anymore, he/she can click on the “Hoàn tác” button to cancel and keep the previous one.

142

Request management pages



Figure 85. Design for request management page

The Figure 85 shows the interface when clicking on the “Danh sách các đề xuất” option, the system will show a table of sent requests. A staff can click on the “Xem thêm...” option to see more details. A staff can find a specific tour by using a search bar.

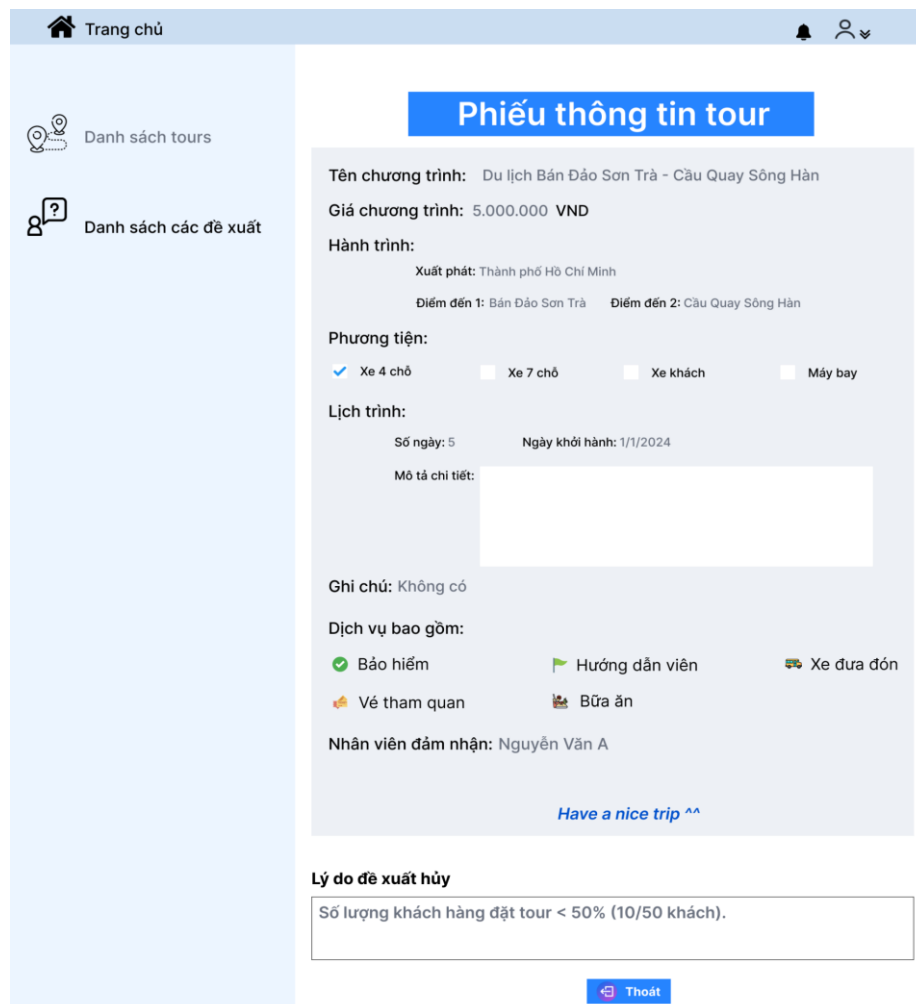


Figure 86. Design for view of a sent request (cancellation request)

The Figure 86 illustrates the interface when clicking on the “Xem thêm...” option of a request cancellation, the system will show the detailed information tour form along with the reason. In this case, a staff can only see the request and cannot have any operations. If a staff does not view anymore, he/she can click on the “Thoát” button to exit.

The screenshot displays a web application interface for managing tour requests. On the left, a sidebar contains navigation links: "Trang chủ" (Home), "Danh sách tours" (Tour List), and "Danh sách các đề xuất" (List of proposals). The main content area features a message box titled "Lý do từ chối" (Reason for rejection) stating: "Hệ thống hàng không đang gặp sự cố, không thể hỗ trợ trong chuyến du lịch này." (The flight system is experiencing a problem, cannot support this trip). Below this is a form titled "Phiếu thông tin tour" (Tour information form). The form includes fields for: "Tên chương trình" (Program name) with a value of "Du lịch Bà Nà Hills - Cầu Rồng - Cầu Quay Sông Hàn"; "Giá chương trình" (Program price) of "6.000.000 VND"; "Hành trình" (Itinerary) starting from "Thành phố Hồ Chí Minh" and ending at "Cầu Quay Sông Hàn"; "Phương tiện" (Transportation) with options for "Xe 4 chỗ", "Xe 7 chỗ", "Xe khách", and "Máy bay" (checked); "Lịch trình" (Schedule) with "Số ngày: 5" and "Ngày khởi hành: 1/1/2024"; "Ghi chú" (Remarks) set to "Không có" (None); "Dịch vụ bao gồm" (Services included) with checkboxes for "Bảo hiểm", "Hướng dẫn viên", "Xe đưa đón", "Vé tham quan", and "Bữa ăn"; and "Nhân viên đảm nhận" (Staff responsible) listed as "Nguyễn Văn A". At the bottom right, there is a "Chỉnh sửa" (Edit) button.

Figure 87. Design for view of a sent request (edit request)

The Figure 87 illustrates the interface when clicking on the “Xem thêm...” option of an edit request, the system will show a detailed information tour form. If there is any response from the manager, it will be shown above the tour form. Moreover, a staff can edit that edit request by clicking on the “Chỉnh sửa” button.

5.1.3. Wireframe for Manager View

5.1.3.1. UI Flow Diagram

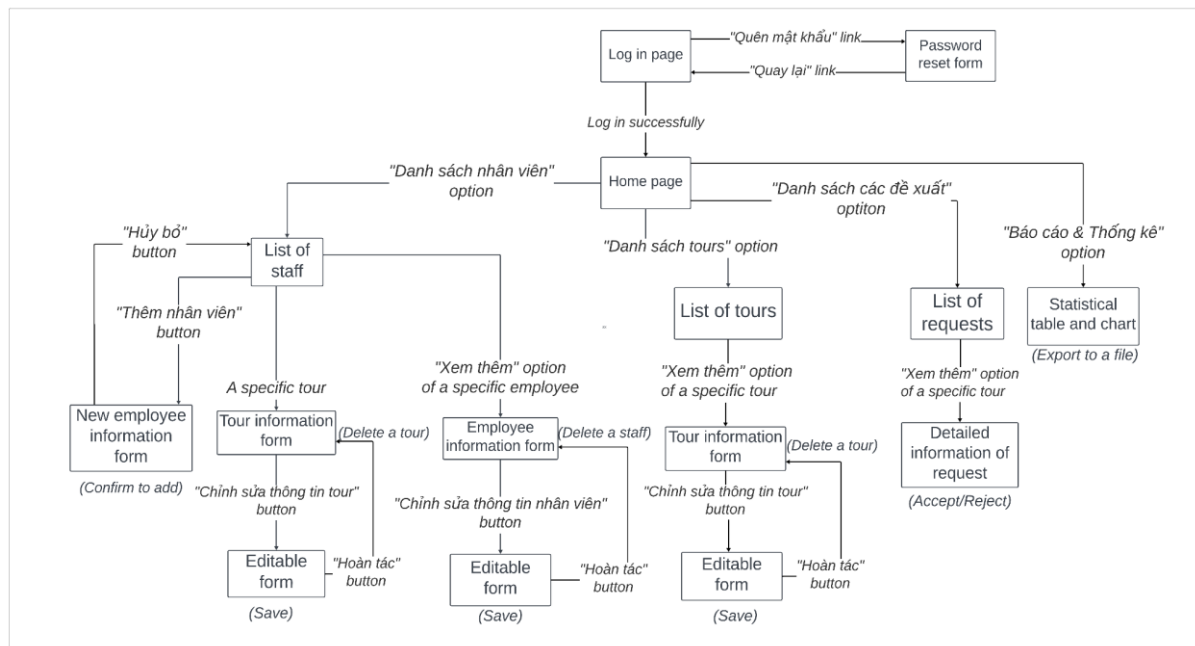


Figure 89. UI Flow Diagram for Manager View

The Figure 89 above illustrates how the manager can interact with the system. When the manager logs in successfully, he/she is brought to the main page view. He/she will have four main options - “Danh sách nhân viên”, “Danh sách tours”, “Danh sách các đề xuất”, and “Báo cáo & Thống kê”. For each different option, there will be different operations with the system. To understand more specifically how those operations are performed, we continue to analyze in the Detailed UI design part below.

5.1.3.2. Detailed UI design

There are two main parts for Manager UI: header and main.

Header



Figure 90. Design for Header in manager view

In Figure 90, from the left-hand side to the right-hand side, we have:

- Home icon: We use this icon to show that the manager is at the main page.
- Bell icon: We use this icon to show received notifications.
- Account icon: The manager can click on this icon and choose to log out of the system.

Main: The main includes the designs for authentication and main page designs (when Manager logins successfully)

- UI designs for Manager Authentication

Login form

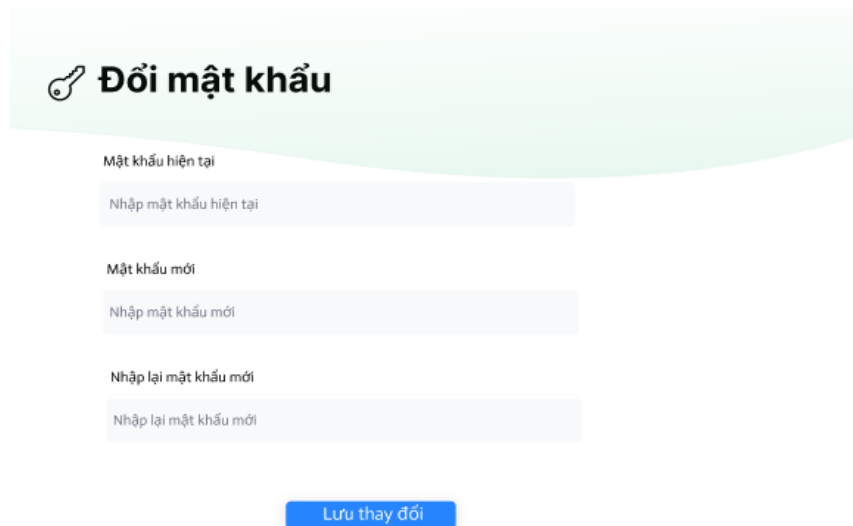
The Figure 76 illustrates the login interface when staff or manager access the web page. When the manager accesses the web page, the system will show the Login form. The manager is required to login successfully in order to use the system. If the manager forgets his/her password, he/she can click on “Quên mật khẩu” option to reset the password.

Password reset form

The Figure 54 illustrates the interface when clicking on the “Quên mật khẩu” option. Similar to the reset password reset form for customers, when the manager clicks on the “Quên mật khẩu” option, the system will show the form. The manager can fill his/her email address in the blank cell named “Email”. After finishing and clicking on the “Gửi” button. He/She will receive a link to reset his/her password via email address that he/she just sent before. If the manager recalls his/her password and does not want to change the password anymore, at the reset password form, he/she can click on the “Quay lại” option to go back.

- Main page designs for Manager (Login successfully)

Reset password page



The image shows a password change form titled "Đổi mật khẩu" (Change Password) with a key icon. It contains three input fields: "Mật khẩu hiện tại" (Current password), "Mật khẩu mới" (New password), and "Nhập lại mật khẩu mới" (Repeat new password). A blue button labeled "Lưu thay đổi" (Save changes) is at the bottom.

Figure 91. Design for password change

The Figure 91 illustrates the interface when the manager chooses the “Đổi mật khẩu” option of the account icon on the header bar. The system will show a password change form. The manager can change his/her password by entering the current password, the new password and re-typing the new password. The manager then clicks on the “Lưu thay đổi” button and the system will check the password if it satisfies the requirements before the password is successfully changed.

Staff management pages



The image shows a staff management page. On the left is a sidebar with icons and labels: "Trang chủ" (Home), "Danh sách nhân viên" (Staff list), "Danh sách tours" (Tours list), "Danh sách các đề xuất" (List of proposals), and "Báo cáo & Thống kê" (Reports & Statistics). The main content area is titled "Danh sách nhân viên" (Staff list) and includes a search bar labeled "Tìm kiếm". Below the search bar is a table with the following data:

ID	Họ	Tên	Các tour đảm nhận	Khác
1	Nguyễn	Văn A	- Tour Bà Nà Hills - Cầu Rồng - Bán... - Tour Bán Đảo Sơn Trà - Cầu Quay... - Tour Ngũ Hành Sơn - Suối khoáng... - Tour Cầu Rồng - Bảo Tàng Điêu K...	Xem thêm...
2	Nguyễn	Văn B	Không có	Xem thêm...

At the bottom of the table area, there is a blue button labeled "Thêm nhân viên" (Add staff) and the text "Tổng số: 2" (Total: 2).

Figure 92. Design for staff management page

The Figure 92 illustrates the interface when the Manager chooses the “Danh sách nhân viên” option. When clicking on the “Danh sách nhân viên” option, the system will show

a table of staff along with tours they take over. The manager can find a specific staff member easily by using a search bar. Furthermore, the manager can also add a new staff member, and view detailed information about a specific staff member/tour by clicking on the “Thêm nhân viên” button, “Xem thêm...” option, and the name of a tour, respectively.

The screenshot shows the 'Thêm nhân viên' (Add Staff) form. On the left is a sidebar with icons and labels: 'Trang chủ' (Home), 'Danh sách nhân viên' (Staff List), 'Danh sách tours' (Tours List), 'Danh sách các đề xuất' (List of suggestions), and 'Báo cáo & Thống kê' (Reports & Statistics). The main content area has a blue header 'Phiếu thông tin nhân viên' (Staff Information Form). Below the header are input fields for 'Họ:' (Last name), 'Tên:' (First name), 'Giới tính:' (Gender) with radio buttons for 'Nam' (Male) and 'Nữ' (Female), 'ID: 2', 'Ngày/tháng/năm sinh:' (Date of birth) with a date picker, 'SDT liên lạc:' (Contact phone number), and 'Email:'. At the bottom are two buttons: 'Xác nhận thêm' (Confirm add) in blue and 'Hủy bỏ' (Cancel) in red.

Figure 93. Design for adding a new staff member function

The Figure illustrates 93 the interface when choosing the “Thêm nhân viên” button. When clicking on the “Thêm nhân viên” button, the system will show a form for the manager to fill in. After finishing, the manager can click on the “Xác nhận thêm” button to add a new staff member. If the manager does not want to add anymore, he/she can click on the “Hủy bỏ” to cancel the form.

The screenshot shows the 'Thông tin nhân viên' (Staff Information) view. The sidebar is identical to Figure 93. The main content area has a blue header 'Thông tin nhân viên' (Staff Information). Below the header, the staff member's details are displayed: 'Họ: Nguyễn', 'Tên: Văn A', 'Giới tính: Nam', 'ID: 1', 'Ngày/tháng/năm sinh: 1/12/1986', 'SDT liên lạc: 0123456789', and 'Email: a.nguyen@kdbulich.vn'. Below the details are two buttons: 'Chỉnh sửa thông tin nhân viên' (Edit staff information) in blue and 'Xóa' (Delete) in red. At the bottom, there is a section 'Các tours đảm nhận:' (Tours assigned:) followed by a list of tours: 'Tour Bà Nà Hills - Cầu Rồng - Bán Đảo Sơn Trà', 'Tour Bán Đảo Sơn Trà - Cầu Quay Sông Hàn', 'Tour Ngũ Hành Sơn - Suối khoáng nóng Thần Tài', and 'Tour Cầu Rồng - Bảo tàng Điêu Khắc Chăm'.

Figure 94. Design for staff information view (case 1)

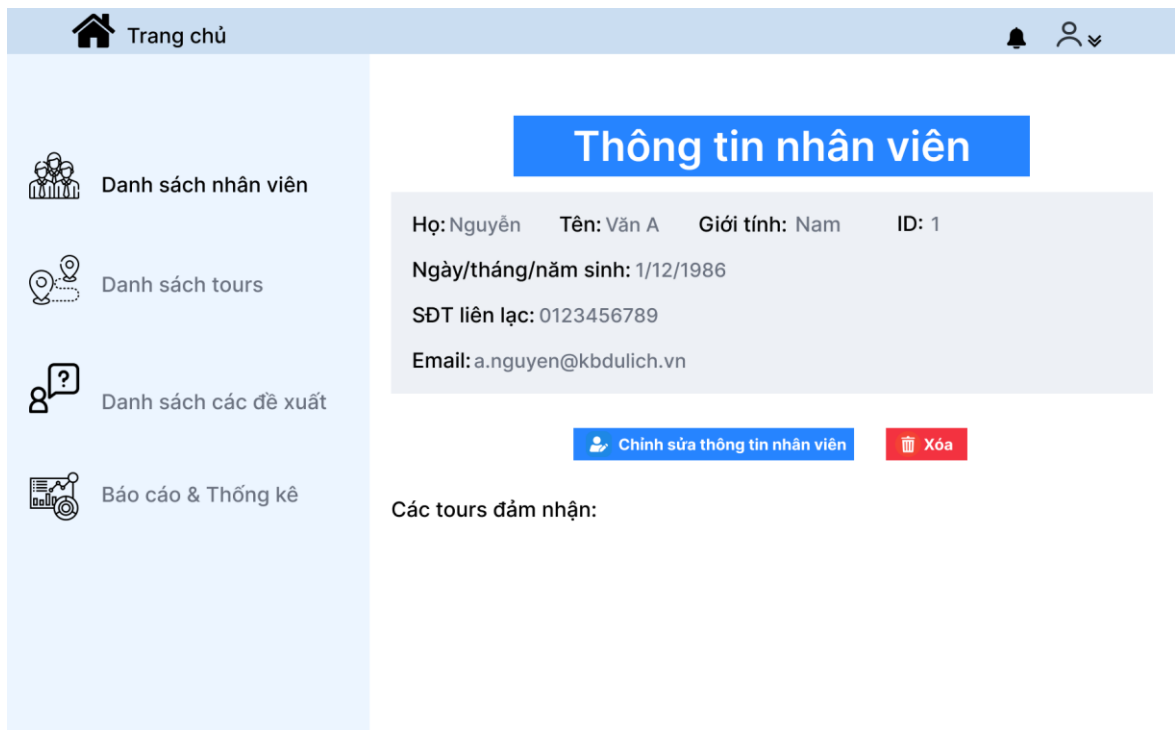


Figure 95. Design for staff information view (case 2)

The Figure 94 and 95 illustrate the interface when clicking on the “Xem thêm” option of a specific staff member. If the chosen staff member takes over no tour, the interface will be *case 2*. If the chosen staff member takes over at least one tour, the interface will be *case 1*. As we can see one more difference between the two cases is that if the chosen staff member takes over at least one tour, the “Xóa” button will be deactivated. The “Xóa” button can only be in use if the chosen staff member takes over no tour. At that time, the manager can click on the “Xóa” button to delete the chosen staff member.

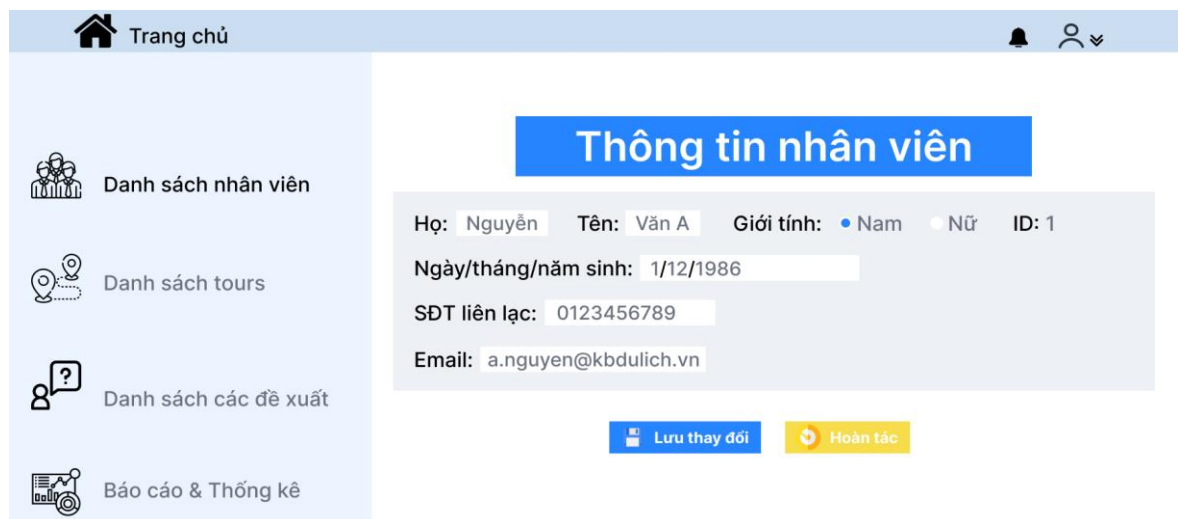


Figure 96. Design for editing staff information

The Figure 96 illustrates the interface when clicking on the “Chỉnh sửa thông tin nhân viên” button, the system will show a editable information form of the chosen staff member. The manager can change anything he/she wants. After finishing, the manager can click on the “Lưu thay đổi” button to save new changes to the system. If the manager does not want to edit anymore and keeps the previous one, he/she can click on the “Hoàn tác” button.

Tour management pages

Mã tour	Tên tour	Nhân viên đảm nhận	Khác
T1	Du lịch Bà Nà Hills - Cầu Rồng - Bán ...	Nguyễn Văn A	Xem thêm...
T2	Du lịch Bán Đảo Sơn Trà - Cầu Quay ...	Nguyễn Văn A	Xem thêm...

Tổng số: 2

Figure 97. Design for tour list management page

The Figure 97 illustrates the interface when clicking on the “Danh sách tours” option, the system will show a table of created tours along with the names of staff who take over them. The manager can find a specific tour easily by using a search bar. Furthermore, the manager can also view detailed information about a tour by clicking on the “Xem thêm” option of a specific tour. All operations after clicking on the “Xem thêm” option are the same as the A specific tour part we just analyzed above.

Trang chủ

Danh sách nhân viên

Danh sách tours

Danh sách các đề xuất

Báo cáo & Thống kê

Phiếu thông tin tour

Tên chương trình: Bà Nà Hills - Cầu Rồng - Bán Đảo Sơn Trà

Giá chương trình: 5.000.000 VND

Hành trình:

Xuất phát: Thành phố Hồ Chí Minh

Điểm đến 1: Bà Nà Hills
Điểm đến 2: Cầu Rồng
Điểm đến 3: Bán Đảo Sơn Trà

Phương tiện:

☒ Xe 4 chỗ
☐ Xe 7 chỗ
☐ Xe khách
☐ Máy bay

Lịch trình:

Số ngày: 5
Ngày khởi hành: 1/1/2024

Mô tả chi tiết:

Ghi chú: Không có

Dịch vụ bao gồm:

☒ Bảo hiểm
☒ Hướng dẫn viên
☒ Xe đưa đón

☒ Vé tham quan
☒ Bữa ăn

Nhân viên đảm nhận: Nguyễn Văn A

Have a nice trip ^^

Chỉnh sửa thông tin tour

Xóa tour

Figure 98. Design for view of tour details

The Figure 98 illustrates the interface when choosing a specific tour option, the system will show detailed information on the tour form. The manager can view the tour or operate with the tour by clicking on the “Chỉnh sửa thông tin tour” or “Xóa tour” button. If the manager determines that the chosen tour is no longer needed and he/she wants to remove it, he/she can click on the “Xóa tour” button.

Trang chủ

Danh sách nhân viên

Danh sách tours

Danh sách các đề xuất

Báo cáo & Thống kê

Phiếu thông tin tour

Tên chương trình: Bà Nà Hills - Cầu Rồng - Bán Đảo Sơn Trà

Gía chương trình: 5.000.000 VND

Hành trình:

Xuất phát: Thành phố Hồ Chí Minh

Điểm đến 1: Bà Nà Hills

Điểm đến 2: Cầu Rồng

Điểm đến 3: Bán Đảo ...

Thêm điểm dừng

Phương tiện:

Xe 4 chỗ

Xe 7 chỗ

Xe khách

Máy bay

Lịch trình:

Số ngày: 5

Ngày khởi hành: 1/1/2024

Mô tả chi tiết:

Thêm mô tả chi tiết

Ghi chú: Không có

Dịch vụ bao gồm:

Bảo hiểm

Bữa ăn

Hướng dẫn viên

Xe đưa đón

Vé tham quan

Nhân viên đảm nhận: Nguyễn Văn A

Have a nice trip ^^

Lưu thay đổi

Hoàn tác

Figure 99. Design for tour details editing form

The Figure 99 illustrates the interface when clicking on the “Chỉnh sửa thông tin tour” button, the system will show an editable form for the manager to edit. The manager can change anything he/she wants. After finishing, the manager can click on the “Lưu thay đổi” button to save new changes and update to the system. If the manager does not want to edit the tour anymore, he/she can click on the “Hoàn tác” button to cancel the editable form and keeps the previous one.

153

Request management pages

Trang chủ

Danh sách nhân viên

Danh sách tours

Danh sách các đề xuất

Báo cáo & Thống kê

Danh sách các đề xuất

Danh sách các đề xuất

Tìm kiếm

Mã tour	Tên tour	Nhân viên đảm nhận	Khác
T1	Tour Bà Nà Hills - Cầu Rồng - Bán ...	Nguyễn Văn A	Xem thêm...
T2	Tour Bán Đảo Sơn Trà - Cầu Quay ...	Nguyễn Văn A	Xem thêm...

Tổng số: 2

Figure 100. Design for requests management page

The Figure 100 illustrates the interface when clicking on the “Danh sách các đề xuất” option, the system will show a table of requests sent from staff members. The manager can find a specific request easily by using a search bar. Furthermore, the manager can also view more detailed information about a request by clicking on the “Xem thêm” option of a specific request.

Trang chủ

Danh sách nhân viên

Danh sách tours

Danh sách các đề xuất

Báo cáo & Thống kê

Phiếu thông tin tour

Tên chương trình: Bà Nà Hills - Cầu Rồng - Bán Đảo Sơn Trà

Giá chương trình: 5.000.000 VND

Hành trình:

Xuất phát: Thành phố Hồ Chí Minh

Điểm đến 1: Bà Nà Hills Điểm đến 2: Cầu Rồng Điểm đến 3: Bán Đảo Sơn Trà

Phương tiện:

☒ Xe 4 chỗ
☐ Xe 7 chỗ
☐ Xe khách
☐ Máy bay

Lịch trình:

Số ngày: 5 Ngày khởi hành: 1/1/2024

Mô tả chi tiết:

Ghi chú: Không có

Dịch vụ bao gồm:

☒ Bảo hiểm
☒ Hướng dẫn viên
☒ Xe đưa đón

☒ Vé tham quan
☒ Bữa ăn

Nhân viên đảm nhận: Nguyễn Văn A

Have a nice trip ^^

Lý do đề xuất hủy:

Số lượng khách hàng đăng kí ít hơn 50% (10/50 khách).

Chấp nhận hủy

Từ chối

Figure 103. Design for request details view (cancellation request)

The Figure 103 illustrates the interface when the manager chooses the “Xem thêm” option of a specific request (request cancellation), the system will show detailed information on the tour along with the reason why the staff request to delete the tour. If the manager accepts to delete the request, he/she can click on the “Chấp nhận hủy” button. If the manager still wants to keep the tour and refuses to delete it, he/she can click on the “Từ chối” button.

Trang chủ

Danh sách nhân viên

Danh sách tours

Danh sách các đề xuất

Báo cáo & Thống kê

Phiếu thông tin tour

Tên chương trình: Bà Nà Hills - Cầu Rồng - Bán Đảo Sơn Trà

Giá chương trình: 5.000.000 VND

Hành trình:

Xuất phát: Thành phố Hồ Chí Minh

Điểm đến 1: Bà Nà Hills Điểm đến 2: Cầu Rồng Điểm đến 3: Bán Đảo Sơn Trà

Phương tiện:

☒ Xe 4 chỗ
☐ Xe 7 chỗ
☐ Xe khách
☐ Máy bay

Lịch trình:

Số ngày: 5 Ngày khởi hành: 1/1/2024

Mô tả chi tiết:

Ghi chú: Không có

Dịch vụ bao gồm:

☒ Bảo hiểm
☒ Hướng dẫn viên
☒ Xe đưa đón

☒ Vé tham quan
☒ Bữa ăn

Nhân viên đảm nhận: Nguyễn Văn A

Have a nice trip ^^

Lý do đề xuất hủy:

Số lượng khách hàng đăng kí ít hơn 50% (10/50 khách).

*Lý do từ chối

Hoàn tất

Hủy

Figure 104. Design for view to reject a cancellation request

The Figure 104 illustrates the interface when the manager chooses the “Từ chối” button of the cancellation request, the next operation will be the same as the operation when the manager chooses the “Từ chối” button of the edit request. He/she also has to fill in the reason why he/she rejected the request. After finishing, the manager clicks on the “Hoàn tất” button to send the response to the staff. If the manager does not want to reject the request anymore, he/she can click on the “Hủy” button.

Dashboard

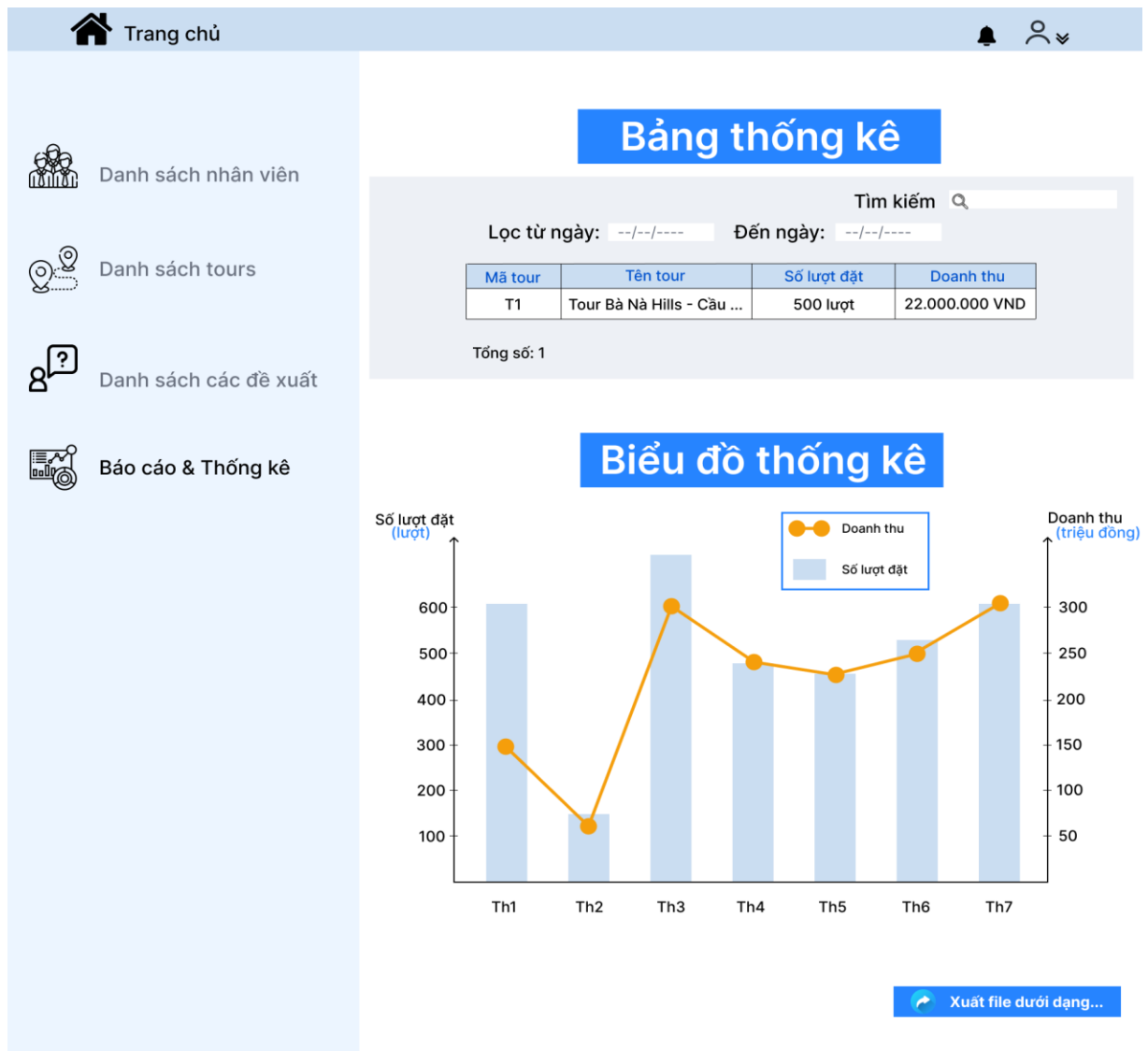


Figure 105. Design for report and statistics page

The Figure 105 illustrates the interface when clicking on the “Báo cáo & Thống kê” option, the system will show a statistical table along with a statistical diagram. The manager can search for a specific tour by using a search bar or uses the filter “Lọc từ ngày” - “Đến ngày” to find many tours at the same time in the same period of time. Furthermore, the manager can export the statistical table and statistical diagram as a hard copy by clicking on the “Xuất file dưới dạng” button.

5.2. Application building

After analyzing and designing the system, we start setting up the code base and implementing some interfaces for authentication, the header, the footer and the homepage for Guest. However, due to time constraints, we will continue to implement it in the next phase.

Authentication interface implementation

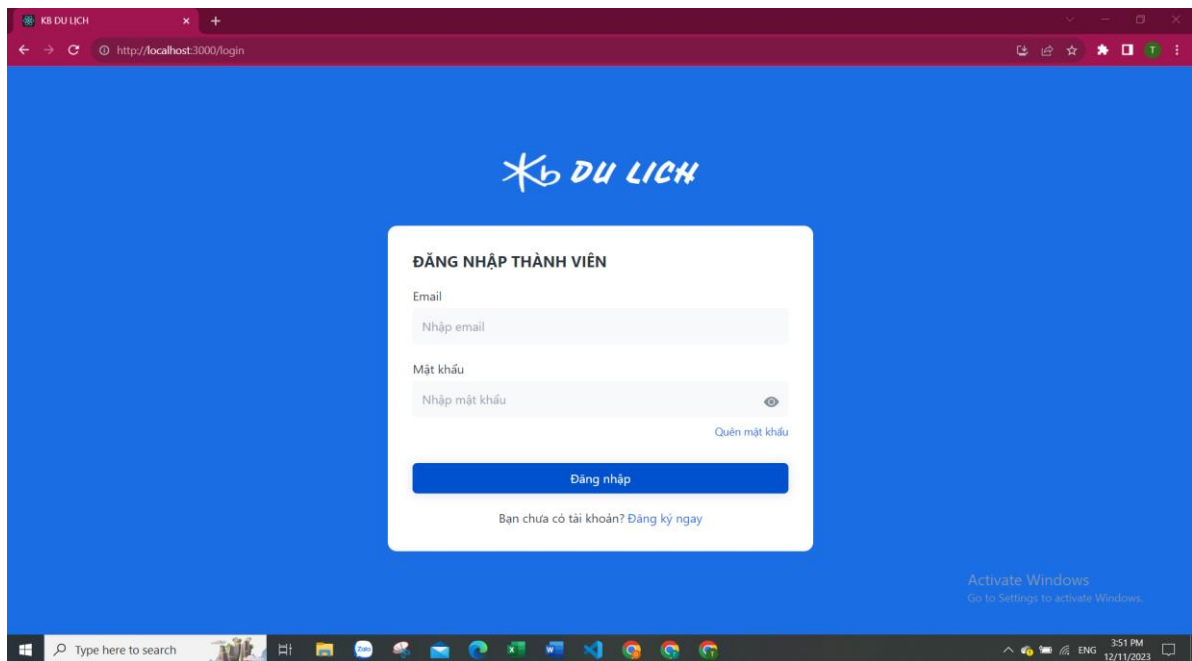


Figure 106. Login form implementation

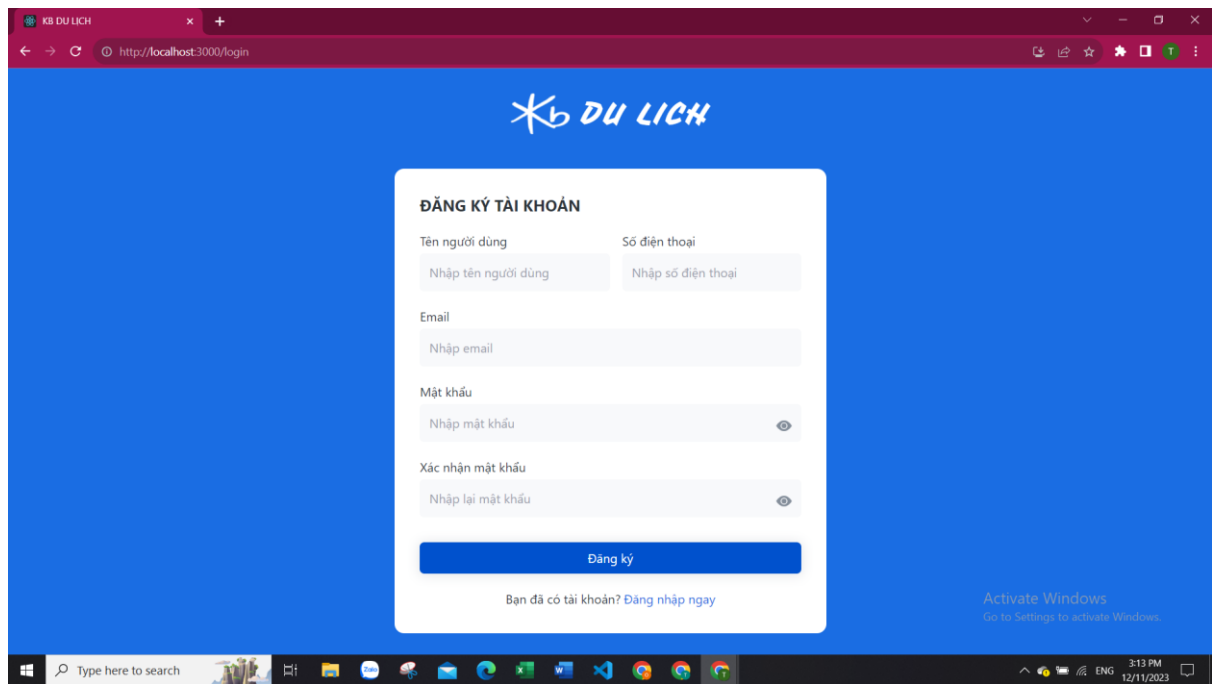


Figure 107. Registration form implementation

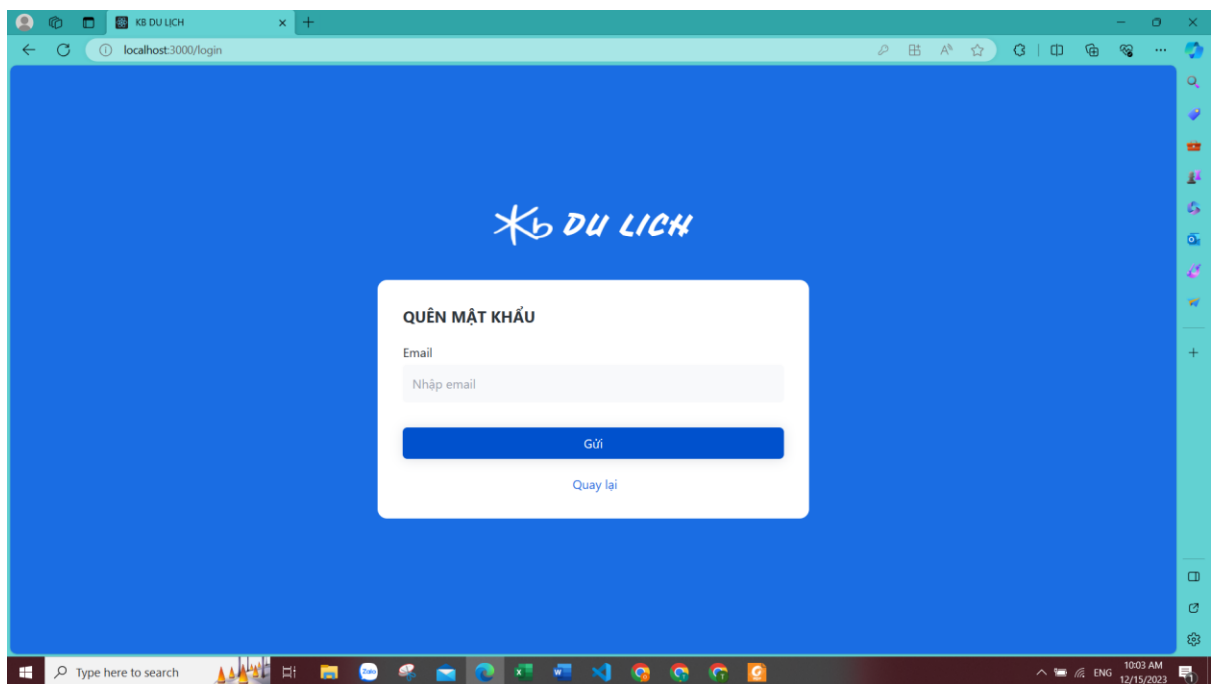


Figure 108. Forgot password form implementation

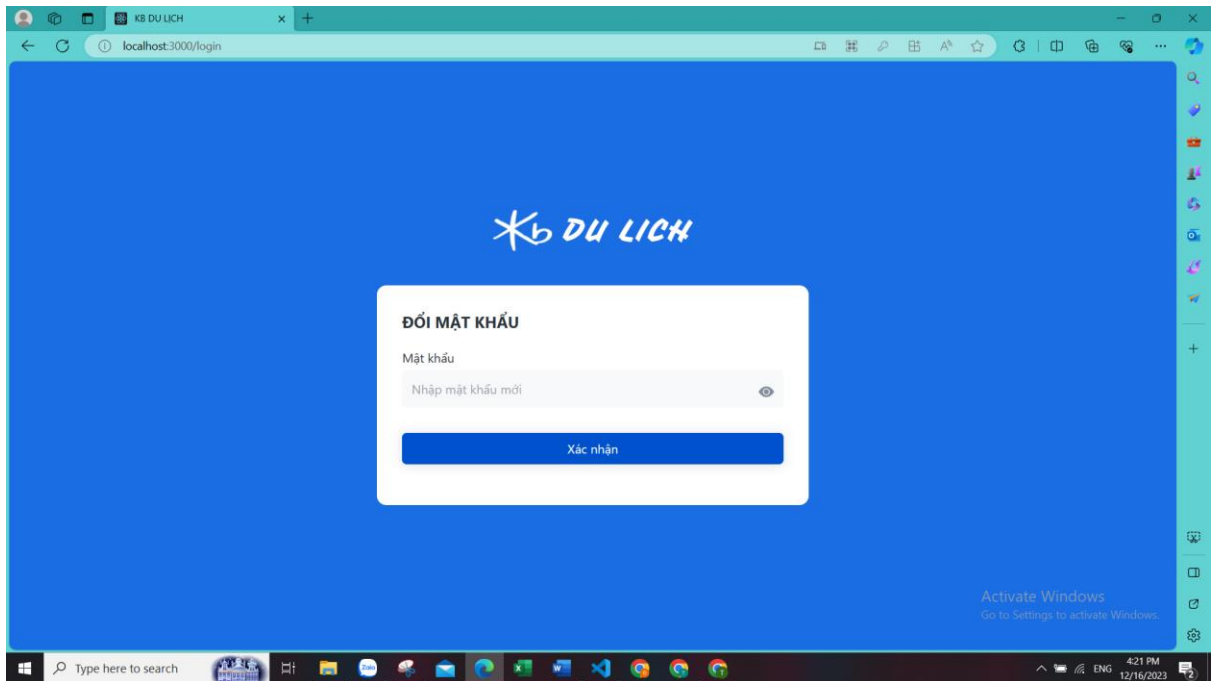


Figure 109. Reset password form implementation for the first login of Staff

Header and Footer implementation

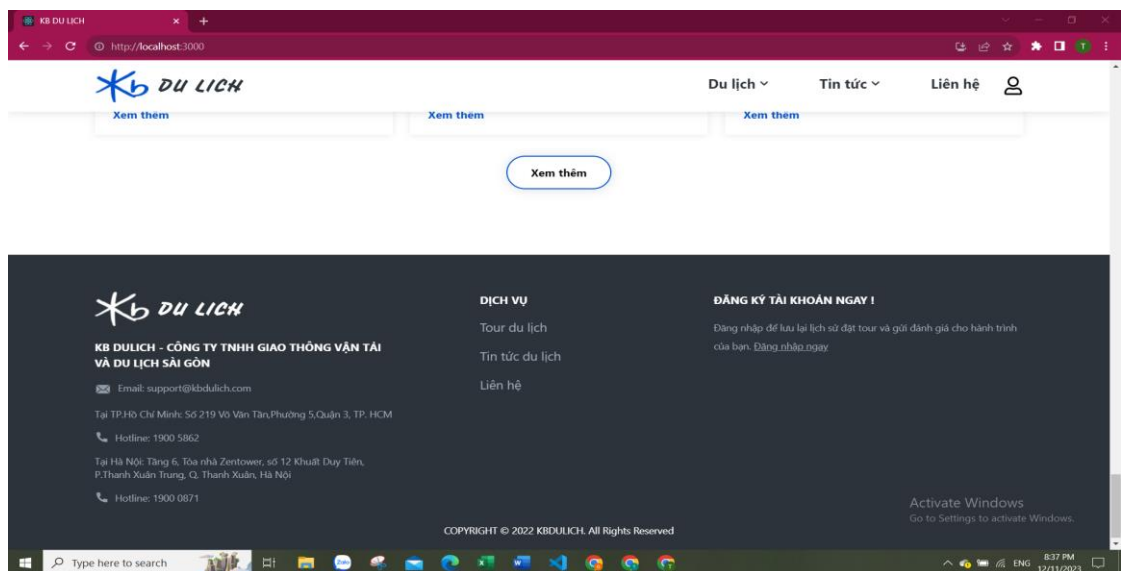


Figure 110. Header and Footer implementation

Homepage for Guest implementation

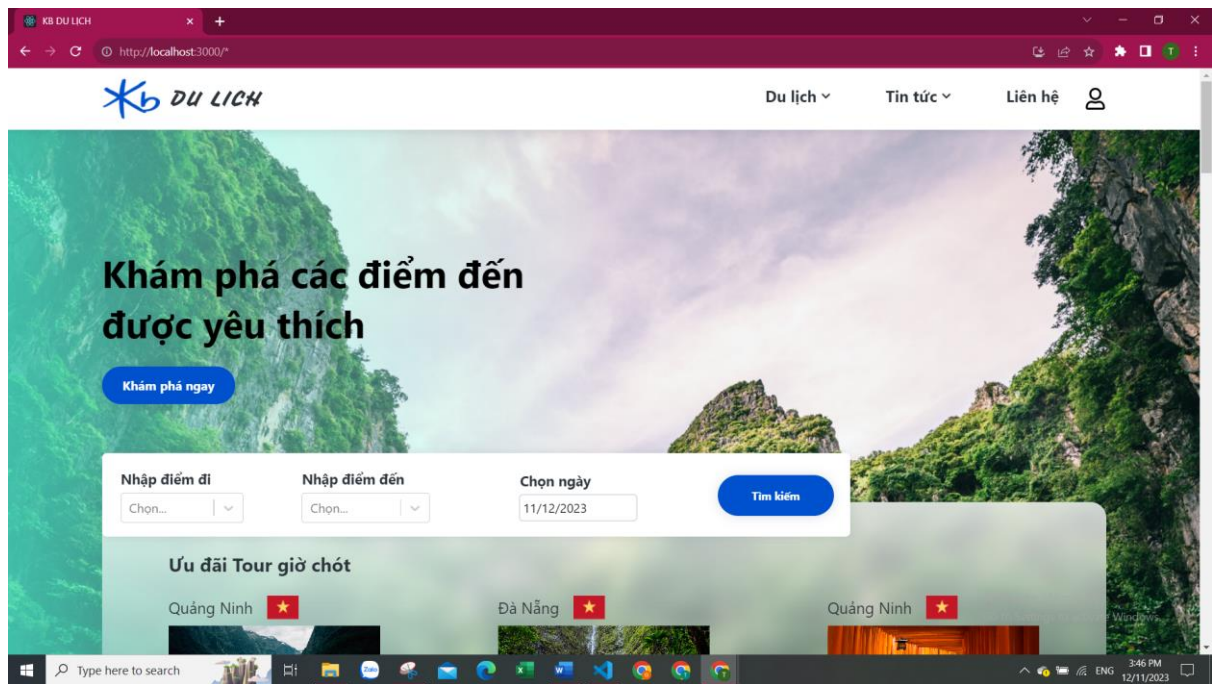


Figure 111. Homepage implementation - First section

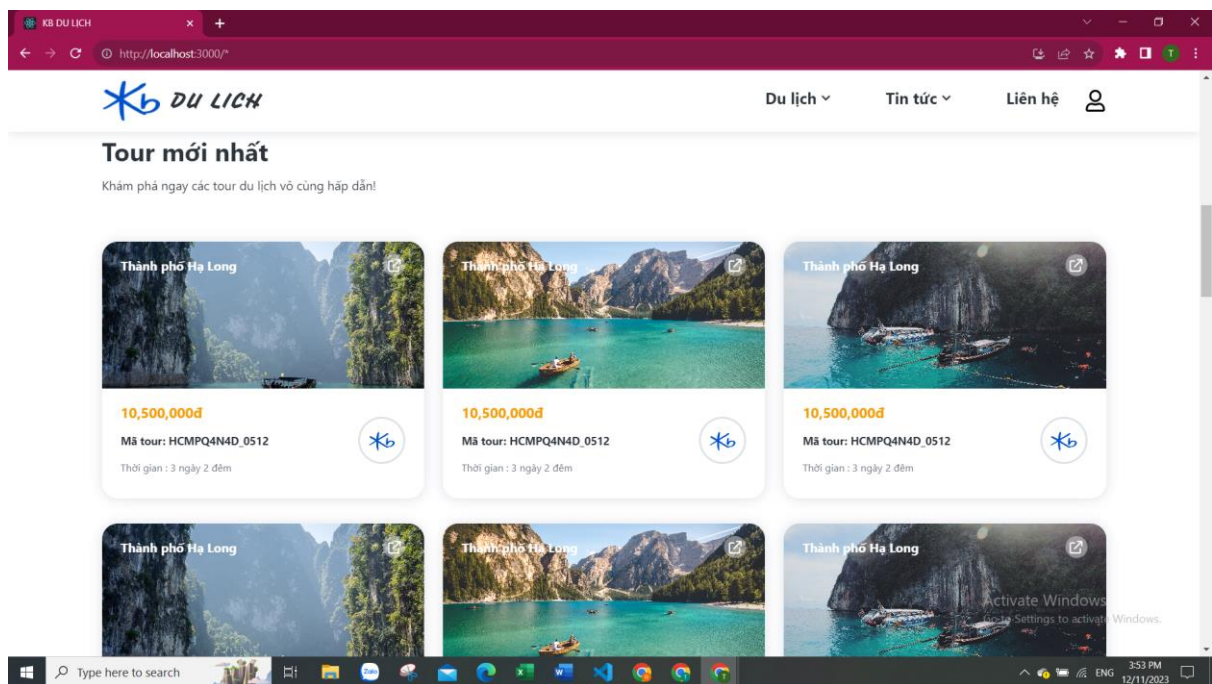


Figure 112. Homepage implementation - Second section

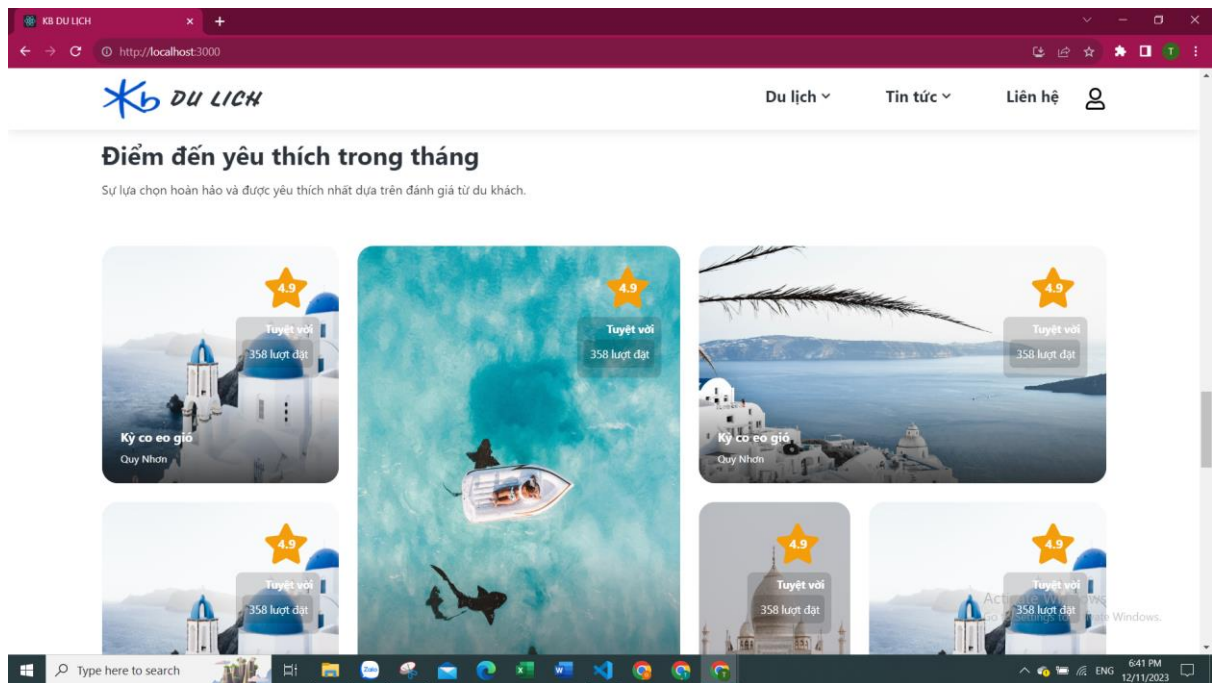


Figure 113. Homepage implementation - Second section

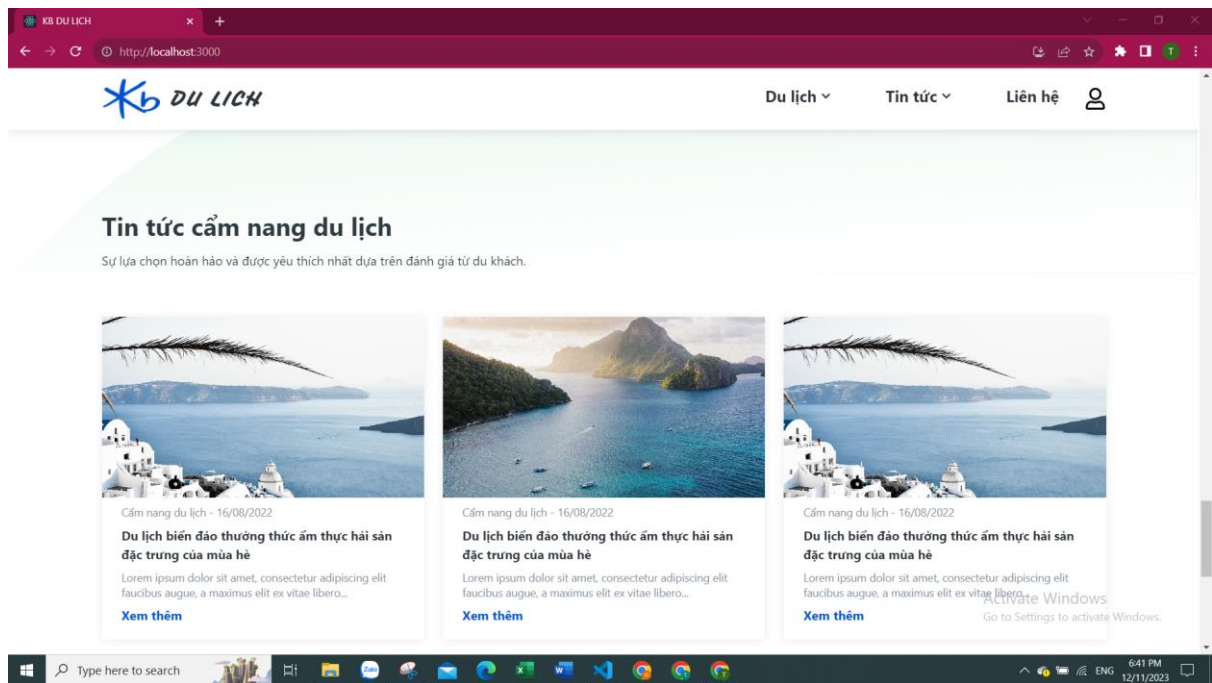


Figure 114. Homepage implementation - Fourth section

Chapter 6

Summary and Future plan

6.1. Task evaluation

During this specialized project, we have achieved the following objectives set by the thesis goals and motivation:

- Core Features for the Tourism application such as Online tour booking, Tourist attraction and tour suggestion, Tour management, etc.
- System architecture to verify that the quality functional and nonfunctional requirements have been mentioned in the design.
- User Interface for every stakeholder in the applications.
- Implementation of some user interfaces

6.2. Future plan

The Table 27 illustrates the project timeline for the next phase. In this phase, we will focus on system implementation and testing. In the first two weeks, we are going to set up the environment for client and server side as well as generate the dataset for our application. The next step is to implement all the interface and features for our system. This step takes around 8 weeks. Finally, we will test if the system meets all the requirements and start deploying the system. In fact, we may have to start testing the system while coding, for example, the API testing. Besides, the report will be updated weekly with completed tasks.

Table 27. Implementation plan for the next phase

		Week														
Task	Assigned to	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Code base initiation and setup																
Client side setup	Trần															
Server side setup	Khanh															
Generate data for database	Trần															
System development																
Create database	Linh															
Application API	Khanh, Trần, Linh															
Authentication UI	Trần															
Implement Authentication + Authorization	Khanh, Trần, Linh															
Customer + Manager UI	Trần, Linh															
Staff UI	Linh															
Search suggestion	Linh															
Implement features	Khanh, Trần, Linh															
Testing and deployment																
Testing and fixing errors	Trần, Linh, Khanh															
Deployment	Khanh															
Writing report	Khanh, Trần, Linh															
Preparing slides	Khanh, Trần, Linh															

References

- [1] MDN, "Responsive design - Learn web development," 5 July 2023. [Online]. Available: https://developer.mozilla.org/en-US/docs/Learn/CSS/CSS_layout/Responsive_Design. [Accessed 5 Nov 2023].
- [2] A. S. Gillis, "Data binding," WhatIs, 6 May 2022. [Online]. Available: <https://www.techtarget.com/whatis/definition/data-binding>. [Accessed 17 Dec 2023].
- [3] A. Kumar, "n Introduction to MVCS architecture," Quantiphi, Inc, 31 March 2021. [Online]. Available: <https://quantiphi.com/an-introduction-to-mvcs-architecture/>. [Accessed 17 Dec 2023].
- [4] M. Kanetscheider, "A Python Implementation of the Unit of Work and Repository Design Pattern using SQLAlchemy," DEV Community, 22 May 2023. [Online]. Available: <https://dev.to/manukanne/a-python-implementation-of-the-unit-of-work-and-repository-design-pattern-using-sqlmodel-3mb5>. [Accessed 12 Dec 2023].
- [5] Selleo, "Fundamentals of Front-End Development | A guide to key technologies and concepts," Selleo, 17 October 2023. [Online]. Available: <https://selleo.com/blog/fundamentals-of-front-end-development>. [Accessed 16 Dec 2023].
- [6] D. Garbar, "Why use Front-end Frameworks," Belitsoft, 4 April 2023. [Online]. Available: <https://belitsoft.com/what-do-front-end-frameworks-do>. [Accessed 16 Dec 2023].
- [7] AngularJS, "AngularJS: Developer Guide: Introduction," AngularJS, [Online]. Available: <https://docs.angularjs.org/guide/introduction>. [Accessed 17 Nov 2023].
- [8] V. Kanade, "Understanding AngularJS, its benefits and challenges," Spiceworks, 14 March 2023. [Online]. Available: <https://www.spiceworks.com/tech/devops/articles/what-is-angular-js/>. [Accessed 17 Dec 2023].

- [9] I. Sharma, "What is Vue js? The Pros and Cons of Vue.JS," Software and Technology Blog - TatvaSoft, [Online].
Available: <https://www.tatvasoft.com/outsourcing/2021/10/what-is-vue-js-and-its-benefits.html>. [Accessed 17 Dec 2023].
- [10] J. Desai, "10 Main advantages of React JS Development," Positiwise, 31 August 2023. [Online]. Available: <https://positiwise.com/blog/10-main-advantages-of-react-js-development>. [Accessed 17 Dec 2023].
- [11] javatpoint, "Pros and Cons of ReactJS," javatpoint, [Online]. Available: <https://www.javatpoint.com/pros-and-cons-of-react>. [Accessed 17 Dec 2023].
- [12] React, "Components and props – react," React, [Online]. Available: <https://legacy.reactjs.org/docs/components-and-props.html>. [Accessed 17 Dec 2023].
- [13] GeeksforGeeks, "ReactJS components," GeeksforGeeks, 21 Nov 2023. [Online]. Available: <https://www.geeksforgeeks.org/reactjs-components/>. [Accessed 17 Nov 2023].
- [14] "What is react-router-dom ?," GeeksforGeeks, 25 May 2023. [Online]. Available: <https://www.geeksforgeeks.org/reactjs-components/>. [Accessed 17 Dec 2023].
- [15] React Router, "Feature Overview v6.21.0," React Router, [Online]. Available: <https://reactrouter.com/en/main/start/overview>. [Accessed 17 Dec 2023].
- [16] React, "API | APIS | memo," React, [Online]. Available: <https://react.dev/reference/react/memo>. [Accessed 17 Nov 2023].
- [17] Redux, "Redux Fundamentals, Part 3: State, Actions, and Reducers," Redux, 25 November 2023. [Online].
Available: <https://redux.js.org/tutorials/fundamentals/part-3-state-actions-reducers>. [Accessed 17 Dec 2023].
- [18] "Axios in React a guide for beginners," GeeksforGeeks, 22 Nov 2023. [Online]. Available: <https://www.geeksforgeeks.org/axios-in-react-a-guide-for-beginners/>. [Accessed 17 Dec 2023].
- [19] "Utility-First Fundamentals - Tailwind CSS," Tailwind CSS, [Online]. Available: <https://tailwindcss.com/docs/utility-first>. [Accessed 17 Dec 2023].

- [20] S. Das, "Top 10 API testing tools in 2023 | BrowserStack.," BrowserStack., 26 July 2023. [Online]. Available: <https://www.browserstack.com/guide/top-api-testing-tools>.
- [21] "Why artillery?," Artillery Docs., [Online]. Available: <https://www.artillery.io/docs/get-started/why-artillery>.
- [22] T. Hamilton, "How to use JMeter for Performance & load testing.," Guru99., 17 November 2023. [Online]. Available: <https://www.guru99.com/jmeter-performance-testing.html>.
- [23] "Top 10 HTTP benchmarking and load testing tools.," thechief.io., [Online]. Available: <https://thechief.io/c/editorial/top-10-http-benchmarking-and-load-testing-tools/>.
- [24] "What is the use of middleware Redux thunk," GeeksforGeeks, 14 May 2022. [Online]. Available: <https://www.geeksforgeeks.org/what-is-the-use-of-middleware-redux-thunk/>. [Accessed 17 Dec 2023].

Appendix

1. Work contribution

In this Specialized project, each member will be responsible for a part of the work. However, after each part is done, other members will review and give opinions to adjust to be the most suitable.

Table 28. Work contribution

No	Full name	Workload
1	Nguyễn Mỹ Khanh	2.1.2. Related works - Service management software 3.1. Draw Use-case diagram 3.3. Sequence diagram 3.4. Draw EERD 3.5.2. Architecture diagram 3.6. Component diagram 4.2. Server development research 4.3. Database management system research
2	Trần Gia Linh	1.1. Motivation 1.2. The Project's Scope and Objectives 1.3. Project structure 2.2. Stakeholders 3.1. Draw and edit Use-case diagram 3.1. Use-case diagram explanation 3.1.1. Guest/ Customer Use Case Table Description 3.1.2. Staff Use Case Table Description 3.1.3. Manager Use Case Table Description 3.4.2. Relational mapping 4.4. Testing technology research 5.1.2. Wireframe for Staff View 5.1.3. Wireframe for Manager View

3	Phạm Quỳnh Trân	Declaration of Authenticity + Acknowledgement Abstract System flow description 2.1.1. Related works – Tourism websites 2.3. Functional & Non-functional requirements 3.1. Draw and edit Use-case diagram 3.2. Activity diagram 3.4. Draw and edit EERD 3.4. EERD explanation 3.5.1. Architectural pattern 3.7. Class diagram 4.1. Front-end development research 5.1.1. Wireframe for Customer View 5.2. Application building 6. Task evaluation and Future plan Appendix Report editing
---	-----------------	--

2. Work process

		Week													
Task	Assigned to	38	39	40	41	42	43	44	45	46	47	48	49	50	
Declaration of Authenticity + Acknowledgement	Trần														
Requirement elicitation															
System flow description	Trần														
Abstract	Trần														
Introduction motivation, scope and objectives	Linh														
Project structure	Linh														
Related work research	Trần, Khanh														
Functional & Non-functional requirements	Trần														

		Week													
Task	Assigned to	38	39	40	41	42	43	44	45	46	47	48	49	50	
System design															
Use-case diagram	Linh, Trần, Khanh														
Use-case diagram explanation	Linh														
Use-case scenarios	Linh, Trần, Khanh														
Activity diagram + explain	Trần														
Sequence diagram + explain	Khanh														
Architectural pattern + explain	Trần														
Class diagram + explain	Trần														
Architecture diagram + explain	Khanh														
Component diagram + explain	Khanh														
Design EERD	Khanh, Trần														
EERD explanation	Trần														
Relational mapping	Linh														
Customer UI flow diagram	Trần														
Design Customer UI + explain	Trần														
Staff and Manager UI flow diagram	Linh														
Design Staff UI + explain	Linh														
Design Manager UI + explain	Linh														

		Week													
Task	Assigned to	38	39	40	41	42	43	44	45	46	47	48	49	50	
TechoLOGY research															
Front-end development research	Trần														
Server development research	Khanh														
Database management system research	Khanh														
Testing research	Linh														
Application building	Trần														
Task evaluation and Future plan	Trần														